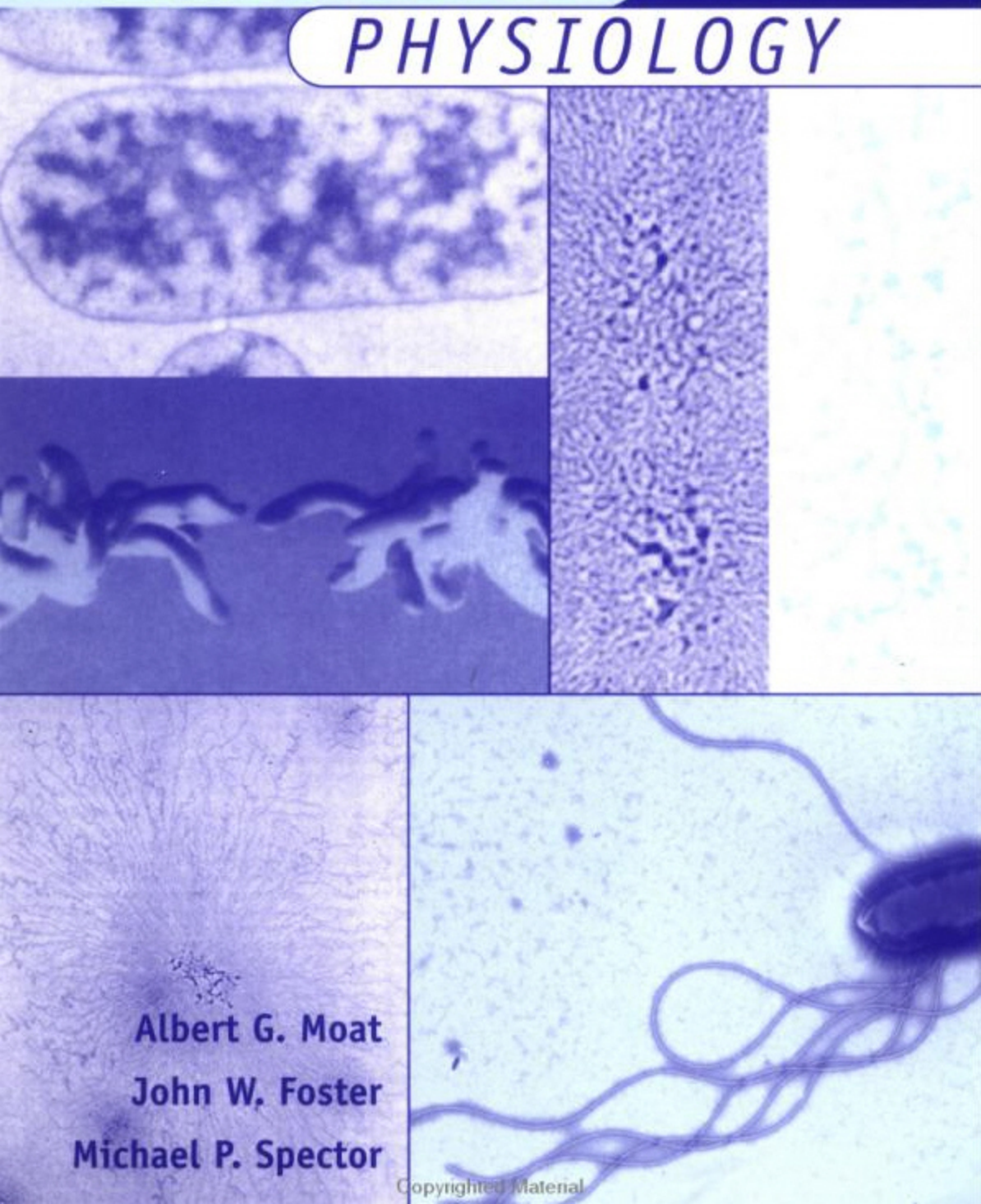




MICROBIAL

fourth edition

PHYSIOLOGY



Albert G. Moat
John W. Foster
Michael P. Spector

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CONTENTS

PREFACE

xix

1 INTRODUCTION TO MICROBIAL PHYSIOLOGY

1

The *Escherichia coli* Paradigm / 1

Cell Structure / 1

 The Cell Surface / 1

Synthesis of DNA, RNA, and Protein / 7

 Metabolic and Genetic Regulation / 10

Microbial Genetics / 11

Chemical Synthesis / 12

 Chemical Composition / 12

 Energy / 13

 Oxidation–Reduction Versus Fermentation / 15

 Nitrogen Assimilation / 18

Special Topics / 19

 Endospores / 19

 Growth / 19

 Continuous Culture / 22

Factors Affecting Growth / 22

 Nutrition / 22

 Oxygen / 24

 Carbon Dioxide / 24

Extremophiles / 25
Microbial Stress Responses / 26
Summary / 26

2 MACROMOLECULAR SYNTHESIS AND PROCESSING: DNA, RNA, AND PROTEIN SYNTHESIS

27

Structure of DNA / 28
Bacterial Nucleoids / 31
REP Elements / 35
DNA Replication / 36
DNA Replication is Bidirectional and Semiconservative / 36
DNA Polymerase Functions as a Dimer / 36
Model of DNA Replication / 39
Initiation of DNA Replication / 42
Termination of DNA Replication and Chromosome Partitioning / 46
RNA Synthesis: Transcription / 47
RNA Synthesis / 47
RNA Turnover / 53
RNA Processing / 54
Protein Synthesis: Translation / 57
Transfer RNA / 59
Charging of tRNA / 62
Ribosome Structure and Synthesis / 62
Initiation of Polypeptide Synthesis / 68
Elongation / 69
Peptide Bond Formation / 71
Translocation / 71
Termination / 72
Posttranslational Processing / 73
When Nonsense Makes Sense / 74
Coupled Transcription and Translation / 74
Protein Folding and Chaperones / 74
Folding Stages / 75
Protein Folding and Chaperone Mechanisms Outside the Cytoplasm / 76
Quality Control / 77
Protein Trafficking / 77
Insertion of Integral Membrane Proteins and Export of Periplasmic Proteins / 77
Secretion of Proteins Across the Outer Membrane / 81

Protein Degradation /	83
Degradation of Abnormal Proteins /	83
Energy-Dependent Proteases /	86
Antibiotics that affect Nucleic Acid and Protein Synthesis /	88
Agents Affecting DNA Metabolism /	88
Agents Affecting Transcription /	91
Agents Affecting Translation /	92
Nucleoids /	98
DNA Replication /	98
Transcription and Translation /	98
Protein Folding, Trafficking, and Degradation /	99
Antibiotics /	100

3 BACTERIAL GENETICS: DNA EXCHANGE, RECOMBINATION, MUTAGENESIS, AND REPAIR

101

Transfer of Genetic Information in Prokaryotes /	101
Plasmids /	102
Partitioning /	102
Incompatibility /	103
Nonconjugative, Mobilizable Plasmids /	103
Resistance Plasmids /	104
Plasmids in Other Bacterial Genera /	104
Plasmid Replication /	104
Addition Modules: Plasmid Maintenance by Host Killing: The <i>ccd</i> Genes /	108
Conjugation /	108
F Factor /	108
cis/trans complementation Test /	115
Conjugation and Pheromones in Enterococci /	116
Conjugation, Cell–Cell Signaling, and Bacterial-Induced Tumors /	117
Transformation /	118
Gram-Positive Transformation /	119
Gram-Negative Transformation /	123
Transfection and Forced Competence /	124
Transduction /	124
Recombination /	127
General Recombination /	128
Genetics of Recombination /	131
Restriction and Modification /	133

Insertion Sequences and Transposable Elements /	138
Transposon <i>Tn10</i> /	140
Transposon <i>Tn3</i> /	143
Conjugative Transposition /	144
Evolutionary Consideration /	144
Integrans /	145
Mutagenesis /	145
Spontaneous Mutations /	147
The Nature of Mutational Events /	147
Suppressor Mutations /	149
DNA Repair Systems /	152
Photoreactivation /	152
Nucleotide Excision Repair /	152
Transcription-Coupled Repair /	155
Methyl-Directed Mismatch Repair /	156
Very Short-Patch Mismatch Repair /	158
DNA Glycosylases and Base Excision Repair /	158
Adaptive Response to Methylating and Ethylating Agents /	160
Postreplication Daughter Strand Gap Repair /	160
SOS-Inducible Repair /	162
Replication Restart /	165
Adaptive Mutations /	166
Plasmids /	167
Transformation /	167
Conjugation /	168
Recombination /	168
Restriction Modification /	169
Transposition /	169
Mutagenesis /	169
Repair Mechanisms /	170

4 MICROBIAL PHYSIOLOGY IN THE GENOMIC ERA: A REVOLUTIONARY TALE

171

Genomic and Proteomic Tools /	172
Cloning a Genome /	172
DNA Sequencing /	172
Web Science: Internet Tools for DNA Sequence Analysis /	173
Gene Replacement /	176
Gene Arrays /	177
Proteomics /	177

Traditional Tools /	181
Mutant Hunts /	181
Transcriptional and Translational Gene Fusions (Reporter Genes) /	182
Polymerase Chain Reaction /	183
DNA Mobility Shifts (Gel Shifts and Supershifts) /	185
Finding Transcriptional Starts by Primer Extension /	186
Detecting DNA, RNA, Protein, and DNA-Binding Proteins by Southern, Northern, Western, and Southwestern Blots /	187
Two-Hybrid Analysis /	190
Summary /	192

5 REGULATION OF PROKARYOTIC GENE EXPRESSION

194

Transcriptional Control /	194
DNA-Binding Proteins /	195
The <i>lac</i> Operon: A Paradigm of Gene Expression /	197
Catabolite Control: Sensing Energy Status /	201
Class I and Class II CRP-Dependent Genes /	204
The Catabolite Repressor/Activator Protein Cra /	204
Catabolite Control: The Gram-Positive Paradigm /	206
The <i>gal</i> Operon: DNA Looping with a Little Help from Hu /	206
The Arabinose Operon: One Regulator, Two Functions /	208
Attenuation Controls /	211
Transcriptional Attenuation Mechanisms /	211
Translational Attenuation Control: The <i>pyrC</i> Strategy /	215
Membrane-Mediated Regulation: The <i>put</i> System /	216
Recombinational Regulation of Gene Expression (Flagellar Phase Variation) /	217
Translational Repression /	219
Anti- σ Regulation by Molecular Hijacking /	220
Titrating a Posttranscriptional Regulator: The CsrA/CsrB Carbon Storage Regulatory Team /	222
Global Control Networks /	223
Communication with the Environment: Two-Component Regulatory Systems /	224
Regulation of Nitrogen Assimilation and Nitrogen Fixation: Examples of Integrated Biochemical and Genetic Controls /	227
Phosphate Uptake: Communication Between Transport and Two-Component Regulatory Systems /	232

Quorum Sensing: How Bacteria Talk to Each Other /	234
Proteolytic Control /	235
Summary /	236

6 BACTERIOPHAGE GENETICS

239

General Characteristics of Bacteriophages /	239
T4 Phage /	245
Structure /	245
General Pattern of T4 Gene Expression /	247
T4 Genome /	250
λ Phage /	256
The Lysis-Lysogeny Decision /	259
Transcription /	259
Function of Cro Versus CI Repressor and the Structure of O_L and O_R /	260
Establishment of Repressor Synthesis /	260
Control of Integration and Excision /	262
Negative Retroregulation of <i>int</i> by <i>sib</i> /	263
λ -Phage Replication /	264
μ Phage: Transposition as a Lifestyle /	266
Φ X174 /	271
Summary /	274
General /	274
T4 Bacteriophage /	274
λ Phage /	275
ϕ X174 /	275
μ Phage /	275

7 CELL STRUCTURE AND FUNCTION

277

The Eukaryotic Nucleus /	277
Bacterial Nucleoids /	279
Nucleosomes /	283
Mitochondria /	287
Microbial Cell Surfaces /	288
Eukaryotic Cell Surfaces /	288
Prokaryotic Cell Surfaces /	289
Surface Layers of Bacteria /	290
Peptidoglycans of Bacterial Cell Walls /	290
Peptidoglycan (Murein) Hydrolases /	295
Peptidoglycan (Murein) Synthesis /	295
Teichoic Acids and Lipoteichoic Acids /	300
Outer Membranes of Gram-Negative Bacteria /	303

Lipopolysaccharide Biosynthesis	/ 308
Enterobacterial Common Antigen	/ 309
Cytoplasmic Membranes	/ 310
Permeability and Transport	/ 313
Periplasm	/ 313
Other Membranous Organelles	/ 314
Capsules	/ 315
Microbial Biofilms	/ 322
Organs of Locomotion	/ 323
Cilia and Flagella of Eukaryotes	/ 323
Bacterial (Prokaryotic) Flagella	/ 325
Chemotaxis	/ 328
Swarming Motility	/ 334
Motility in Spirochetes	/ 337
Gliding Motility	/ 339
Pili or Fimbriae	/ 340
Nucleus, Nucleosomes, and Nucleoids	/ 343
Mitochondria	/ 344
Eukaryotic Cell Surface	/ 344
Surface (S) layers	/ 344
Bacterial Cell Wall Peptidoglycan (Murein)	/ 345
Teichoic and Lipoteichoic Acids	/ 346
Outer Membrane	/ 346
Cytoplasmic Membrane	/ 346
Periplasm	/ 346
Capsules	/ 346
Biofilms	/ 347
Cilia and Flagella of Eukaryotes	/ 347
Bacterial Flagella	/ 347
Chemotaxis	/ 348
Swarming Motility	/ 348
Gliding Motility	/ 348
Motility in Spirochetes	/ 348
Pili or Fimbriae	/ 349

8 CENTRAL PATHWAYS OF CARBOHYDRATE METABOLISM

350

Alternate Pathways of Carbohydrate Metabolism	/ 351
Fructose Bisphosphate Aldolase Pathway	/ 351
Alternate Pathways of Glucose Utilization	/ 354
Entner-Doudoroff or Ketogluconate Pathway	/ 354
Phosphoketolase Pathway	/ 356
Oxidative Pentose Phosphate Cycle	/ 358

Gluconeogenesis / 360
Regulation / 360
Glycogen Synthesis / 361
Tricarboxylic Acid Cycle / 361
Glyoxylate Cycle / 365

9 ENERGY PRODUCTION AND METABOLITE TRANSPORT 368

Energy Production / 368
Substrate-Level Phosphorylation / 369
Oxidative Phosphorylation / 371
Measurement of PMF / 372
Electron Transport Systems / 373
Anaerobic Respiration / 376
Conversion of PMF to Energy / 377
Structure of F_1F_0 and the <i>atp</i> Operon / 379
Energy Yield / 380
Generating ATP in Alkalophiles / 380
Energetics of Chemolithotrophs / 380
pH Homeostasis / 382
Metabolite Transport / 383
Facilitated Diffusion / 383
Mechanosensitive Channels / 385
ATP-Binding Cassette Transporter Family / 385
Chemiosmotic-Driven Transport / 385
Establishing Ion Gradients / 387
Specific Transport Systems / 387
ATP-Linked Ion Motive Pumps / 387
The Histidine Permease / 389
Iron / 389
Phosphotransferase System / 390
Summary / 392
Energy Production / 392
Metabolite Transport / 392

10 METABOLISM OF SUBSTRATES OTHER THAN GLUCOSE 394

Utilization of Sugars other than Glucose / 394
Lactose / 394
Galactose / 396
Maltose / 396

Mannitol / 396	
Fucose and Rhamnose / 397	
Melibiose, Raffinose, Stachyose, and Guar Gum / 399	
Pectin and Aldohexuronate Pathways / 400	
Cellulose Degradation / 402	
Starch, Glycogen, and Related Compounds / 403	
Metabolism of Aromatic Compounds / 407	
Pectin Utilization / 410	
Cellulose Utilization / 410	
Utilization of Starch, Glycogen, and Related Compounds / 411	
Utilization of Aromatic Hydrocarbons / 411	

11 FERMENTATION PATHWAYS 412

Fermentation Balances / 412	
Yeast Fermentation / 415	
Lactic Acid–Producing Fermentations / 417	
Butyric Acid—and Solvent-Producing Fermentations / 423	
Fermentations of the Mixed-Acid Type / 425	
Propionic Acid Fermentation / 428	
Acetic Acid Fermentation / 430	
Fermentation Pathways / 431	
Yeast Fermentation / 431	
Lactic Acid Fermentation / 432	
Butyric Acid and Solvent-Producing Fermentations / 432	
Mixed-Acid Fermentations / 433	
Propionic Acid Fermentation / 433	
Acetic Acid Fermentation / 433	

12 PHOTOSYNTHESIS AND INORGANIC METABOLISM 434

Characteristics and Metabolism of Autotrophs / 434	
Photosynthetic Bacteria and Cyanobacteria / 434	
Autotrophic CO ₂ Fixation and Mechanisms of Photosynthesis / 437	
Hydrogen Bacteria / 440	
Nitrifying Bacteria / 442	
Sulfur Bacteria / 442	
Iron Bacteria / 443	
Methyloolithotrophs / 444	
Methanogens / 446	

13 LIPIDS AND STEROLS 450

- Lipid Composition of Microorganisms / 451
 - Straight-Chain Fatty Acids / 451
 - Branched-Chain Fatty Acids / 453
 - Ring-Containing Fatty Acids / 454
 - Alk-1-enyl Ethers (Plasmalogens) / 455
 - Alkyl Ethers / 456
 - Phospholipids (Phosphoglycerides) / 457
 - Glycolipids / 458
- Biosynthesis of Fatty Acids / 459
- Biosynthesis of Phospholipids / 464
- Degradation of Fatty Acids / 466
- Biosynthesis of Isoprenoids / 468

14 NITROGEN METABOLISM 475

- Biological Nitrogen Fixation / 475
- The Nitrogen Fixation Process / 479
 - Components of the Nitrogenase System / 480
- Symbiotic Nitrogen Fixation / 483
- Inorganic Nitrogen Metabolism / 487
- Assimilation of Inorganic Nitrogen / 492
- General Reactions of Amino Acids / 494
 - Amino Acid Decarboxylases / 494
 - Amino Acid Deaminases / 495
 - Amino Acid Transaminases (Aminotransferases) / 497
 - Amino Acid Racemases / 498
 - Role of Pyridoxal-5'-Phosphate in Enzymatic Reactions with Amino Acids / 499
- The Stickland Reaction / 500
 - Nitrogen Fixation / 501
 - Inorganic Nitrogen / 502
 - Urease / 502
 - Assimilation of Inorganic Nitrogen / 502

15 BIOSYNTHESIS AND METABOLISM OF AMINO ACIDS 503

- The Glutamate or α -Ketoglutarate Family / 503
 - Glutamine and Glutathione Synthesis / 503
 - The Proline Pathway / 504
 - Aminolevulinate Synthesis / 504
 - The Arginine Pathway / 504

Polyamine Biosynthesis / 509	
The α -Ketoadipate Pathway to Lysine / 510	
The Aspartate and Pyruvate Families / 513	
Asparagine Synthesis / 513	
The Aspartate Pathway / 514	
The Bacterial Pathway to Lysine / 515	
Threonine, Isoleucine, and Methionine Formation / 516	
Isoleucine, Valine, and Leucine Biosynthesis / 518	
Regulation of the Aspartate Family / 518	
The Serine-Glycine Family / 520	
Aminolevulinate and the Pathway to Tetrapyrroles / 523	
The Aromatic Amino Acid Pathway / 523	
Phenylalanine, Tyrosine, and Tryptophan / 523	
The Common Aromatic Amino Acid Pathway / 525	
Pathways to Tyrosine and Phenylalanine / 526	
<i>p</i> -Aminobenzoate and Folate Biosynthesis / 531	
Enterobactin Biosynthesis / 533	
The Pathway to Ubiquinone / 534	
Menaquinone (Vitamin K) Biosynthesis / 534	
Biosynthesis of Nicotinamide Adenine Dinucleotide (NAD) / 534	
Histidine Biosynthesis / 539	
Amino Acids / 541	
Glutamate (α -Ketoglutarate) Family / 541	
Aspartate and Pyruvate Families / 542	
Serine-Glycine Family / 543	
Aromatic Amino Acid Family / 543	
Histidine / 544	

16 PURINES AND PYRIMIDINES

545

Biosynthesis of Purines / 545
Biosynthesis of Pyrimidines / 550
Interconversion of Nucleotides, Nucleosides, and Free Bases: Salvage Pathways / 554
Regulation of Purine and Pyrimidine Biosynthesis / 555
Purines and Pyrimidines / 559
Riboflavin Biosynthesis / 560
Thiamine Biosynthesis / 560

17 BACTERIAL CELL DIVISION

561

Cell Division in Gram-Negative Rods / 561
Cell Division in Gram-Positive Cocci / 570

Cell Division in Gram-Positive Bacilli / 575	
General Reviews / 578	
Cell Division in Gram-Negative Rods / 579	
Cell Division in Gram-Positive Cocci / 580	
Cell Division in Gram-Positive Bacilli / 581	

18 MICROBIAL STRESS RESPONSES 582

Osmotic Stress and Osmoregulation / 582	
High Osmolality / 583	
Low Osmolality / 584	
Osmotic Control of Gene Expression / 585	
Aerobic to Anaerobic Transitions / 587	
Formate Nitrate Regulation / 589	
Nitrate Response / 589	
ArcAB System / 591	
Oxidative Stress / 592	
Regulation of the Oxidative Stress Response / 594	
pH Stress and Acid Tolerance / 596	
Thermal Stress and the Heat Shock Response / 597	
Nutrient Stress and the Starvation—Stress Response / 601	
Starvation—Stress Response / 601	
Stringent Control / 602	
Extremophiles / 605	
Summary / 608	
Osmotic Stress and Osmoregulation / 608	
Aerobic to Anaerobic Transitions / 609	
Oxidative Stress / 609	
pH Stress and Acid Tolerance / 610	
Thermal Stress and the Heat Shock Response / 610	
Nutrient Stress and the Starvation Stress Response / 611	
Stringent Control / 611	
Extremophiles / 611	

19 BACTERIAL DIFFERENTIATION 612

<i>Bacillus</i> Endospore Formation / 612	
Life Cycle of <i>Bacillus</i> / 613	
Stages of Sporulation / 614	
Physiological and Genetic Aspects of Sporulation / 616	
Sporulation Genes / 616	
Initiation / 617	

Transition from Stage II to Stage III	/ 619
Forespore Development	/ 620
Final Stages of Sporulation	/ 621
Spore Cortex Synthesis	/ 622
Spore Coat Protein Synthesis	/ 622
Activation, Germination, and Outgrowth of Bacterial Endospores	/ 623
Activation	/ 624
Germination	/ 624
Outgrowth	/ 627
Myxobacterial Developmental Cycle	/ 628
Life Cycle of Myxobacteria	/ 628
Aggregation and Fruiting Body Formation	/ 629
Genetics of <i>Myxococcus xanthus</i> Development	/ 632
<i>Caulobacter</i> Differentiation	/ 637
Life Cycle of <i>Caulobacter crescentus</i>	/ 637
The Stalk, the Holdfast, and the Flagellum: Structure, Genetics, and Regulation	/ 638
Regulation and Checkpoints of the Cell Cycle of <i>C. crescentus</i>	/ 642
Endospore Formation	/ 644
Germination and Outgrowth of Endospores	/ 645
Myxobacterial Developmental Cycle	/ 646
<i>Caulobacter</i> Differentiation	/ 647

20 HOST–PARASITE INTERACTIONS

648

Overview of Host–Parasite Relationships	/ 648
Structures and Functions Involved in Host–Parasite Interactions	/ 650
Adherence/Colonization	/ 650
Virulence Factor Secretion Systems	/ 653
Exotoxins	/ 658
Quorum Sensing	/ 664
Paradigms of Bacterial Pathogenesis	/ 669
Enteropathogenic <i>Escherichia coli</i>	/ 669
<i>Salmonella Enterica</i> Serovars	/ 669
<i>Listeria Monocytogenes</i>	/ 670
<i>Chlamydia</i> spp	/ 672
Overview	/ 672
Adherence/Colonization	/ 672
Virulence Factor Secretion Systems	/ 673

xviii CONTENTS

Exotoxins / 673

Quorum Sensing / 674

Paradigms of Bacterial Pathogenesis / 675

INDEX

676

PREFACE

The field of microbial physiology has expanded at an incredibly rapid pace since the last edition of this text. The development and implementation of new, highly sophisticated, techniques to study the molecular genetics and physiology of an ever broadening range of microbes has prompted us to write a fourth edition to this book. To give full measure to the extraordinary advances made in microbial physiology we have found it necessary to reorder, separate, and add new material. However, in doing so we have attempted to remain true to the goal of the first edition of “Moat’s Notes” and each subsequent edition by targeting discussions to undergraduate and beginning graduate students while providing sufficient detail useful to established microbial physiologists. This new edition continues the tradition of addressing the physiology of a variety of microbes and not just *Escherichia coli*. We have updated chapters on bacterial structures, intermediary metabolism, genetics and growth; and added chapters discussing the genomic and proteomic methodologies employed by the new breed of microbial physiologist. We have reorganized, updated and expanded chapters on microbial stress responses and bacterial differentiation and have added a chapter on host-parasite interactions that correlates microbial physiology with microbial pathogenesis. We hope that the reader, be they an advanced undergraduate entering the field or a professor who has been in the field for forty years, will come to better appreciate the elegant simplicities and the intricate complexities of microbial physiology, while at the same time realizing that there is still much to be learned.

The authors would like to thank the many students, colleagues, and family who provided help and encouragement as we compiled this new edition. We are particularly

xx PREFACE

thankful to those who granted us permission to use figures or illustrations and/or provided us with original materials for this purpose.

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CHAPTER 1

INTRODUCTION TO MICROBIAL PHYSIOLOGY

THE *ESCHERICHIA COLI* PARADIGM

Microbial physiology is an enormous discipline encompassing the study of thousands of different microorganisms. It is, of course, foolhardy to try to convey all that is known on this topic within the confines of one book. However, a solid foundation can be built using a limited number of organisms to illustrate key concepts of the field. This text helps set the foundation for further inquiry into microbial physiology and genetics. The gram-negative organism *Escherichia coli* is used as the paradigm. Other organisms that provide significant counterexamples to the paradigm or alternative strategies to accomplish a similar biochemical goal are also included. In this chapter we paint a broad portrait of the microbial cell with special focus on *E. coli*. Our objective here is to offer a point of confluence where the student can return periodically to view how one aspect of physiology might relate to another. Detailed treatment of each topic is provided in later chapters.

CELL STRUCTURE

As any beginning student of microbiology knows, bacteria come in three basic models: spherical (coccus), rod (bacillus), and spiral (spirillum). They do not possess a membrane-bound nucleus as do eukaryotic microorganisms; therefore, they are prokaryotic. In addition to these basic types of bacteria, there are other more specialized forms described as budding, sheathed, and mycelial. Figure 1-1 presents a schematic representation of a typical (meaning *E. coli*) bacterial cell.

The Cell Surface

The interface between the microbial cell and its external environment is the cell surface. It protects the cell interior from external hazards and maintains the integrity of the cell

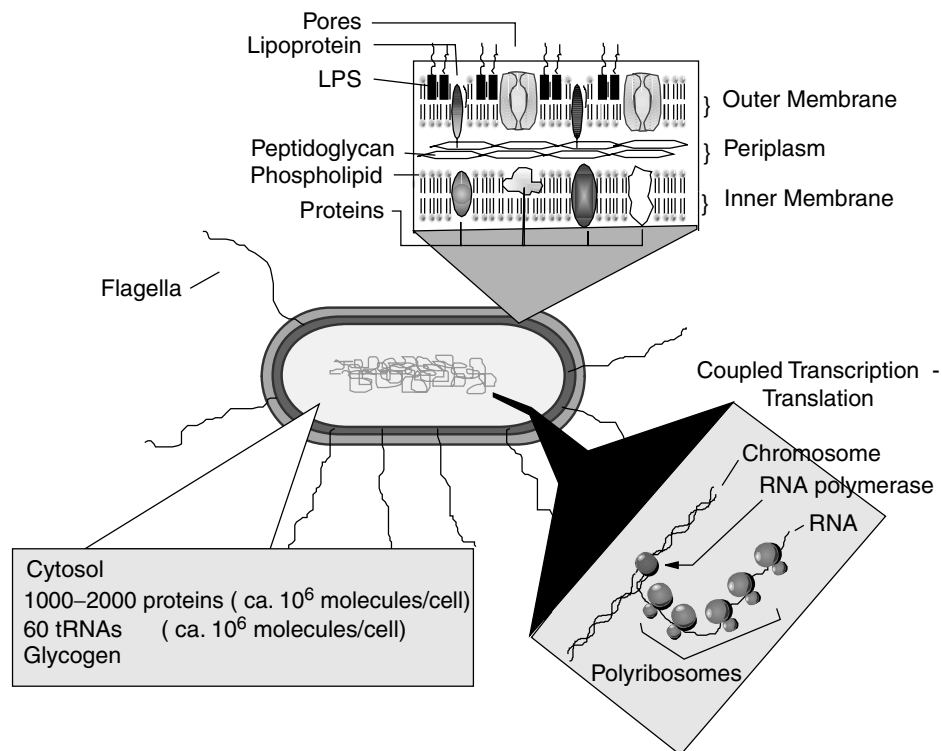


Fig. 1-1. Diagrammatic representation of a “typical” bacterial cell (*Escherichia coli*). Portions of the cell are enlarged to show further details.

as a discrete entity. Although it must be steadfast in fulfilling these functions, it must also enable transport of large molecules into and out of the cell. These large molecules include carbohydrates (e.g., glucose), vitamins (e.g., vitamin B₁₂), amino acids, and nucleosides, as well as proteins exported to the exterior of the cell. The structure and composition of different cell surfaces can vary considerably depending on the organism.

Cell Wall. In 1884, the Danish investigator Christian Gram devised a differential stain based on the ability of certain bacterial cells to retain the dye crystal violet after decoloration with 95% ethanol. Cells that retained the stain were called gram positive. Subsequent studies have shown that this fortuitous discovery distinguished two fundamentally different types of bacterial cells. The surface of gram-negative cells is much more complex than that of gram-positive cells. As shown in the schematic drawings in Figure 1-2, the gram-positive cell surface has two major structures: the cell wall and the cell membrane. The cell wall of gram-positive cells is composed of multiple layers of peptidoglycan, which is a linear polymer of alternating units of N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM). A short peptide chain is attached to muramic acid. A common feature in bacterial cell walls is cross-bridging between the peptide chains. In a gram-positive organism such as *Staphylococcus aureus*, the cross-bridging between adjacent peptides may be close to 100%. By contrast, the frequency of cross-bridging in *Escherichia coli* (a gram-negative organism) may be as low as 30% (Fig. 1-3). Other components—for example, lipoteichoic acid (only

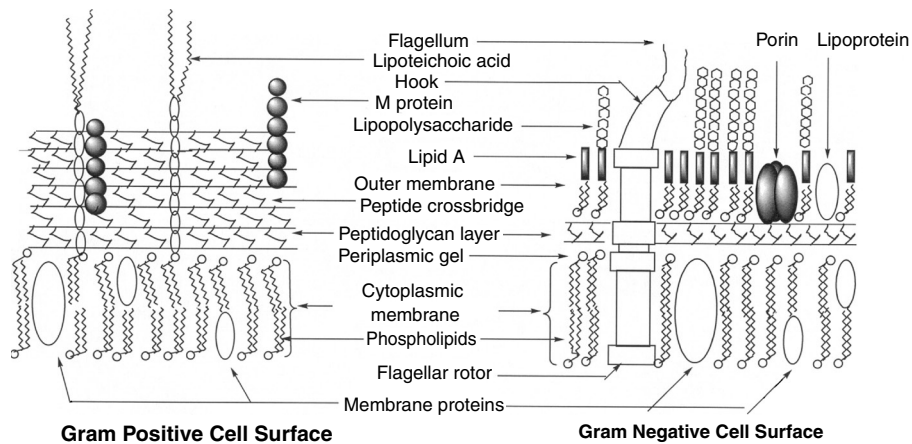


Fig. 1-2. Composition of the cell surfaces of gram-positive and gram-negative bacteria. Not all structures shown are found in all organisms. For example, M protein is only used to describe a structure in some of the streptococci. Also, not all organisms have flagella.

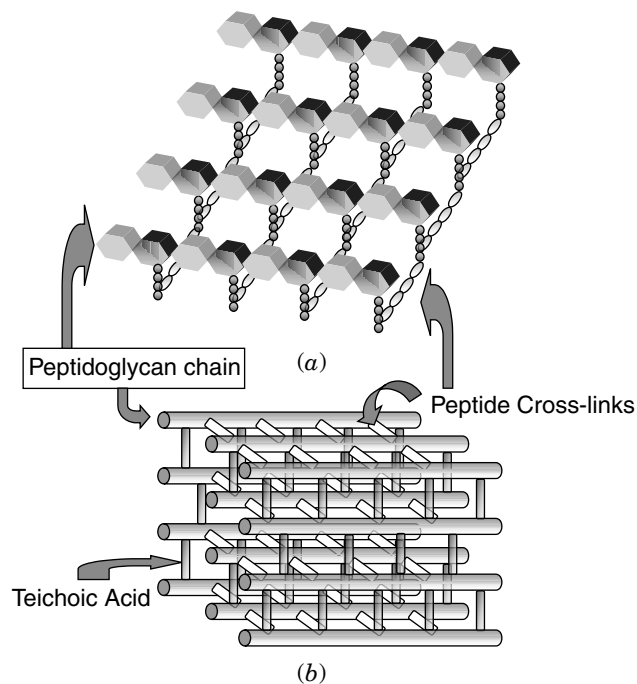


Fig. 1-3. Diagrammatic views of bacterial peptidoglycan. (a) Monolayer of peptidoglycan. Lightly shaded hexagons represent N-acetylglucosamine; darkly shaded hexagons represent N-acetylmuramic acid; vertically arranged spheres represent the peptide side chains; horizontal ovals represent the amino acid cross-bridges between peptide chains. (b) Diagrammatic representation of the multilayered peptidoglycan in the gram-positive cell wall. Long horizontal bars denote the chains of N-acetylglucosamine and N-acetylmuramic acid. Short horizontal bars indicate peptide cross-bridges and vertical bars represent teichoic acid.

present in gram-positive organisms)—are synthesized at the membrane surface and may extend through the peptidoglycan layer to the outer surface.

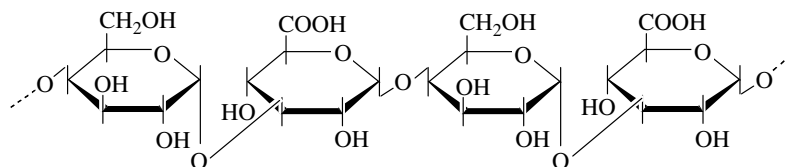
The peptidoglycan layer of a gram-negative cell is generally a single monolayer. An outer membrane surrounding the gram-negative cell is composed of phospholipids, lipopolysaccharides, enzymes, and other proteins, including lipoproteins. The space between this outer membrane and the inner membrane is referred to as the **periplasmic space**. It may be traversed at several points by various enzymes and other proteins (Fig. 1-2).

Membranes. The cytoplasmic membrane of both gram-positive and gram-negative cells is a lipid bilayer composed of phospholipids, glycolipids, and a variety of proteins. The proteins in the cytoplasmic membrane may extend through its entire thickness. Some of these proteins provide structural support to the membrane while others function in the transport of sugars, amino acids, and other metabolites.

The outer membrane of gram-negative cells contains a relatively high content of **lipopolysaccharides**. These lipid-containing components represent one of the most important identifying features of gram-negative cells: the **O antigens**, which are formed by the external polysaccharide chains of the lipopolysaccharide. This lipid-containing component also displays **endotoxin** activity—that is, it is responsible for the shock observed in severe infections caused by gram-negative organisms. Bacterial cell surfaces also contain specific carbohydrate or protein receptor sites for the attachment of **bacteriophages**, which are viruses that infect bacteria. Once attached to these receptor sites, the bacteriophage can initiate invasion of the cell.

Gram-positive and gram-negative cells have somewhat different strategies for transporting materials across the membrane and into the cell. The cytoplasmic membrane of gram-positive organisms has immediate access to media components. However, chemicals and nutrients must first traverse the outer membrane of gram-negative organisms before encountering the cytoplasmic membrane. Gram-negative cells have **pores** formed by protein triplets in their outer membrane that will permit passage of fairly large molecules into the periplasmic space. Subsequent transport across the inner or cytoplasmic membrane is similar in both gram-positive and gram-negative cells.

Capsules. Some bacterial cells produce a capsule or a **slime layer** (Fig. 1-4) of material external to the cell. Capsules are composed of either polysaccharides (high-molecular-weight polymers of carbohydrates) or polymers of amino acids called polypeptides (often formed from the D- rather than the L-isomer of an amino acid). The capsule of *Streptococcus pneumoniae* type III is composed of glucose and glucuronic acid in alternating β -1, 3- and β -1, 4- linkages:



This capsular polysaccharide, sometimes referred to as pneumococcal polysaccharide, is responsible for the virulence of the pneumococcus. *Bacillus anthracis*, the anthrax

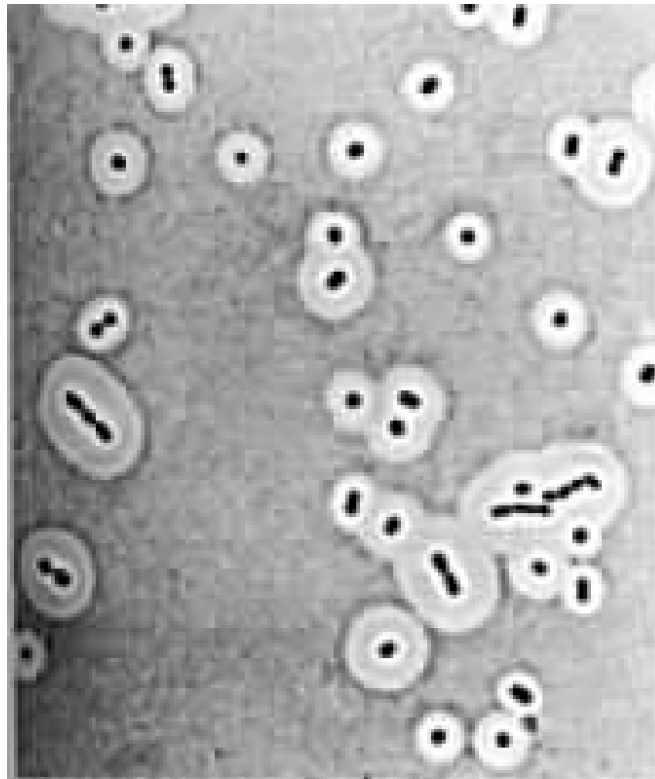


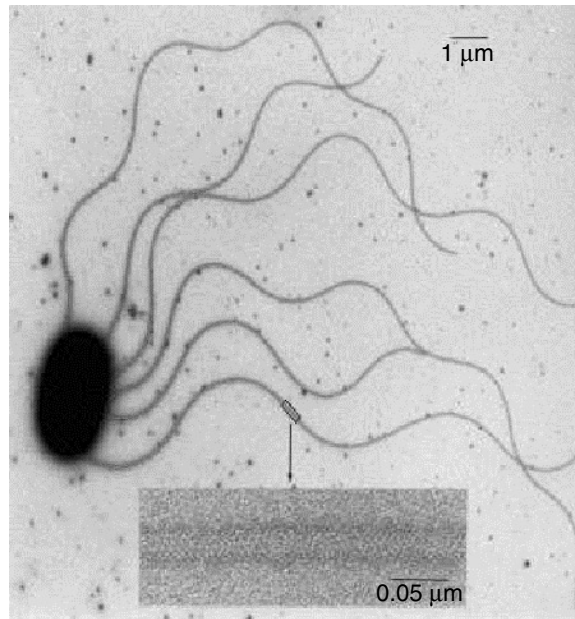
Fig. 1-4. Capsules of *Streptococcus pneumoniae*.

bacillus, produces a polypeptide capsule composed of D-glutamic acid subunits, which is a virulence factor for this organism.

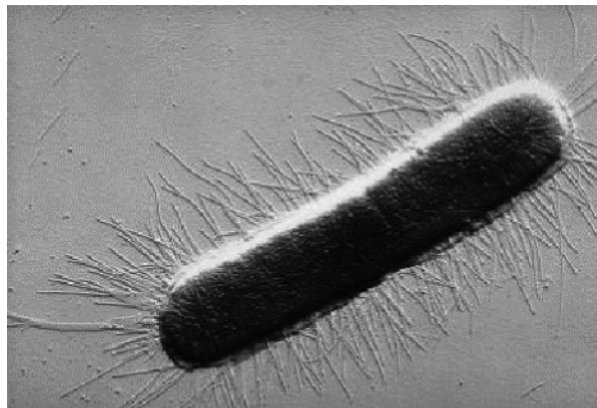
Organs of Locomotion. Many microorganisms are motile—that is, able to move from place to place in a concerted manner—especially in an aqueous environment. In the case of bacteria, this motility is accomplished by means of simple strands of protein (flagellin) woven into helical organelles called flagella. The bacterial flagellum is attached at the cell surface by means of a basal body (Fig. 1-5a). The basal body contains a motor that turns the flagellum, which propels the organism through the liquid environment.

Pili or Fimbriae. Many bacteria possess external structures that are shorter and more rigid than flagella. These structures have been termed pili (from Latin meaning “hair”) or fimbriae (from Latin meaning “fringe”). These appendages also appear to arise from a basal body or granule located either within the cytoplasmic membrane or in the cytoplasm immediately beneath the membrane (Fig. 1-5b). Generalized or common pili play a role in cellular adhesion to surfaces or to host cells.

Ribosomes. The cytoplasm of all cells has a fine granular appearance observed in many electron micrographs. Tiny particles called ribosomes are responsible for this look. Ribosomes contain approximately 65% RNA and 35% protein (see Fig. 1-1).



(a)



(b)

Fig. 1-5. Microbial appendages. (a) Flagella of *Salmonella typhimurium*. (b) Pili of *Escherichia coli*. (Source: Pili Image courtesy Indigo Instruments. Visit <http://www.indigo.com>.) Reprint permission is granted with this footer included.

The ribosome orchestrates the polymerization of amino acids into proteins (i.e., protein synthesis). At higher magnification under the electron microscope the ribosome particles are spherical. In properly prepared specimens the ribosomes are observed as collections or chains held together on a single messenger RNA (mRNA) molecule and are referred to as **polyribosomes** or simply polysomes.

The more or less spherical ribosome particle, when examined by sucrose gradient sedimentation, has been found to have a svedberg coefficient of 70S. (A svedberg unit denotes the rate of sedimentation of a macromolecule in a centrifugal field and

is related to the molecular size of that macromolecule.) The prokaryotic ribosome may be separated into two lower-molecular-weight components: one of 50S and another of 30S. Only the complete 70S particle functions in polypeptide synthesis. By comparison, the ribosomes of eukaryotic cells are associated with the endoplasmic reticulum, are larger (80S), and are composed of 40S and 60S subunits. The function of both 70S and 80S ribosomes in protein synthesis is identical. Curiously, eukaryotic mitochondria characteristically display 70S ribosomes—not the 80s particles that you would expect—because mitochondria probably evolved from endosymbiotic prokaryotic cells, a hypothesis supported by extensive analyses comparing bacterial and mitochondrial genomes.

SYNTHESIS OF DNA, RNA, AND PROTEIN

The chromosome of *E. coli* is a single, circular, double-stranded DNA molecule whose nucleotide sequence encodes all the information required for cell growth and structure. The major molecular events required for propagating the species start with the chromosome and include DNA replication, transcription, and translation. In bacteria, replication involves the accurate duplication of chromosomal DNA and the formation of two daughter cells by **binary fission**. In binary fission the cell grows until a certain mass-to-DNA ratio is achieved, at which point new DNA is synthesized and a centrally located cross-wall is constructed that will ultimately separate the two daughter cells.

A simplified view of **DNA replication** in *E. coli* is shown in the diagram in Figure 1-6. The double-stranded DNA molecule unwinds from a specific starting point (**origin**). The new DNA is synthesized opposite each strand. The enzyme involved in

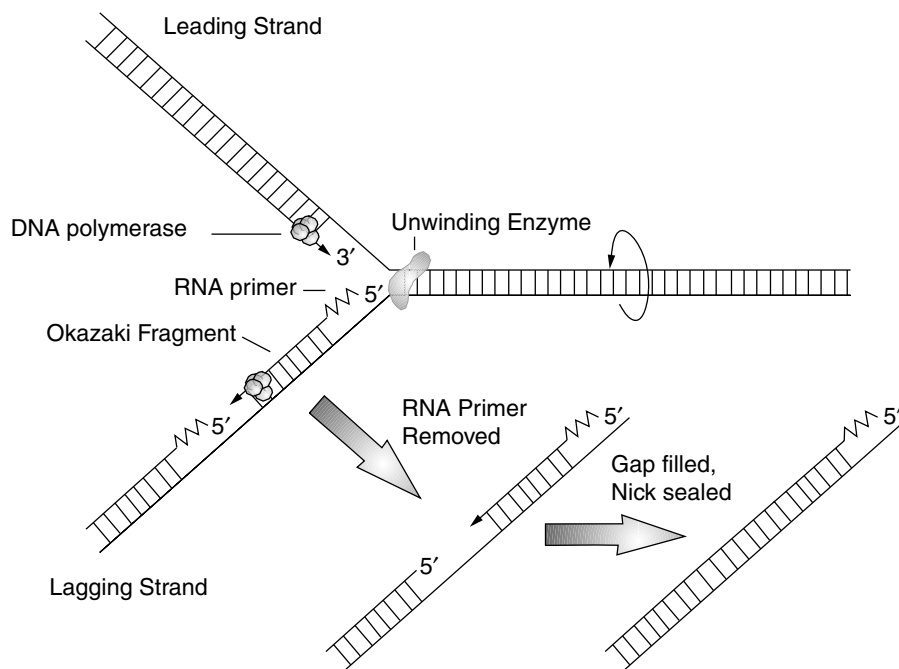


Fig. 1-6. Simplified depiction of DNA replication.

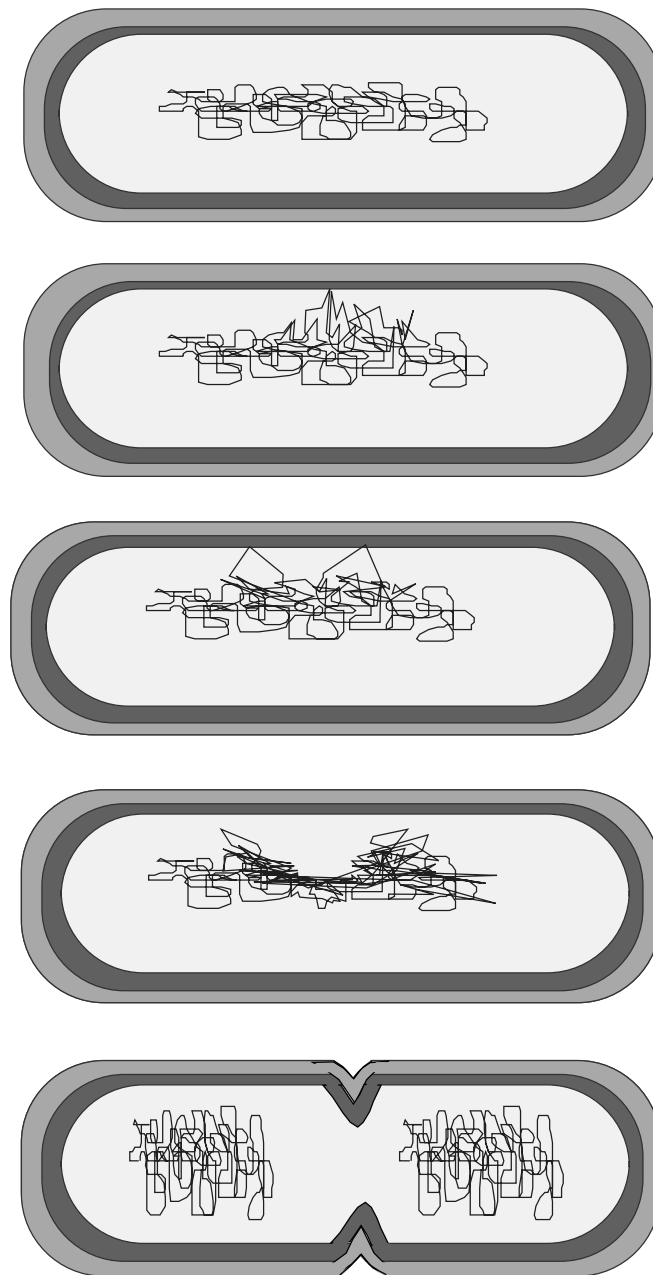


Fig. 1-7. Segregation of the bacterial chromosome.

replication (**DNA polymerase**) uses a parent strand as a template, placing adenine residues opposite thymine, and cytosine residues opposite guanine. New DNA is synthesized in both directions from the origin and continues until both replication forks meet at the **terminus** 180° from the origin. At this point, cell division proceeds with cross-wall formation occurring between the two newly synthesized chromosomes

(Fig. 1-7). Note that the chromosome appears attached to the cell membrane as the daughter chromosomes begin to separate. At some point about midway to the ends of the cell, the nascent chromosomes separate from the membrane but continue to move toward the cell poles by a still undefined mechanism. Chromosomal segregation into pre-daughter cells must occur before the cell completes construction of the cross-wall that will separate the two offspring (see “Termination of DNA Replication and Chromosome Partitioning” in Chapter 2).

The genetic information contained within DNA is processed in two steps to produce various proteins. Protein synthesis (**translation**) is depicted in Figure 1-8. The enzyme **RNA polymerase** (DNA-dependent RNA polymerase) first locates the beginning of a gene (**promoter**). This area of the chromosome then undergoes a localized unwinding, allowing RNA polymerase to transcribe RNA from the DNA template. Before the RNA—called messenger RNA (mRNA)—is completely transcribed, a ribosome will attach to the beginning of the message.

As already noted, the ribosome contains of two subunits, 30S and 50S, each composed of special ribosomal proteins and **ribosomal ribonucleic acids (rRNA)**. rRNA molecules do not, by themselves, code for any protein but form the architectural scaffolding that directs assembly of the proteins to form a ribosome. The ribosome translates mRNA into protein by reading three nucleotides (known as a **triplet codon**) as a specific amino acid. Each amino acid used by the ribosome must first be attached to an adaptor or **transfer RNA (tRNA)** molecule specific for that amino acid. tRNA containing an attached amino acid is referred to as a **charged tRNA molecule**. A part of the tRNA molecule called the **anticodon** will base-pair with the codon in mRNA.

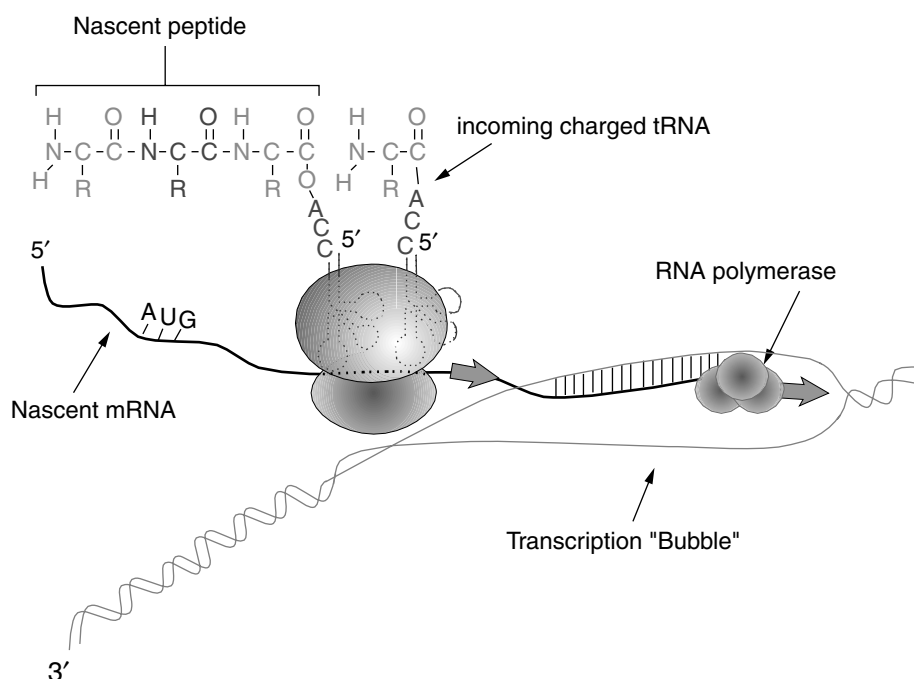


Fig. 1-8. Sequence of events involved in transcription and translation.

When two such charged tRNA molecules simultaneously occupy adjacent sites on the ribosome, the ribosome catalyzes the formation of a peptide bond between the two amino acids.

At this point, the two amino acids are attached to one tRNA while the other tRNA is uncharged and eventually released from the ribosome. The ribosome is then free to move along the message to the next codon. The process continues until the ribosome reaches the end of the message, at which point a complete protein has been formed. Notice that synthesis of the protein begins with the N-terminal amino acid and finishes with the C-terminal amino acid. Also note that the ribosome begins translating at the 5' end of the mRNA while the DNA strand encoding the mRNA is read by RNA polymerase starting at the 3' end. Although the beginning of a gene is usually called the 5' end, this doesn't refer to the strand that is actually serving as a template for RNA polymerase. It refers to the complementary DNA strand whose sequence is the same as the mRNA (except for containing T instead of U). The details of replication, transcription, and translation are discussed in Chapter 2.

Metabolic and Genetic Regulation

For a cell to grow efficiently, all the basic building blocks and all the macromolecules derived from them have to be produced in the correct proportions. With complex metabolic pathways, it is important to understand the manner by which a microbial cell regulates the production and concentration of each product. Two common mechanisms of metabolic and genetic regulation are

1. **Feedback inhibition** of enzyme activity (**metabolic regulation**)
2. **Repression** or **induction** of enzyme synthesis (**genetic regulation**)

In feedback inhibition, the *activity* of an enzyme already present in the cell is inhibited by the end product of the reaction. In genetic repression, the *synthesis* of an enzyme (see previous discussion of transcription and translation) is inhibited by the end product of the reaction. Induction is similar except the substrate of a pathway stimulates synthesis of the enzyme. Hypothetical pathways illustrating these concepts are presented in Figure 1-9. In Figure 1-9a, excessive production of intermediate B results in the inhibition of enzyme 1 activity, a phenomenon known as feedback or end-product inhibition. Likewise, an excess of end-product C may inhibit the activity of enzyme 1 by feedback inhibition.

In contrast to feedback inhibition, an excess intracellular concentration of end-product C may cause the cell to stop synthesizing enzyme 1, usually by inhibiting transcription of the genes encoding the biosynthetic enzymes (Fig. 1-9b). This action is referred to as **genetic repression**. The logic of this control is apparent when considering amino acid biosynthesis. If the cell has more than enough of a given amino acid, that amino acid will activate a repressor protein, which then blocks any further transcription of the biosynthesis genes. In contrast, substrates such as carbohydrates can stimulate the transcription of genes whose protein products consume that carbohydrate. This genetic process is called **induction** (Fig. 1-9c). Different organisms may employ quite different combinations of feedback inhibition, repression, and induction to regulate a metabolic pathway. In Chapter 5, these and other regulatory mechanisms are discussed in greater detail.

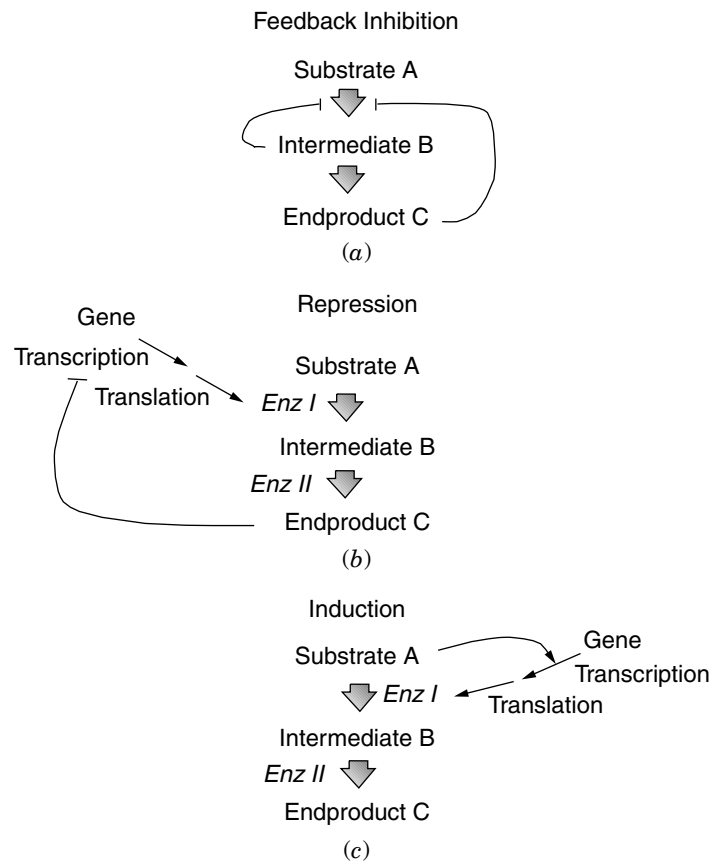


Fig. 1-9. Diagrammatic presentation of feedback inhibition of enzyme activity and end-product repression of enzyme synthesis. *a*, *b* and *c* are chemical intermediates in the hypothetical pathway. Arrow indicates activation, line with cross indicates inhibition.

MICROBIAL GENETICS

Having just outlined the processes of transcription, translation, and replication, it is now possible to define several genetic terms. The gene may be defined as a heritable unit of function composed of a specific sequence of purine and pyrimidine bases, which in turn determines the base sequence in an RNA molecule, and, of course, the sequence of bases in an RNA molecule specifies the sequence of amino acids incorporated into a polypeptide chain. The **genotype** of an organism is the sum total of all of the hereditary units of genes. The observed expression of the genetic determinants—that is, the structural appearance and physiological properties of an organism—is referred to as its **phenotype**.

An individual gene can exist in different forms as a result of nucleotide sequence changes. These alternative gene forms are referred to as **alleles**. Genetic material is not absolutely stable but can change or mutate. The process of change is known as **mutagenesis**. Altered genes are referred to as mutant alleles in contrast to the normal or **wild-type** alleles. **Spontaneous mutations** are thought to arise during replication,

repair, and recombination of DNA as a result of errors made by the enzymes involved in DNA metabolism. Mutations may be increased by the activity of a number of environmental influences. Radiation in the form of X rays, ultraviolet (UV) rays, or cosmic rays may affect the chemical structure of the gene. A variety of chemicals may also give rise to mutations. Physical, chemical, or physicochemical agents capable of increasing the frequency with which mutations occur are referred to as **mutagens**. The resulting alterations are **induced mutations** in contrast to *spontaneous mutations*, which appear to occur at some constant frequency in the absence of intentionally applied external influences. Since bacterial cells are haploid, mutants are usually easier to recognize because the altered character is more likely to be expressed, particularly if the environment is favorable to mutant development.

The use of mutants has been a tremendous tool in the study of most, if not all, biochemical processes. Genes are usually designated by a three-letter code based on their function. For example, genes involved in the biosynthesis of the amino acid arginine are called *arg* followed by an uppercase letter to indicate different *arg* genes (e.g., *argA*, *argB*). A gene is always indicated by lowercase italic letters (e.g., *arg*), whereas an uppercase letter in the first position (e.g., ArgA) indicates the gene product. At this point, we need to expose a common mistake made by many aspiring microbial geneticists concerning the interpretation of mutant phenotypes. Organisms such as *E. coli* can grow on basic minimal media containing only salts, ammonia as a nitrogen source, and a carbon source such as glucose or lactose because they can use the carbon skeleton of glucose to synthesize all the building blocks necessary for macromolecular synthesis. The building blocks include amino acids, purines, pyrimidines, cofactors, and so forth. A mutant defective in one of the genes necessary to synthesize a building block will require that building block as a supplement in the minimal medium (e.g., an *arg* mutant will require arginine in order to grow). Microorganisms also have an amazing capacity to catabolically use many different compounds as carbon sources. However, a mutation in a carbon source utilization gene (e.g., *lac*) does not mean it requires that carbon source. It means the mutant will *not grow* on medium containing that carbon source if it is the only carbon source available (e.g., a *lac* mutant will not grow on lactose).

The chromosome of our reference cell, *E. coli*, is 4,639,221 base pairs long. Gene positions on this map can be given in base pairs starting from the gene *thrL*, or in minutes based on the period of time required to transfer the chromosome from one cell to another by conjugation (100-minute map with *thrL* at 0).

CHEMICAL SYNTHESIS

Chemical Composition

Our paradigm cell (the gram-negative cell *E. coli*) can reproduce in a minimal glucose medium once every 40 minutes. As we proceed through a detailed examination of all the processes involved, the amazing nature of this feat will become increasingly obvious. It is useful to discuss the basic chemical composition of our model cell. The total weight of an average cell is 9.5×10^{-13} g, with water (at 70% of the cell) contributing 6.7×10^{-13} g. The total dry weight is thus 2.8×10^{-13} g. The components that form the dry weight include protein (55%), ribosomal RNA (16.7%), transfer RNA (3%), messenger RNA (0.8%), DNA (3.1%), lipids (9.1%),

lipopolysaccharides (3.4%), peptidoglycans (2.5%), building block metabolites, vitamins (2.9%), and inorganic ions (1.0%). It is interesting to note that the periplasmic space forms a full 30% of the cell volume, with total cell volume being approximately 9×10^{-13} ml (0.9 femtoliters). An appreciation for the dimensions of the cell follows this simple example. One teaspoon of packed *E. coli* weighs approximately 1 gram (wet weight). This comprises about one trillion cells—more than 100 times the human population of the planet. When calculating the concentration of a compound within the cell, it is useful to remember that there are 3 to 4 microliters of water per 1 milligram of dry weight. Our reference cell, although considered haploid, will contain two copies of the chromosome when growing rapidly. It will also contain 18,700 ribosomes and a little over 2 million total molecules of protein, of which there are between 1000 and 2000 different varieties. As you might gather from these figures, the bacterial cell is extremely complex. However, the cell has developed an elegant strategy for molecular economy that we still struggle to understand. Some of what we have learned is discussed throughout the remaining chapters.

In just 40 minutes an *E. coli* cell can make a perfect copy of itself growing on nothing more than glucose, ammonia, and some salts. How this is accomplished seems almost miraculous! All of the biochemical pathways needed to copy a cell originate from just 13 precursor metabolites. To understand microbial physiology, you must first discover what the 13 metabolites are and where they come from.

The metabolites come from glucose or some other carbohydrate. The catabolic dissimilation of glucose not only produces them but also generates the energy needed for all the work carried out by the cell. This work includes biosynthetic reactions as well as movement, transport, and so on. Figure 1-10 is a composite diagram of major pathways for carbohydrate metabolism with the 13 metabolites highlighted. Most of them are produced by the Embden-Meyerhof route and the tricarboxylic acid cycle (see Fig. 1-10). Three are produced by the pentose phosphate pathway. Figure 1-11 illustrates how these compounds are siphoned off from the catabolic pathways and used as starting material for the many amino acids, nucleic acid bases, and cofactors that must be produced. Subsequent chapters deal with the specifics of each pathway, but this figure presents an integrated picture of cell metabolism.

Energy

Another mission of carbohydrate metabolism is the production of energy. The most universal energy transfer compound found in living cells is adenosine triphosphate (ATP) (Fig. 1-12). The cell can generate ATP in two ways: (1) by **substrate-level phosphorylation** in which a high-energy phosphate is transferred from a chemical compound (e.g., phosphoenol pyruvate) to adenosine diphosphate (ADP) during the course of carbohydrate catabolism; or (2) by **oxidative phosphorylation** in which the energy from an electrical and chemical gradient formed across the cell membrane is used to drive a membrane-bound ATP hydrolase complex to produce ATP from ADP and inorganic phosphate.

The generation of an electrical and chemical gradient (collectively called the **proton motive force**) across the cell membrane requires a complex set of reactions in which H^+ and e^- are transferred from chemical intermediates of the Embden-Meyerhof and TCA cycles to a series of membrane-associated proteins called **cytochromes**. As the e^- is passed from one member of the cytochrome chain to another, the energy released

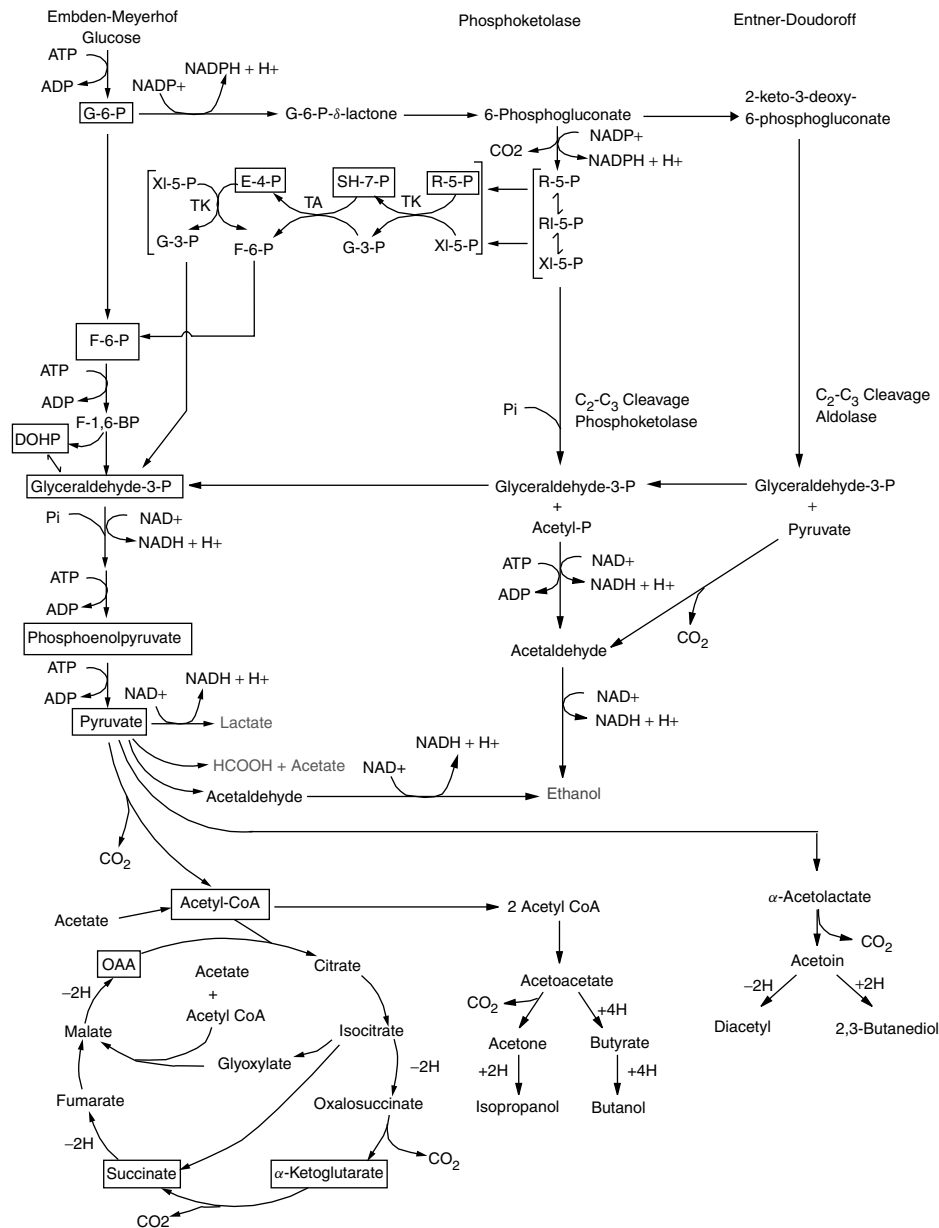


Fig. 1-10. Composite diagram of major pathways of carbohydrate metabolism. The 13 key metabolites are boxed. G, glucose; E, erythrose; R, ribose; XI, xylulose; SH, sedoheptulose; DOHP, dihydroxyacetone phosphate; OAA, oxaloacetate; TK, transketolase; TA, transaldolase.

is used to pump H^+ out of the cell. The resulting difference between the inside and outside of the cell in terms of charge (**electrical potential**) and pH (**chemical potential**) can be harnessed by the cell to generate ATP. Of course, in order for the cytochrome system to work, there must be a terminal electron acceptor molecule. Under aerobic conditions, oxygen will serve that function, but under anaerobic conditions, *E. coli*

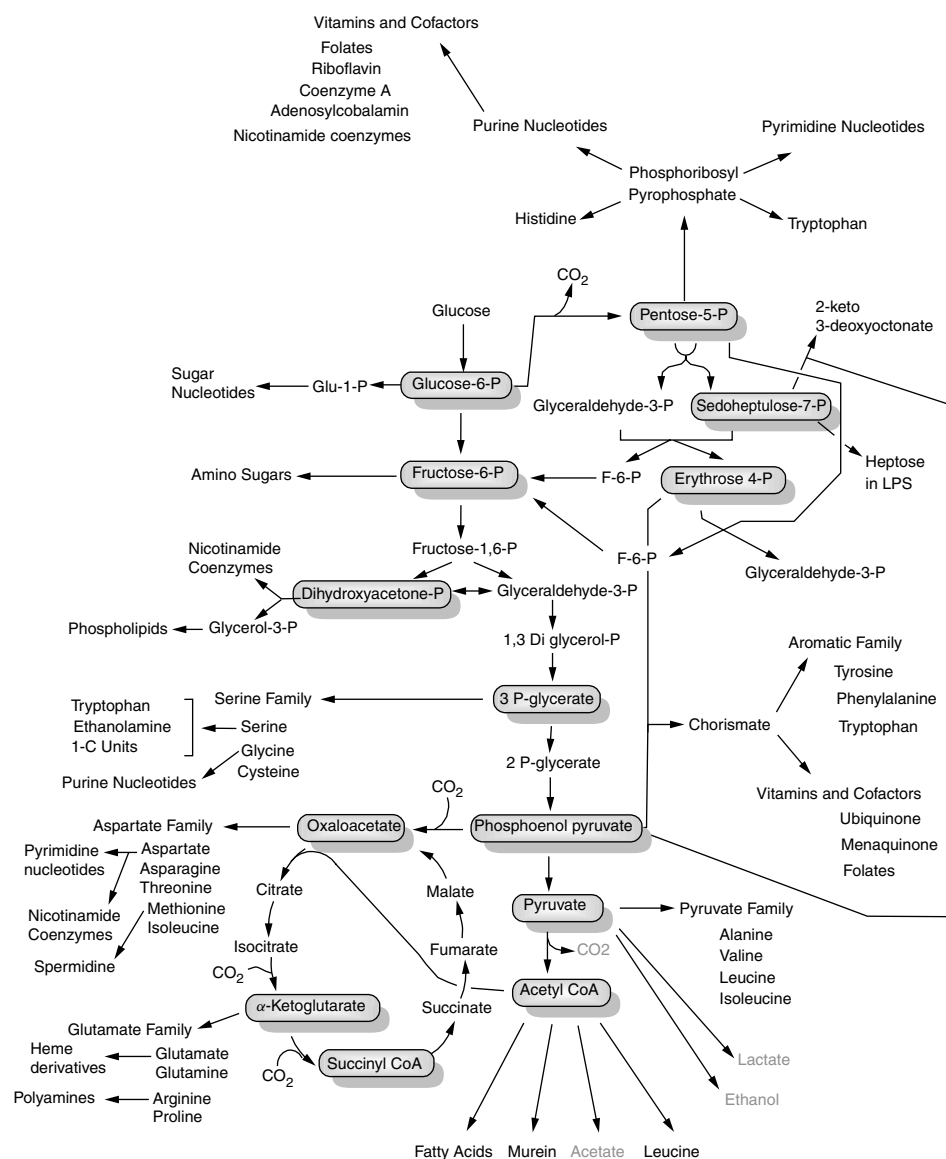


Fig. 1-11. Biosynthetic pathways leading to the amino acids and related compounds. The oblong-circled intermediates are the 13 key compounds that serve as biosynthetic precursors for a variety of essential end products.

has a menu of alternate electron acceptors from which it can choose depending on availability (e.g., nitrate). A more detailed accounting of this process is discussed in Chapter 9.

Oxidation–Reduction Versus Fermentation

Carbohydrate metabolism is the progressive oxidation of a sugar in which hydrogens are transferred from intermediates in the pathway to hydrogen-accepting molecules.

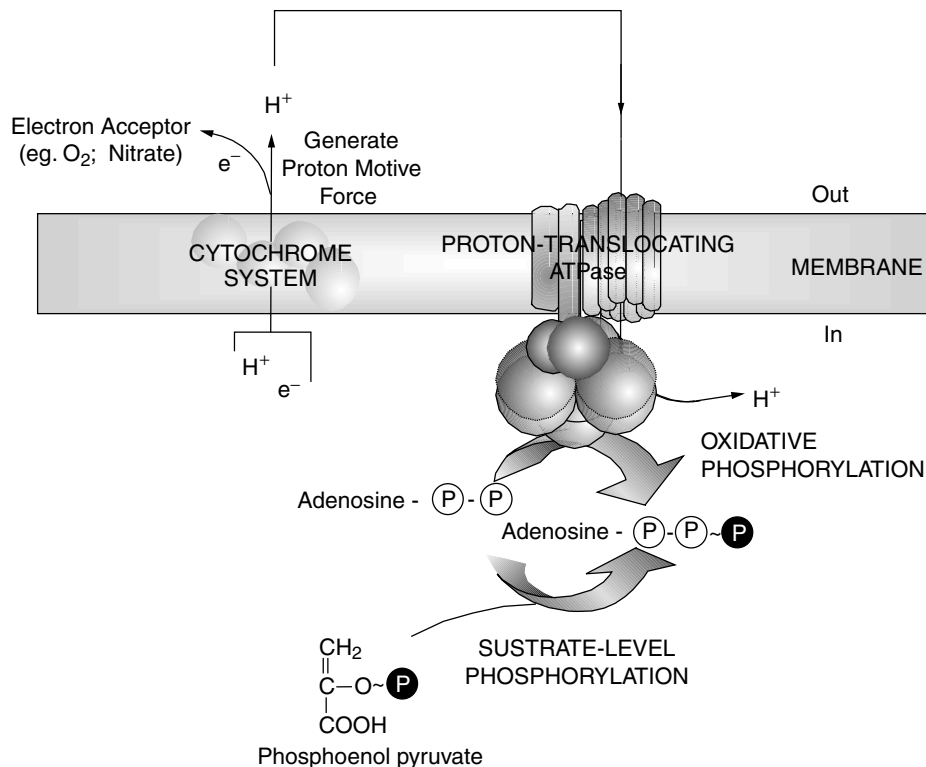


Fig. 1-12. Reactions essential to energy production. Oxidative phosphorylation. The energy that comprises the proton motive force can be harnessed and used to generate ATP when protons from outside the cell pass through the membrane-associated proton-translocating ATPase. The energy released will run the ATPase in reverse. It is estimated that passage of three H^+ through the ATPase is required to generate one ATP. Substrate-level phosphorylation. Energy contained within high-energy phosphate bonds of certain glycolytic intermediates can be transferred to ADP, forming ATP. The example shows phosphoenolpyruvate.

The most commonly used hydrogen acceptor compound is **nicotinamide adenine dinucleotide (NAD)** (Fig. 1-13). It is the reduced form of NAD (NADH) that passes the H^+ and e^- to the cytochrome system. However, a problem can develop when a cell is forced to grow in an anaerobic environment without any alternate electron acceptors. This situation could lead to a complete depletion of NAD^+ , with all of the NAD pool converted to NADH. NADH, produced during the early part of glycolysis, would not be able to pass its H along and therefore the cell could not regenerate NAD^+ . If this situation were allowed to develop, the cell would stop growing because there would be no NAD^+ to continue glycolysis! To avoid this problem, many microorganisms, including *E. coli*, can regenerate NAD^+ by allowing NADH to transfer H to what would otherwise be dead-end intermediates in the glycolytic pathway (e.g., pyruvate or acetyl CoA). The process, known as **fermentation**, produces lactic acid, isopropanol, butanol, ethanol, and so on, depending on the organism. *E. coli* does not perform all of these fermentation reactions. It is limited to lactate, acetate, formate, ethanol, CO_2 , and H_2 production (Fig. 1-10). Table 1-1 lists the fermentation patterns for some other common organisms.

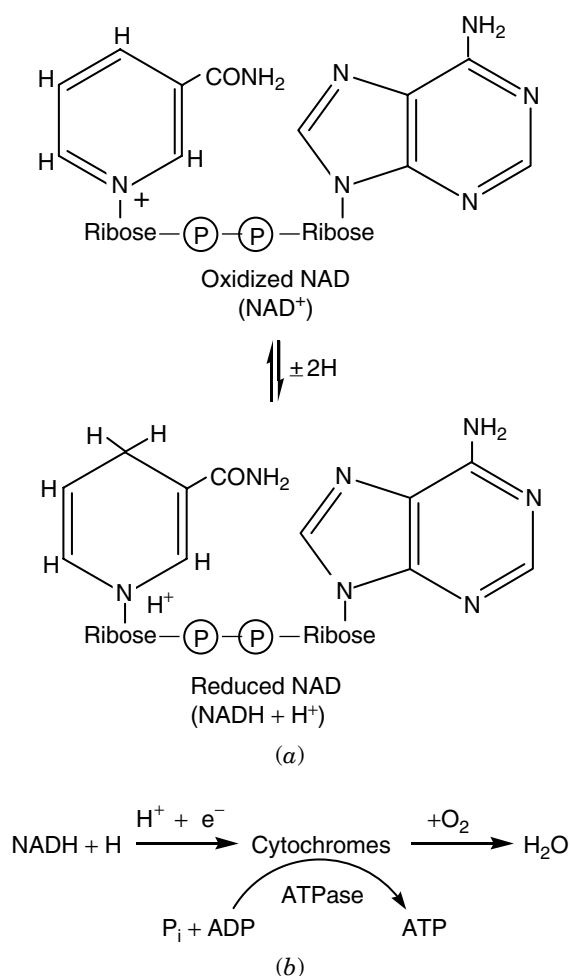


Fig. 1-13. Nicotinamide adenine dinucleotide (NAD). Function of NAD in oxidation–reduction reactions. (a) Hydrogen atoms removed from a hydrogen donor are transferred to the nicotinamide portion of NAD. (b) The hydrogen atoms can be transferred from NAD to an acceptor such as cytochrome pigments.

TABLE 1-1. Variation in Fermentation Products Formed from Pyruvate

Organism	Product(s)
<i>Saccharomyces</i> (yeast)	Carbon dioxide, ethanol
<i>Streptococcus</i> (bacteria)	Lactic acid
<i>Lactobacillus</i> (bacteria)	Lactic acid
<i>Clostridium</i> (bacteria)	Acetone, butyric acid, isopropanol, butanol
<i>Enterobacter</i> (bacteria)	Acetone, carbon dioxide, ethanol, lactic acid
<i>E. coli</i> (bacteria)	Lactic acid, acetic acid, H_2 , ethanol, formic acid, carbon dioxide

The cell does not only catabolize glucose via glycolysis. There are alternate metabolic routes available for the dissimilation of glucose. One use for alternate pathways of carbohydrate metabolism (e.g., the phosphoketolase pathway; see Fig. 1-10) is the generation of biosynthetic **reducing power**. The cofactor NAD is actually divided functionally into two separate pools. NAD(H) is used primarily for catabolic reactions, whereas a derivative, NAD phosphate (NADP), and its reduced form, NADPH, are involved in biosynthetic (anabolic) reactions. The phosphoketolase pathway is necessary for the generation of the NADPH that is essential for biosynthetic reactions.

Nitrogen Assimilation

A major omission in our discussion to this point involves the considerable amount of nitrogen (N) needed by microorganisms. Every amino acid, purine, pyrimidine, and many other chemicals in the cell include nitrogen in their structures. Since glucose does not contain any nitrogen, how do cells acquire it? Some microorganisms can fix atmospheric nitrogen via nitrogenase to form ammonia (NH_4^+) and then assimilate the ammonia into amino acids (e.g., *Rhizobium*). Other organisms such as *E. coli* must start with NH_4^+ . The assimilation of N involves the amidation of one of the 13 key metabolites, α -ketoglutarate, to form glutamic acid (Fig. 1-14). After assimilation into glutamate, the amino nitrogen is passed on to other compounds by transamination reactions. For example, glutamate can pass its amino group to oxaloacetate to form aspartate. From Figure 1-11, it can be seen that aspartate, like glutamate, is a precursor for several other amino acids. The subject of nitrogen assimilation is covered in depth in Chapter 14.

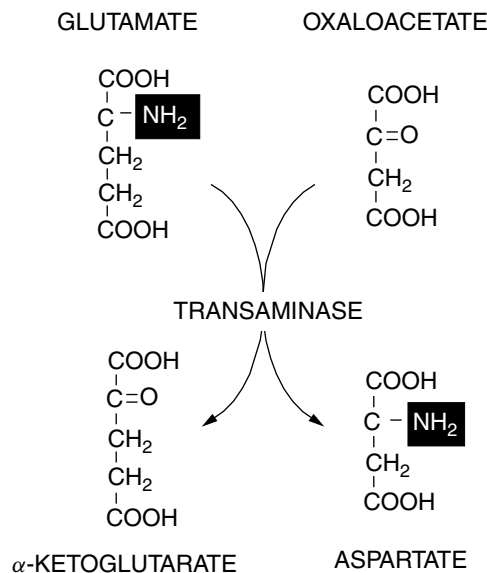


Fig. 1-14. Transamination. In this example, the amine group from glutamic acid is transferred to oxaloacetate, forming aspartic acid.

SPECIAL TOPICS

Endospores

A few bacteria such as *Bacillus* and *Clostridium* produce specialized structures called endospores. Endospores are bodies that do not stain with ordinary dyes and appear as unstained highly refractile areas when seen under the light microscope. They provide resistance to heat, desiccation, radiation, and other environmental factors that may threaten the existence of the organism. Endospores also provide a selective advantage for survival and dissemination of the species that produce them. Under the electron microscope, spores show a well-defined multilayered exosporium, an electron-dense outer coat observed as a much darker area, and a thick inner coat. In the spore interior, the darkly stained ribosomes and the nuclear material may also be visible (Fig. 1-15).

Growth

Growth of a cell is the culmination of an ordered interplay among all of the physiological activities of the cell. It is a complex process involving

1. Entrance of basic nutrients into the cell
2. Conversion of these compounds into energy and vital cell constituents
3. Replication of the chromosome
4. Increase in size and mass of the cell
5. Division of the cell into two daughter cells, each containing a copy of the genome and other vital components

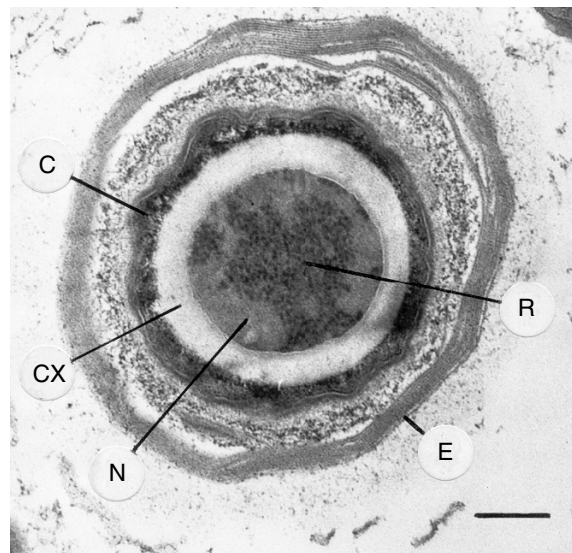


Fig. 1-15. Mature spore of *Clostridium botulinum*. Shown is a well-defined, multilayered exosporium (E), an electron-dense outer coat layer, a thick inner coat (C) and a less dense cortex (CX). The darkly stained ribosomes (R) and nucleoid areas (N) are clearly differentiated in the spore interior. Bar equals 0.2 μm . (Source: From Stevenson, K. E., R. H. Vaughn, and E. V. Crisan, 1972. *J. Bacteriol.* **109**:1295.)

Microbiologists usually consider the phenomenon of growth from the viewpoint of population increase, since most current techniques do not allow the detailed study of individual cells. A study of the increase in population implies that each cell, as it is produced, is capable of producing new progeny.

Growth Cycle. Under ideal circumstances in which cell division commences immediately and proceeds in unhampered fashion for a protracted period of time, prokaryotic cell division follows a geometric progression:

$$\begin{array}{ccccccccccc} 2^0 & \longrightarrow & 2^1 & \longrightarrow & 2^2 & \longrightarrow & 2^3 & \longrightarrow & 2^4 & \longrightarrow & 2^5 & \longrightarrow & 2^6 & \longrightarrow & 2^7 \\ & & & & & & & & & & & & & & \\ & \longrightarrow & 2^8 & \longrightarrow & 2^9 & \longrightarrow & \text{etc.} & & & & & & & & \end{array}$$

This progression may be expressed as a function of 2 as shown in the line above. The number of cells (b) present at a given time may be expressed as

$$b = 1 \times 2^n$$

The total number of cells (b) is dependent on the number of generations (n = number of divisions) occurring during a given time period. Starting with an inoculum containing more than one cell, the number of cells in the population can be expressed as

$$b = a \times 2^n$$

where a is the number of organisms present in the original inoculum. Since the number of organisms present in the population (b) is a function of the number 2, it becomes convenient to plot the logarithmic values rather than the actual numbers. Plotting the number of organisms present as a function of time generates a curvilinear function. Plotting the logarithm of the number, a linear function is obtained as shown in Figure 1-16. For convenience, logarithms to the base 10 are used. This is possible because the logarithm to the base 10 of a number is equal to 0.3010 times the logarithm to the base 2 of a number.

Up to this point it has been assumed that the individual generation time (i.e., the time required for a single cell to divide) is the same for all cells in the population. However, in a given population, the generation times for individual cells vary, so the term **doubling time** encompasses the doubling time for the total population. As shown in Figure 1-16, the cells initially experience a period of adjustment to the new environment, and there is a lag in the time required for all of the cells to divide. Actually, some of the cells in the initial inoculum may not survive this **lag phase** and there may be a drop in the number of viable cells. The surviving cells eventually adjust to the new environment and begin to divide at a more rapid rate. This rate will remain constant until conditions in the medium begin to deteriorate (e.g., nutrients are exhausted). Since plotting the cell number logarithm during this period results in a linear function, this phase of growth is referred to as the **logarithmic (log) phase** or, more correctly, the **exponential phase**.

All cultures of microorganisms eventually reach a maximum population density in the **stationary phase**. Entry into this phase can result from several events. Exhaustion of essential nutrients, accumulation of toxic waste products, depletion of oxygen, or development of an unfavorable pH are the factors responsible for the decline in the

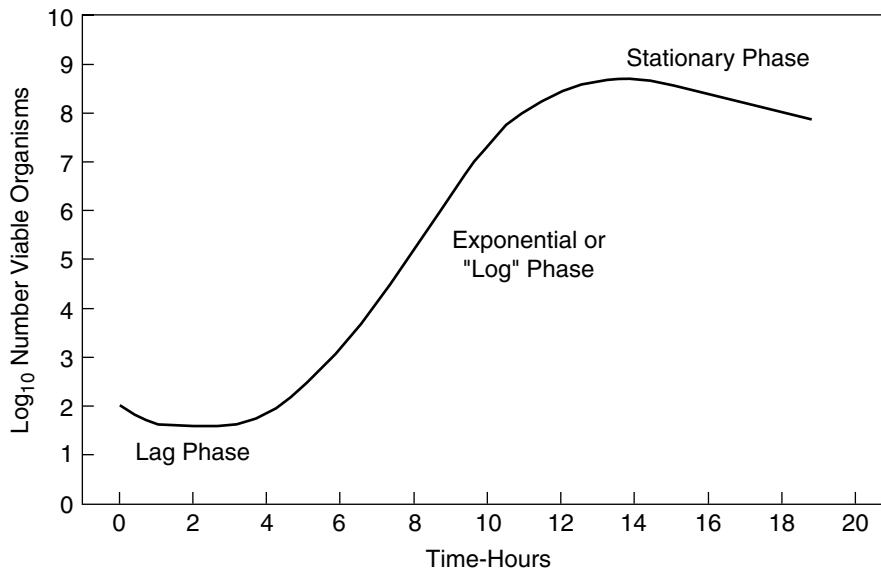


Fig. 1-16. A typical growth curve for a bacterial culture.

growth rate. Although cell division continues during the stationary phase, the number of cells that are able to divide (viable cells) are approximately equal to the number that are unable to divide (nonviable cells). Thus, the stationary phase represents an equilibrium between the number of cells able to divide and the number that are unable to divide.

Eventually, the death of organisms in the population results in a decline in the viable population and the **death phase** ensues. The exact shape of the curve during the death phase will depend on the nature of the organism under observation and the many factors that contribute to cell death. The death phase may assume a linear function such as during heat-induced death where viable cell numbers decline logarithmically.

Some additional considerations of the growth curve are important in assessing the effect of various internal as well as external factors on growth. Since the number of cells in a population (b) is equal to the number of cells in the initial inoculum (a) $\times 2^n$,

$$b = a \times 2^n$$

Then

$$\log_2 b = \log_2 a + n$$

$$\log_{10} b = \log_{10} a + n \log_{10} 2$$

$$\log_{10} b = \log_{10} a + (n \times 0.3010)$$

Solving the equation for n , the number of generations that occurred between the time of inoculation and the time of sampling is

$$n = \frac{\log_{10} b - \log_{10} a}{0.3010}$$

The generation time (t_g) or doubling time may be determined by dividing the time elapsed (t) by the number of generations (n):

$$t_g = t/n$$

Continuous Culture

Usually bacteria are grown in “batch” culture in which a flask containing media is inoculated and growth is allowed to occur. This is a closed system where it is actually very difficult to manipulate growth rate. In batch cultures, growth rate is determined internally by properties of the bacteria themselves. A batch culture can be used to grow bacteria at different rates as long as the nutrient added is at a concentration that does not support maximal growth. But, to accomplish this, the cell density, and thus the cell number, will be too low for certain analyses. To grow bacteria at slow growth rates and at high cell density, a chemostat is used. In this apparatus, fresh medium containing a limiting nutrient is added from a reservoir to the culture vessel at a set rate. The volume in the culture vessel is kept constant by an overflow device that removes medium and cells at the same rate as fresh medium is added. In a chemostat, growth rate is determined externally by altering the rate-limiting nutrient added to the culture vessel. The faster the limiting nutrient is added, the faster the growth rate.

FACTORS AFFECTING GROWTH

Nutrition

All living organisms have certain basic nutritional requirements: sources of carbon, nitrogen, energy, and essential growth factors (minerals and vitamins) are needed to support growth. Microorganisms vary widely in their nutritional requirements. Two main groups of organisms are classified on the basis of their ability to gain energy from certain sources and the manner in which they satisfy their carbon and nitrogen requirements for growth:

1. **Lithotrophs** utilize carbon dioxide as the sole source of carbon and gain energy through the oxidation of inorganic compounds (**chemolithotrophs** or “rock eaters”) or light (**photolithotrophs**). Inorganic nitrogen is utilized for the synthesis of organic compounds.

2. **Organotrophs** generally prefer organic substrates as a source of energy and carbon. **Photoorganotrophs** utilize light as a source of energy for the assimilation of carbon dioxide as well as organic compounds. **Chemoorganotrophs** utilize organic compounds for growth.

Although their nutritional requirements are remarkably simple, chemolithotrophic bacteria must be metabolically complex since they synthesize all of their cellular components and provide the energy for this activity through the oxidation of inorganic compounds. One fundamental characteristic of strict chemolithotrophs is that they are unable to grow on or assimilate exogenous organic compounds. Facultative chemolithotrophs can utilize exogenous organic carbon sources. Chemolithotrophs possess unique mechanisms for carbon dioxide fixation such as the ribulose biphosphate (Calvin-Benson) cycle and the reductive carboxylic acid (Campbell-Evans) cycle (see Chapter 9).

Some organotrophic organisms utilize carbon dioxide as a source of carbon, but most prefer organic carbon sources and generally cannot subsist on carbon dioxide as the sole carbon source. Organotrophs may use inorganic nitrogen, but most members of the group grow better when supplied with organic nitrogen compounds. For example, *E. coli*, *Enterobacter aerogenes*, yeasts, and molds grow luxuriantly on glucose as the only organic nutrient. Other organotrophs such as streptococci and staphylococci also exhibit specific requirements for one or more nitrogen sources as amino acids, purines, or pyrimidines (see Table 1-2).

Fatty acids are required by some organisms, particularly in the absence of certain B vitamins. Replacement of a growth factor requirement by the addition of the end product of a biosynthetic pathway in which the vitamin normally functions is referred to as a **sparing action**. This type of activity has been reported for many growth factors, including amino acids, purines, pyrimidines, and other organic constituents. If a vitamin can completely replace a particular organic nutrient in a defined medium, that nutrient cannot be regarded as a true growth requirement since it can be synthesized in the presence of the requisite vitamin.

TABLE 1-2. Nutritional Requirements of Some Organotrophs

	<i>Escherichia coli</i>	<i>Salmonella typhi</i>	<i>Staphylococcus aureus</i> ^a	<i>Leuconostoc paramesenteroides</i> ^b
Basic Nutrients				
Glucose	}			Required by all for maximum growth in defined medium
NH ₄ ⁺				
Mn ²⁺				
Mg ²⁺				
Fe ²⁺				
K ⁺				
Cl ⁻				
SO ₄ ²⁻				
PO ₄ ³⁻				
Additional Requirements				
	None	Tryptophan	Nicotinic acid Thiamine 10 amino acids	Nicotinic acid Thiamine Pantothenate Pyridoxal Riboflavin Cobalamin Biotin <i>p</i> -Aminobenzoate Folate Guanine Uracil 16 Amino acids Sodium acetate Tween 80

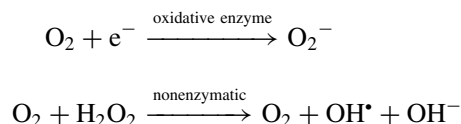
^aFrom Gladstone, G. P. 1937. *Br. J. Exp. Pathol.* **18**:322.

^bFrom Garvie, E. I. 1967. *J. Gen. Microbiol.* **48**:429.

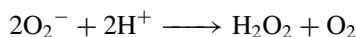
Although most bacterial membranes do not contain sterols, they are required in the membranes of some members of the Mycoplasmataceae. (These organisms do not possess a cell wall.) *Mycoplasma* require sterols for growth. *Acholeplasma* do not require sterols; however, they produce terpenoid compounds that function in the same capacity as sterols. Fungi (yeasts and molds) contain sterols in their cell membranes but in most cases appear to be capable of synthesizing them.

Oxygen

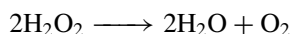
Microorganisms that require oxygen for their energy-yielding metabolic processes are called **aerobes**, while those that cannot utilize oxygen for this purpose are called **anaerobes**. **Facultative organisms** are capable of using either respiratory or fermentation processes, depending on the availability of oxygen in the cultural environment. Aerobic organisms possess cytochromes and cytochrome oxidase, which are involved in the process of oxidative phosphorylation. Oxygen serves as the terminal electron acceptor in the sequence and water is one of the resultant products of respiration. Some of the oxidation–reduction enzymes interact with molecular oxygen to give rise to superoxide (O_2^-), hydroxyl radicals ($\text{OH}^\bullet$), and hydrogen peroxide (H_2O_2), all of which are extremely toxic:



The enzyme **superoxide dismutase** dissipates superoxide:



Superoxide dismutase is present in aerobic organisms and those that are aerotolerant, but not in strict anaerobes. Many, but not all, aerobes also produce catalase, which can eliminate the hydrogen peroxide formed:



Aerotolerant organisms generally do not produce catalase. Hence, growth of these organisms is frequently enhanced by culture on media containing blood or other natural materials that contain catalase or peroxidase activity. Organisms that do not utilize oxygen may tolerate it because they do not interact in any way with molecular oxygen and do not generate superoxide or peroxide.

Anaerobic bacteria from a variety of genera are present in the normal flora of the animal and human body as well as in a number of natural habitats such as the soil, marshes, and deep lakes. A number of the more widely known genera of anaerobic organisms are listed in Table 1-3.

Carbon Dioxide

Many organisms are dependent on the fixation of carbon dioxide. Certain organisms thrive better if they are grown in an atmosphere containing increased carbon dioxide.

TABLE 1-3. Genera of Anaerobic Bacteria

Bacilli		Cocci	
Gram Positive	Gram Negative	Gram Positive	Gram Negative
<i>Clostridium</i>	<i>Bacteroides</i>	<i>Peptococcus</i>	<i>Veillonella</i>
<i>Actinomyces</i>	<i>Fusobacterium</i>	<i>Peptostreptococcus</i>	
<i>Bifidobacterium</i>	<i>Vibrio</i>	<i>Ruminococcus</i>	
<i>Eubacterium</i>	<i>Desulfovibrio</i>		
<i>Lactobacillus</i>			
<i>Propionibacterium</i>			

Haemophilus, *Neisseria*, *Brucella*, *Campylobacter*, and many other bacteria require at least 5 to 10% carbon dioxide in the atmosphere to initiate growth, particularly on solid media. Even organisms such as *E. coli* use carbon dioxide to replenish intermediates in the TCA (tricarboxylic acid) cycle that have been siphoned off as precursors for amino acid synthesis. These anapleurotic reactions include pyruvate carboxylase, phosphoenolpyruvate carboxylase, or malic enzyme (see Chapter 8).

Extremophiles

Microorganisms vary widely in their ability to initiate growth over certain ranges of temperature (Table 1-4), hydrogen ion concentration (Table 1-5), and salt concentration. Organisms that function best under extreme environmental conditions are called **extremophiles**. Examples include bacteria found in hot springs and in the thermal vents on the ocean floor. These organisms prefer to grow at extremely high temperatures. Some microorganisms prefer to live in an acidic environment (**acidophilic** organisms) while others prefer an alkaline pH (**alkaliphilic** organisms). *E. coli* prefers a neutral pH environment and thus is classified as **neutrophilic**. (The older term, neutrophilic,

TABLE 1-4. Temperature Ranges of Bacterial Growth

Type of Organism	Growth Temperature (°C)		
	Minimum	Optimum	Maximum
Psychrophilic	−5–0	5–15	15–20
Mesophilic	10–20	20–40	40–45
Thermophilic	25–45	45–60	>80

TABLE 1-5. pH Limits for Growth of Various Microorganisms

Organism	Minimum	Optimum	Maximum
Bacteria	2–5	6.5–7.5	8–11
Yeasts	2–3	4.5–5.5	7–8
Molds	1–2	4.5–5.5	7–8

is not consistent with the nomenclature of the other two groups and can be confused with neutrophils, a form of white blood cell, and thus should not be used.) The ability of certain organisms to grow in extreme environments can be linked to the possession of unique membrane compositions and/or enzymes with unusual temperature or pH optima that are more suitable to their environment.

Microbial Stress Responses

For **normalophiles**, meaning organisms that prefer to grow under conditions of 37°C, pH 7, and 0.9% saline, variations in pH and temperature have a marked impact on enzyme activity and, ultimately, viability. Outside their optimal parameters, enzymes function poorly or not at all, membranes become leaky, and the cell produces compounds (e.g., superoxides) that damage DNA and other macromolecular structures. All of these factors contribute to cell death when the cell is exposed to suboptimal environments. However, many, if not all, microorganisms have built-in stress response systems that sense when their environment is deteriorating, such as when medium acidifies to dangerous levels. At this point, signal transduction systems perceive the stress and transmit instructions to the transcription/translation machineries to increase expression of specific proteins whose job is to protect the cell from stress. The various genetic regulatory systems and protection strategies used by the cell to survive stress are discussed in Chapters 5 and 18.

SUMMARY

This chapter is a highly condensed version of the remainder of this book, provided to build a coherent picture of microbial physiology from the start. Too often textbooks present a student with excruciatingly detailed treatments of one specific topic after another without ever conveying the “big picture.” As a result, the information overload is so great that the student, lost in the details, never develops an integrated view of the cell and what makes it work. Our hope is that the framework in this chapter will be used to build a detailed understanding of microbial physiology and an appreciation of its future promise.

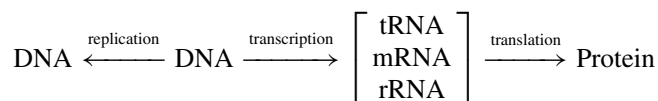
CHAPTER 2

MACROMOLECULAR SYNTHESIS AND PROCESSING: DNA, RNA, AND PROTEIN SYNTHESIS

The way nature has designed and interwoven thousands of biochemical reactions to form the cell is still not fully understood. Yet, what we have managed to decipher is nothing short of amazing. Our discussion starts with the molecular epicenter of the cell—DNA—and builds layer upon layer, illustrating what is known about the biochemical basis of life and how perturbing just one small aspect of a cell's physiology can have a massive ripple effect on hundreds of seemingly unrelated biochemical processes.

This chapter focuses on macromolecules including peptidoglycan, the rigid, cross-bridged molecule that provides cell shape and structural integrity; proteins that carry out biochemical activities; and nucleic acids (DNA and RNA), which contain the information necessary to coordinate cell activities into a balanced system of orderly growth and development. Our discussion of the bacterial cell begins with what are arguably the core processes of all life: nucleic acid and protein syntheses. Peptidoglycan synthesis is described in Chapter 7.

During growth, the continuous synthesis of various building blocks and their proper assembly into the structures of the cell are dependent on the fidelity with which DNA is replicated. The chain of events emanating from DNA follows the general scheme:



Knowledge of these processes is fundamental to understanding the biological processes of life including energy production, building block biosynthesis, growth and its regulation. Throughout the remainder of the book we refer to numerous genetic loci from *Escherichia coli* and *Salmonella enterica*. More detail on each gene can be found by referring to [Berlyn, 1998] or on the Internet at www.ucalgary.ca/%7Ekesander/

(*Salmonella*), cgsc.biology.yale.edu/(*Escherichia*), or genolist.pasteur.fr/ (*E. coli*, *Mycobacterium*, *Bacillus*, *Helicobacter*).

STRUCTURE OF DNA

In order to appreciate the processes of replication, transcription, and translation of the information contained within a DNA molecule, it is necessary to describe the structure of DNA and to illustrate its ability to direct the events occurring in the cell. DNA is composed of the four deoxyribonucleosides [purine (adenine or guanine) and pyrimidine (cytosine or thymine) bases + deoxyribose] arranged in a chain with phosphodiester bonds connecting the 5'-carbon on the deoxyribose of one nucleoside to the 3'-carbon of the deoxyribose of the adjacent nucleoside as shown in Figure 2-1. The property of polarity (illustrated by the lower part of Fig. 2-1) is maintained by the orientation of the phosphodiester bonds on the 5' and 3' positions of the deoxyribose in the DNA strand. In the bacterial chromosome, DNA is composed of two such strands arranged in apposition; that is, it is double stranded (dsDNA). In dsDNA, the two strands have opposite **polarities** as shown in Figure 2-2 and are described as being antiparallel. One strand will end with a 3'-hydroxyl group (the 3' terminus), while the other will end with a 5'-phosphate (the 5' terminus). Notice, too, that DNA is a negatively charged molecule under physiological conditions.

The purine and pyrimidine bases are planar structures that pair with one another through the forces of hydrogen bonding, usually depicted as dashed or dotted lines in diagrammatic presentations. These base pairs are stacked parallel to one another and lie perpendicular to the phosphodiester backbone of the structure. As a result of covalent

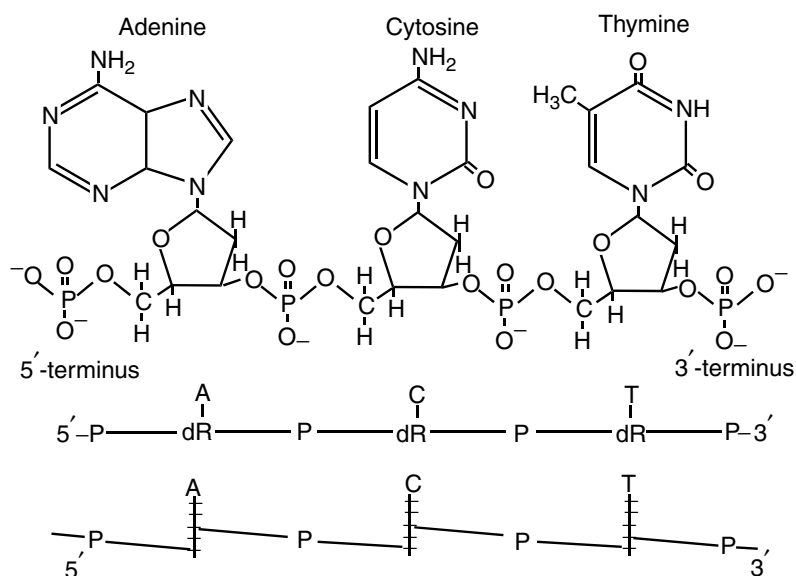


Fig. 2-1. Structure of DNA showing alignment of adjacent nucleotides and some common methods of depicting polarity in abbreviated structures. The crossbars on the vertical lines represent the carbon atoms in deoxyribose.

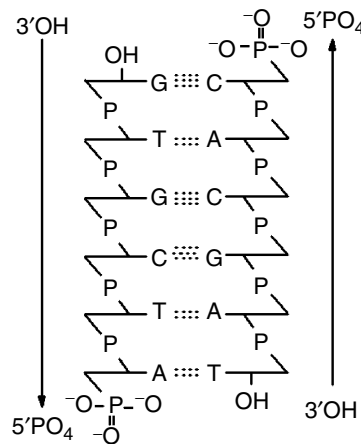


Fig. 2-2. Polarity of the double-stranded DNA molecule. This abbreviated diagrammatic view uses the conventions described in Figure 2-1.

bond angles and noncovalent forces, native DNA assumes a helical structure that exists in different forms depending on the composition of the DNA and the environment. In an aqueous environment, DNA exists primarily in a stable conformation (B form) in which there are 3.4 Å between the stacked bases and 34 Å per right-handed helical turn. Therefore, there are 10 base pairs per turn, and the bases will be arranged so that they are perpendicular to the helical axis as shown in Figure 2-3.

Under certain conditions, DNA can shift into either the **A** conformation (11 rather than 10 base pairs per turn and tilted bases) or become **Z-DNA** (left-handed rather than right-handed helix having 12 base pairs per turn). Native DNA (in vivo) very likely exists in a variety of conformations. Different areas of the bacterial chromosome may well differ topologically as a result of variations in the local environment. Structural differences of this kind may be essential to certain functions of DNA, particularly with regard to the binding of regulatory proteins to specific areas of the chromosome.

Z-DNA is most often formed as a result of extended runs of deoxyguanosine-deoxycytosine [poly(dG-dC)]. In vitro this also requires a fairly high salt environment because poly(dG-dC) will assume a B structure in low salt. Over the years there has been some controversy regarding the in vivo existence and significance of Z-DNA conformations. Proof that Z-DNA can exist in vivo was provided by R. D. Wells and colleagues by placing segments of poly(dG-dC) immediately next to a sequence of DNA that is specifically methylated only when in the B form. This sequence was dramatically undermethylated in vivo when the Z-DNA sequence was included. Evidence that Z-DNA naturally exists in vivo was based on the finding that Z-DNA, unlike B-form DNA, is a reasonable antigen. Antibodies to Z-DNA have been used to demonstrate that the *E. coli* genome has an average of one Z-DNA segment for every 18 kilobars of DNA.

Under certain circumstances, regions of double-stranded DNA may be converted from the double-stranded helix into a nonbase-paired random coil bubble. Both of these states may exist in different regions of the chromosome at the same time, and the process of conversion may be freely reversible. This helical-random coil transition may be essential for the function of DNA in replication and transcription.

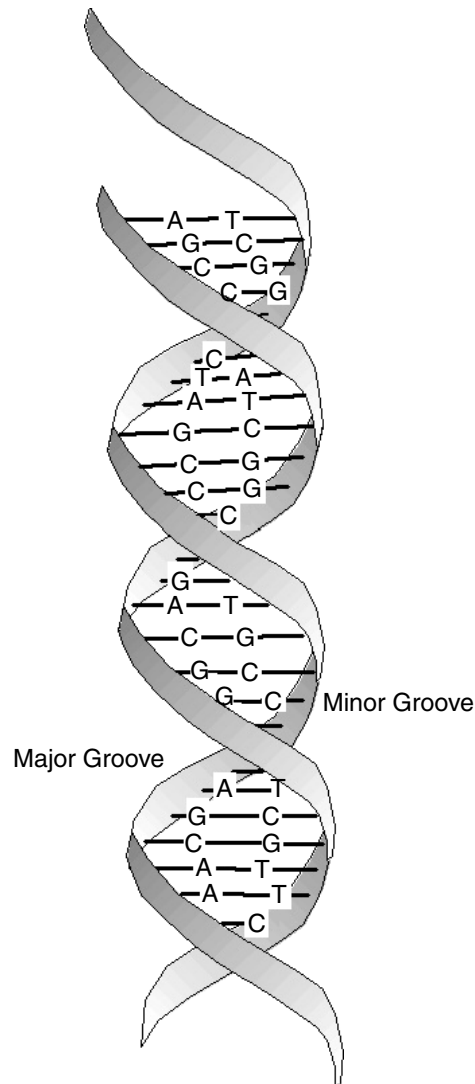


Fig. 2-3. Helical DNA structures.

An interesting biological question concerns how regulatory DNA-binding proteins recognize their proper DNA-binding sequences, since DNA duplexes present a fairly uniform surface to potential regulatory proteins (i.e., a uniform phosphodiester backbone that faces outward). There are two potential mechanisms for nucleic acid–protein interaction. The first model predicts that the specificity of interaction relies on groups present in the major or minor grooves of dsDNA (see Fig. 2-3). Thymine and adenine have groups that will protrude into the grooves. Proteins interact with these groups to bind to specific regions of DNA. The second model is that direct access to bases may occur by a protein interacting with DNA in a randomly melted region. DNA-binding proteins can have one of several characteristic structures that will aid in sequence-specific binding. They are discussed in Chapter 5.

Bacterial Nucleoids

In all living cells, DNA takes on a highly compacted tertiary structure. The first stage of compaction for eukaryotic cells involves the winding of DNA around an octomeric assembly of histone proteins to form the nucleosome structure (see Chapter 7). At this stage, a linear stretch of DNA takes on a bead-like appearance. The second stage is thought to be the solenoidal organization of these nucleosomes into a helical network.

The packaging of DNA in prokaryotic cells is understood less completely, but the problem is obvious. The following example illustrates the extent to which the chromosome of *E. coli* must be compacted to fit in the cell. The *E. coli* genome is about 1.5 mm long (4.2×10^{-15} g). In exponentially growing cells, the DNA comprises 3 to 4% of the cellular dry mass, or about $12\text{--}15 \times 10^{-15}$ g per cell. Thus, there are around three genomes in the average growing cell. If chromosomal DNA were homogeneously distributed over the entire cell volume (1.4×10^{-12} ml), the packing density would be 10 mg/ml. However, DNA is confined to ribosome-free areas of the cell, bringing its packing density to 15–30 mg/ml. The inevitable conclusion is that the chromosome is packaged into what is loosely referred to as a “compactosome.” One component in this process involves supercoiling the DNA molecule. The bacterial chromosome is a covalently closed, circular, double-stranded DNA molecule that is actually *underwound* in the cell, resulting in a negatively supercoiled structure. DNA supercoiling occurs when a circular dsDNA molecule becomes twisted so that the axis of the helix is itself helical (Fig. 2-4). If DNA is overwound (tighter coils made by twisting DNA opposite to the direction of the helix) or underwound (twisted in the same direction as the helix), a significant amount of torsional stress is introduced into the molecule. (Try this with a coiled telephone cord.) In order to relieve this stress, the DNA double helix will twist upon itself. Underwound DNA will form negative supercoils, whereas overwinding results in positive supercoils.

Topoisomerases. Topoisomerases are enzymes that alter the topological form (supercoiling) of a circular DNA molecule (Table 2-1). There are two classic types—I and II—named for whether they cleave one or both strands of DNA, respectively. Type I topoisomerases are further subdivided as type IA (e.g., *E. coli* TopA), which nicks one strand of DNA and passes the other strand through the break; and type IB (e.g., *E. coli* Top III) enzymes, which relax negatively supercoiled DNA by breaking one of the phosphodiester bonds in dsDNA, subsequently allowing controlled rotational diffusion (“swivel”) of the 3′-hydroxyl end around the 5′-phosphoryl end. Both subtypes then reseal the “nicked” phosphodiester backbone (Fig. 2-5).

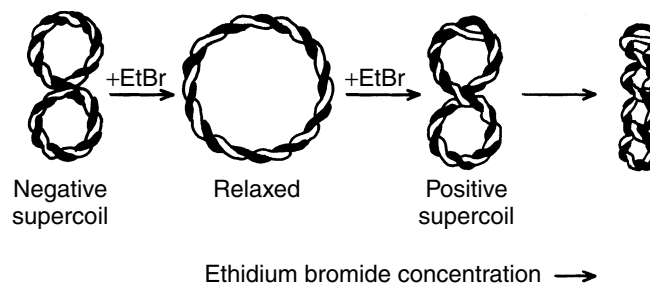


Fig. 2-4. Superhelicity of a closed circular DNA molecule changing from negative supercoil to positive supercoil with increasing concentrations of ethidium bromide.

TABLE 2-1. DNA Topoisomerases

Enzyme	Type	Relaxation of (–) or (+) Twists	Introduction of (–) Twists	Linking of Duplex DNA Circles	ATP Requirement
<i>E. coli</i> topoisomerase I (<i>topA</i> , (omega protein)	IA	(–)	No	Yes	No
Phage λ int	?	(–), (+)	No	Yes	No
Bacterial DNA gyrase (Top II)	II	(–)	Yes	Yes	Yes
<i>E. coli</i> topoisomerase III (<i>topB</i>)	IB	(–), (+)	No	Yes	No
Phage T4 topoisomerase	II	(–), (+)	No	Yes	Yes

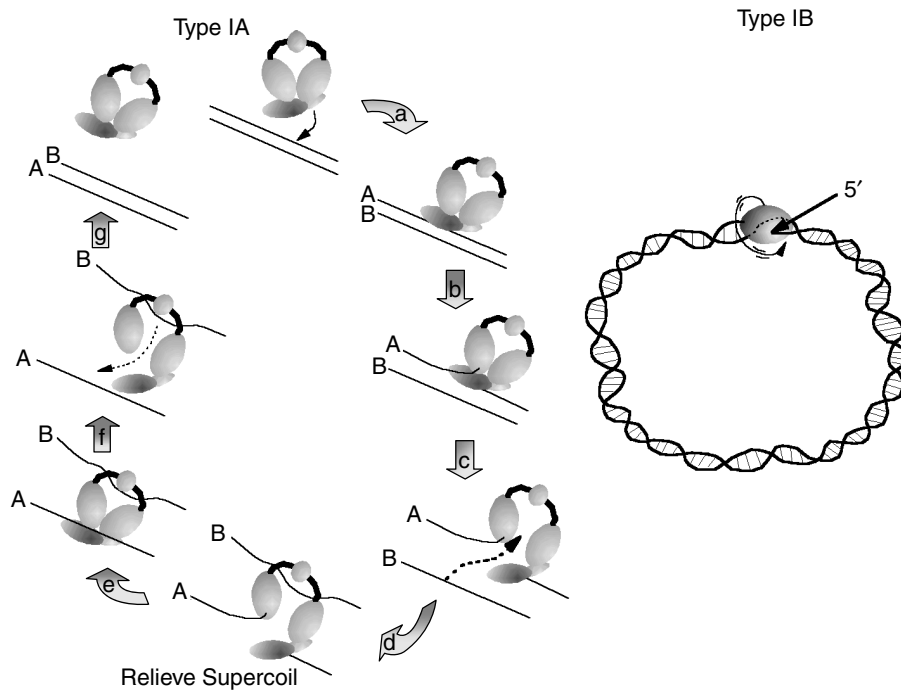


Fig. 2-5. Action of type I topoisomerases. Type I topoisomerases (e.g., *E. coli* TopA) relax negatively supercoiled DNA apparently by one of two mechanisms. Both types IA and IB break one of the phosphodiester bonds in one strand of a DNA molecule. Type IA topoisomerases catalyze a cleavage and transport cycle to relieve supercoils. Domain III of TopA binds, cleaves, and opens an ssDNA segment. Separation is achieved by domain III lifting away from domains I and IV while remaining tethered to the rest of the protein through the domain II arch. After the other strand passes through the break, domain II reassociates with domains I and IV and religates the cleaved strand. Following religation, domain III again lifts away from the rest of the protein, allowing exit of the segment. Notice strands A and B have switched places. Type IB enzymes such as TopB also cleave a single strand but then allow the protein-bound 3'-hydroxyl end of the DNA to swivel around the 5'-phosphoryl end. The topoisomerase then reseals the nicked phosphodiester backbone and the DNA is partially unwound.

Type II topoisomerases, in contrast, require energy to underwind DNA molecules and introduce negative supercoils. Type II topoisomerases, such as DNA gyrase (GyrA and GyrB, called topoisomerase II) and topoisomerase IV (ParC, E), can also relax negative supercoils in the absence of energy. These enzymes utilize the mechanism described in Figure 2-6. The model requires the passage of one dsDNA molecule through a second molecule, which has a double-stranded break in its phosphodiester

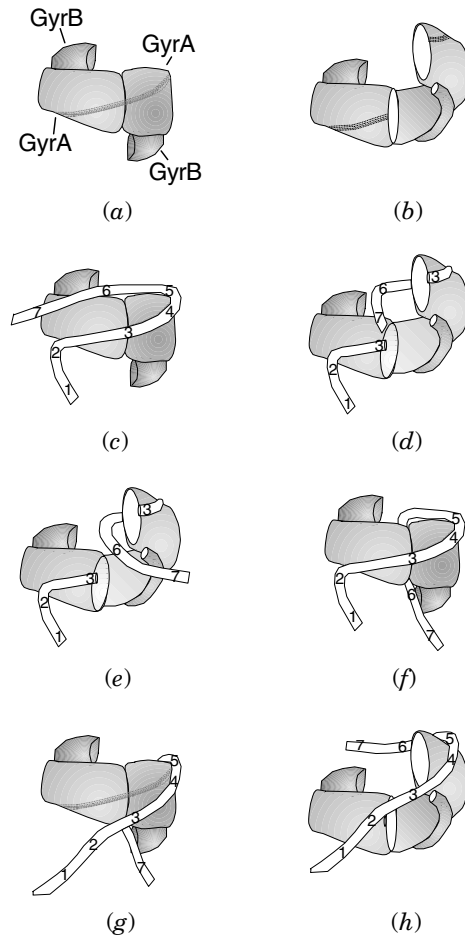


Fig. 2-6. Model for DNA gyrase. (a) Tetrameric gyrase molecule in closed conformation. (b) Gyrase molecule in open conformation. Illustrations c through h represent steps in a scheme for negative supercoiling by inversion of a right-handed DNA loop (c) into a left-handed one (f). (c) A small section of a circular double-stranded DNA molecule is represented as a tube (numbered). The DNA is bound to gyrase in a right-handed coil. Three is the point of DNA cleavage and covalent attachment of subunit A polypeptides. Sections 1–3 and 3–5 are wrapped around the left and right A subunits, respectively, while DNA at 6 contacts a B subunit. (d) Gyrase cleaves DNA at 3 and opens. (e) DNA (point 6) is transferred from the left to the right B subunit through the cleavage point. (f) Gyrase closes and reseals DNA at point 3. The DNA between points 3 and 7 now describes a left-handed coil. (g and h) DNA partially unwraps from gyrase. The gyrase opens, releasing the DNA, and then recloses. (Modified From Wang et al., in Alberts, Mechanistic Studies of DNA replication and genetic recombination 1980.)

backbone. The steady-state level of DNA supercoiling *in vivo* appears to involve interplay between Top I and Top III to relax negative supercoils and DNA gyrase, which introduces negative supercoils. As is discussed later, this supercoiling is required for efficient replication and transcription of procaryotic DNA.

Nucleoid domains and Nucleoid-Associated proteins. As already noted, the bacterial nucleoid is highly compacted, but supercoiling alone cannot account for the level of compaction. A cell 1 μm in length must accommodate a chromosome 1500 μm in length, and it must do this in a way that does not entangle the molecule so that transcription and replication are not impeded! The solution is that the bacterial chromosome actually contains between 30 and 200 negatively supercoiled loops or domains (Fig. 2-7). Each domain represents a separate topological unit, the boundaries of which may be defined by sites on DNA (see “REP Elements” in this chapter) that bind anchoring proteins. Gathering the DNA loops at their bases would compact the chromosome to a radius of less than 1 μm —small enough to fit in the cell. These loops are supercoiled, compacted, and centrally located within the cell. A single-stranded nick in one loop leads to its relaxation without disturbing the other supercoiled loops.

Four different histone-like proteins isolated from *E. coli* contribute to compactosome structure (Table 2-2). They are all basic, heat-stable, DNA-binding proteins present in high proportions on the *E. coli* nucleoid. Their similarity to eukaryotic histones was used to extrapolate a role in forming prokaryotic nucleosomes. However, there is some controversy as to the existence of nucleosomes in *E. coli*. For example, the level of histone-like proteins does not appear sufficient to generate a nucleosome structure. The most abundant histone-like protein, HU, is not even associated with the bulk of the DNA. Instead, it is situated where transcription and translation occur and may play a more direct role in those processes. However, the proteins that associate with the nucleoid are distributed equally over the DNA.

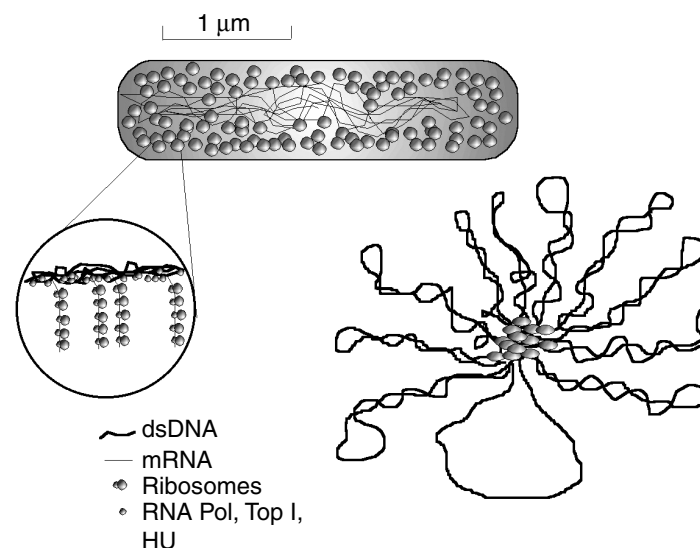


Fig. 2-7. Schematic representation of *E. coli* nucleoids in situ and after isolation and spreading. The insert illustrates an area active in transcription and translation.

TABLE 2-2. Histone-like Proteins of *Escherichia coli*

Name	Subunits	Molecular Weight	Function
Hu (<i>hupA</i>) (<i>hupB</i>)	Hu(α)[HLP _{IIa}] Hu(β)[HLP _{IIb}] (associated as heterologous dimers)	9000 9000	Can form nucleosome-like structures similar to eukaryotic histone H2B
H-NS (H1, <i>hns</i>)		15,000	May selectively modulate in vivo transcription, nucleoid compaction
StpA		15,000	H-NS analog

The most abundant proteins associated with the nucleoid include HU, H-NS, Fis, and RNA polymerase. Recently it has been shown that H-NS, which forms dimers or tetramers, can actually compact DNA in vitro. Two H-NS dimers bound to different regions of a DNA molecule will bind to each other and condense DNA. The tethering points for each domain are suspected of being DNA gyrase, as is the case in eukaryotes, and perhaps HU. In contrast to eukaryotic nucleosomes, the prokaryotic DNA would be in continuous movement such that transcriptionally active segments would be located at the ribosome–nucleoid interface. The inactive parts would be more centrally located. An alternative hypothesis is that the coupled transcription/translation/secretion of exported proteins serves to tether regions of the chromosome to the membrane, thereby defining different chromosomal domains (see Protein Trafficking).

Another role of histones in eukaryotes is charge neutralization of DNA. Could this also be the case in *E. coli*? The data suggest that this is not the situation. Because there is so little histone-like protein in *E. coli*, neutralization of DNA charge must be accomplished by other molecules such as polyamines and Mg^{2+} .

What role might DNA supercoiling have besides nucleoid compaction? There is mounting evidence that the expression of many genes can vary with the extent of negative supercoiling. The cell may even regulate certain sets of genes by differentially modulating domain supercoiling. Evidence suggests that this may occur in response to environmental stresses such as an increase in osmolarity or change in pH. The subsequent, coordinated increase in expression of some genes and decreased expression of others would enable the cell to adapt and survive the stress. Further detail on nucleoid structure of both prokaryotic and eukaryotic microbes is presented in Chapter 7.

REP Elements

One sequence element present in prokaryotic genomes that may play a role in the organization of the nucleoid is called the repetitive extragenic palindrome (REP) or palindromic unit (PU) sequence. It is a 38 base pair palindromic consensus sequence capable of producing a stem-loop structure. A consensus sequence is defined by comparing several like sequences and determining which bases are conserved among them. A palindromic sequence is one in which the 5'-3' sequence of the top strand is the same as the 5'-3' sequence read on the bottom strand. As a result, intrastrand base pairing can occur, resulting in hairpin structures. The consensus REP sequence for *E. coli* is 5'A(AT)TGCC(TG)GATGCG(GA)CG(CT)NNNN(AG)CG(TC)CTTATC

(AC)GGCCTAC(AGX). N indicates any base can be found at that position. Two bases within parentheses indicate one or the other base can be found at that position. The imperfect palindromic region can be observed by starting at the NNNN region and reading in both directions, noting that the bases are complementary (see underlined regions). The REP elements are located in different orientations and arrays between genes within an operon or at the end of an operon. Although there are between 500 and 1000 REP copies, they are always found outside of structural genes. The suggestion that REP or a complex of REP sequences called BIMEs (bacterial-interspersed mosaic elements) play a part in nucleoid structure comes from in vitro studies showing specific interactions between REPs, DNA polymerase I, and DNA gyrase. However, the actual function for these elements is not known.

DNA REPLICATION

DNA Replication is Bidirectional and Semiconservative

For all organisms, the production of viable progeny depends on the faithful replication of DNA by DNA polymerase. DNA replication proceeds in a semiconservative manner where one strand of the parental molecule is contributed to each new (daughter) dsDNA molecule as shown in Figure 2-8. During the first round of DNA replication, each strand is duplicated, so at the end of this step each of the resulting DNA molecules will contain one old and one new strand. Only after the second round of replication will two completely new strands form. Autoradiographic studies provide evidence that the bacterial chromosome remains circular throughout replication and that the ring is doubled between the point of initiation and the growing point (replication fork). If the replication process goes to completion, the entire cyclic chromosome will be duplicated and the two new daughter chromosomes will separate.

Enzymes called DNA polymerases catalyze the polymerization of deoxyribonucleotides in an ordered sequence dictated by a preexisting DNA template. *E. coli* contains five DNA polymerase activities: Pol I (*polA*), Pol II (*dinA*), Pol III (*dnaE*), Pol IV (*dinB*), and Pol V (*umuDC*). We focus on Pol I and III in this section. All DNA polymerases synthesize DNA in the $5' \rightarrow 3'$ direction and require the presence of a preexisting 3'-hydroxyl primer (see Figs. 2-9 and 2-10). Pol I has a bound zinc ion and, in addition to the polymerase activity, possesses a $3' \rightarrow 5'$ exonuclease activity used for "proofreading" (see below) and a $5' \rightarrow 3'$ exonuclease activity. Pol II and Pol III also have $3' \rightarrow 5'$ exonuclease activities.

DNA Polymerase Functions as a Dimer

The fact that polymerases synthesize DNA only in the $5' \rightarrow 3'$ direction raises a logistics problem when you consider that replication of both strands occurs simultaneously in what appears to be the same direction. The appearance is that one of the newly synthesized strands violates the $5' \rightarrow 3'$ directive and elongates in the $3' \rightarrow 5'$ direction. DNA Pol III, the polymerase primarily responsible for replication, manages to replicate both strands simultaneously and maintains the $5' \rightarrow 3'$ rule because it works as a **dimer** at the replication fork. Two polymerase holoenzyme complexes are held together by the θ protein (Fig. 2-10). One member of the dimer continuously synthesizes one strand in a $5' \rightarrow 3'$ direction while the other

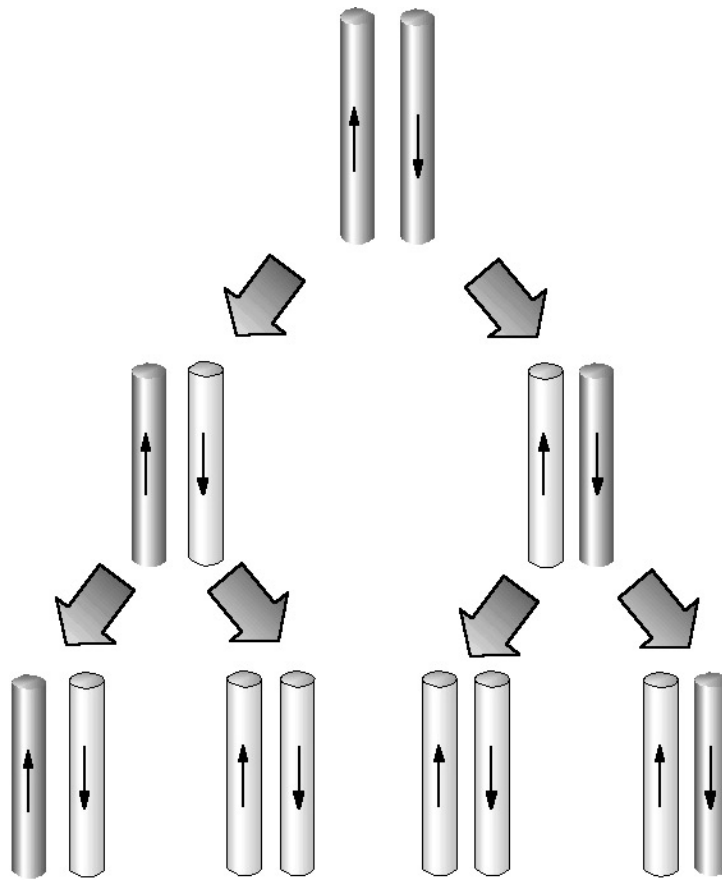


Fig. 2-8. Semiconservative DNA replication. Note that after the first round of replication, each strand of DNA is composed of one new and one old strand. Only after the second round of replication are two completely new strands formed.

polymerase replicates the other strand discontinuously in small fragments of 1000 to 2000 nucleotides (Okazaki fragments). Thus, the template strands are referred to as the leading and lagging strands, respectively. Figure 2-10 illustrates how this is accomplished. The lagging strand loops out and around one of the polymerases. The Pol III dimer can now synthesize the two new DNA strands in the same direction within the moving replication fork while actually moving in opposite directions along the template strands. To avoid difficulties that would result from the generation of very long DNA loops, lagging strand synthesis proceeds discontinuously.

Since DNA polymerases cannot initiate new chain synthesis in the absence of an RNA primer, a primer-generating polymerase (DNA primase) is required for the initiation of each new Okazaki fragment. The primase in *E. coli* is the product of the *dnaG* locus. The short (11 bp) RNA primers are elongated by DNA polymerase III, “erased” by the 5′ → 3′ exonuclease activity of RNase H (an RNA-degrading enzyme that recognizes DNA : RNA hybrids), and resynthesized against the template as DNA by DNA polymerase I. The single-strand nick in the phosphodiester backbone is then resealed by DNA ligase as shown in Figure 2-11.

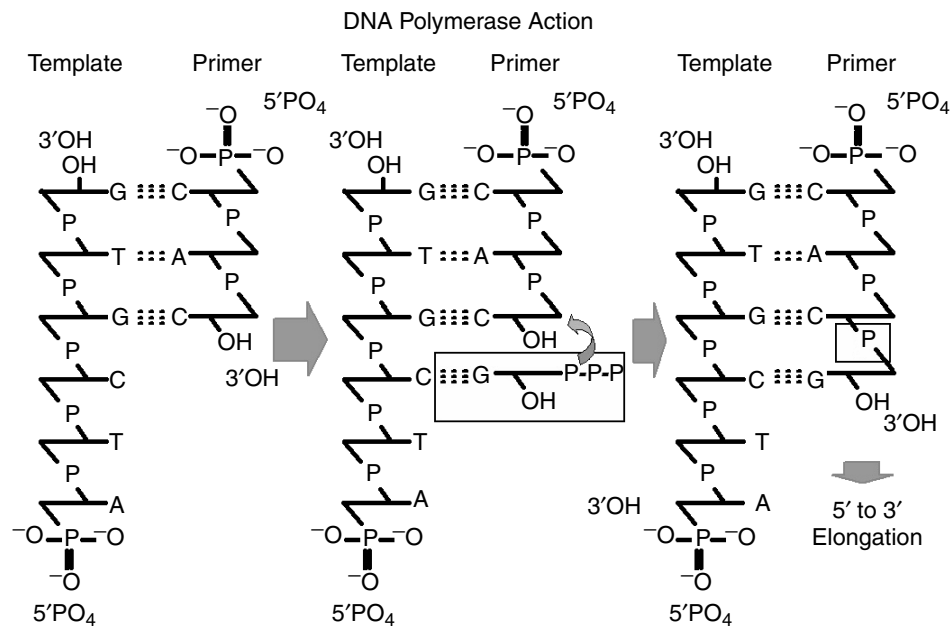


Fig. 2-9. Schematic representation of the action of DNA polymerase in the synthesis of DNA. A DNA template, a complementary RNA primer with a free 3'-hydroxyl end, and Mg^{2+} are required. Replication by DNA polymerase takes place in the 5' → 3' direction.

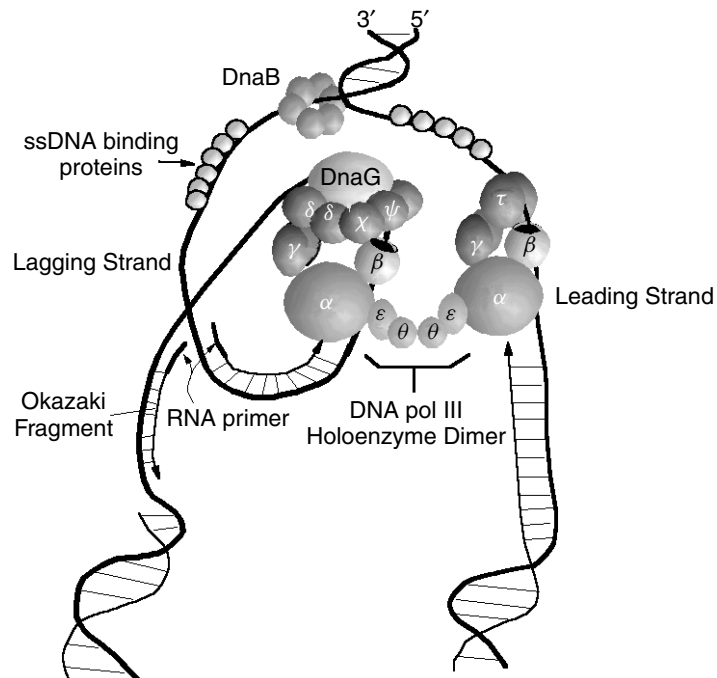


Fig. 2-10. Diagrammatic representation of the replicating fork in DNA synthesis in microbial cells.

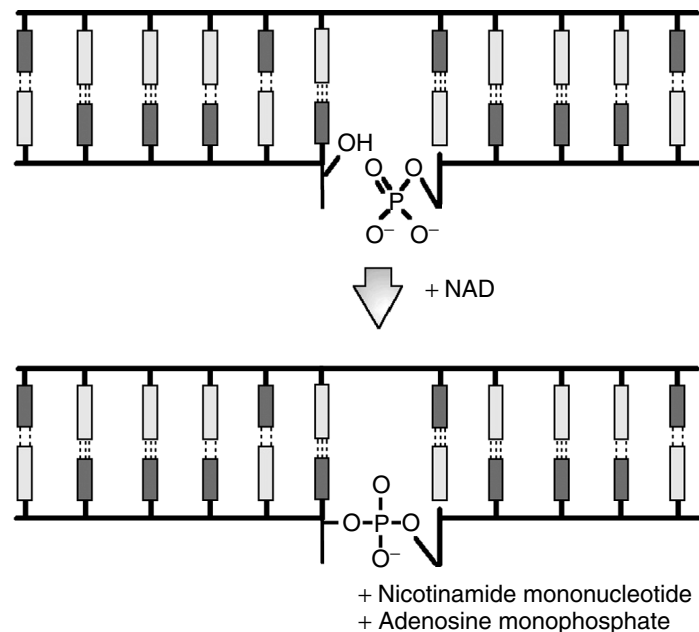


Fig. 2-11. Action of DNA ligase in joining adjacent 3'-hydroxyl and 5'-phosphoryl groups in duplex DNA.

Model of DNA Replication

It is amazing to consider that the bacterial chromosome (4,639,221 bp) replicates at a rate of 800 to 1000 nucleotides per second, yet the frequency of error only amounts to 1 in 10^{10} base pairs replicated. Thus, a high degree of fidelity is maintained under normal circumstances. To bring some perspective to the magnitude of this feat, consider the following. If the DNA duplex was 1 meter in diameter, the replication fork would have to move at about 600 km/hour (375 mph) and the replication machinery would be about the size of a truck. Each replication machine—remember, there are two—would only make a mistake every 170 km (106 miles) over a 400 km (250 mile) trip.

Once initiated, replication is an elaborate process requiring a large number of proteins (Table 2-3). Figure 2-10 presents a detailed model of the replication fork. DNA synthesis is actually a collaboration of two polymerases: primases (DnaG), which start chains, and replicative polymerases, which add to those chains, synthesizing the whole DNA molecule. The following steps comprise the current model of how replication occurs:

1. Prepriming (Primosome). The first step in the replication process is to separate the DNA strands to allow access to the DNA polymerase complex. The DnaB hexameric helicase is required for this process. A DnaC hexamer (loading factor) escorts and loads a DnaB helicase to the origin of replication (or, in combination with preprimosome proteins η , η' , η'' , ι , to sites where replication has arrested). The DnaB hexamer encircles one strand of DNA, and, using the energy derived from ATP hydrolysis, unwinds the DNA duplex to form a replication fork. Once the primase and DNA polymerase complex are on board, interactions between DnaB helicase and

TABLE 2-3. Genes and Proteins Involved in DNA Replication

Protein	Gene	Map Location	Molecular Weight	Function
	<i>oriC</i>	83.5	245 bp	Origin of replication
Protein i (X)	<i>dnaT</i>	99	19,000	Prepriming
Protein n (Z)	<i>priB</i>	95	11,300	Prepriming
Protein n' (Y)	<i>priA</i>	89	81,500	Prepriming (DNA-dependent ATPase)
Protein n''	<i>priC</i>	10	20,200	Prepriming
DnaA	<i>dnaA</i>	83.6	52,300	Initiation, binds <i>oriC</i> , DnaB loading
DnaB	<i>dnaB</i>	91.9	52,200	helicase (hexamer), prepriming priming, DNA-dependent rNTPase
DnaC	<i>dnaC</i>	99.1	27,800	Loading factor for DnaB
Pol III (α)	<i>dnaE</i>	4.4	129,700	DNA Pol III holoenzyme, elongation
Primase	<i>dnaG</i>	69.2	65,400	Priming, RNA primer synthesis, rifampin resistant
γ and τ subunits	<i>dnaX</i>	10.6	47,500;	Synthesis, part of the γ
			71,000	complex, promotes dimerization of Pol III
β (EFI)	<i>dnaN</i>	82.5	40,400	β -clamp, processivity
Helix destabilizing	<i>ssb</i>	92.1	19,000	Single-stranded binding protein
Helix unwinding	<i>rep</i>	85	76,800	Strand separation; not essential for chromosome replication
δ (EFIII)	<i>holA</i>	14.4	38,500	Part of the γ complex, clamp loading
δ'	<i>holB</i>	24.9	36,700	Part of the γ complex, clamp loading
ψ	<i>holD</i>	99.3	15,000	Part of the γ complex, clamp loading
χ	<i>holC</i>	99.6	16,500	Part of the γ complex, clamp loading
θ	<i>holE</i>	41.5	8,700	Pol III dimerization
ε subunit	<i>dnaQ</i> (<i>mutD</i>)	5.1	27,000	Proofreading, 3' to 5' exonuclease
DNA Pol I	<i>polA</i>	87.2	103,000	Gap filling
Ligase	<i>ligA</i>	54.5	73,400	Ligation of single-strand nicks in the phosphodiester backbone
DNA gyrase (subunit α)	<i>gyrA</i> (<i>nalA</i>)	50.3	97,000	Supertwisting
DNA gyrase (subunit β)	<i>gyrB</i>	83.5	89,800	Relaxation of supercoils
DNA Pol II	<i>polB</i>	1.4	120,000	Repair
RNA Pol, β subunit	<i>rpoB</i>	90	150,000	Initiation

components of the polymerases stimulate the unwinding rate 10-fold. The helicase moves into the fork 5' to 3' on the lagging strand (opposite direction to elongation on the lagging strand). Strand separation enables replication to proceed and requires helix destabilizing proteins (single-strand DNA-binding proteins) that prevent unwound DNA from reannealing and protect ssDNA from intracellular nucleases

2. Unwinding. This results in the introduction of positive supercoils in the dsDNA ahead of the fork. Since excessive supercoiling can slow fork movement, DNA gyrase removes these positive supercoils by introducing negative ones.

3. Priming. In addition to unwinding DNA, the DnaB protein is thought to “engineer” ssDNA into a hairpin structure that could serve as a signal for the primase (*dnaG* gene product) to synthesize a RNA primer (beginning with ATP) of about 10 nucleotides. This RNA primer is needed because, unlike RNA polymerases, no DNA polymerase can synthesize DNA de novo (i.e., without a preexisting 3'-OH priming end). The RNA polymerase (primase) activity of DnaG must be activated by interacting with DnaB helicase. Unlike DnaB and Pol III holoenzyme, DnaG does not travel as a stable component of the protein complex at the fork. It is recruited from solution for each priming event.

4. β -clamp loading. Once the RNA primer is formed, DNA polymerase III can bind to the 3'-hydroxyl terminus of the primer and begin to synthesize new DNA. However, this is not a straightforward process. Before the Pol III- α subunit (polymerase activity proper) can bind, a β -clamp protein is loaded onto the primed, single-stranded DNA. The β -clamp is a ring-shaped homodimer that encircles the DNA (Fig. 2-10). It is loaded onto DNA by the γ -complex clamp loader composed of five different polypeptides with the stoichiometry $\gamma_2\delta\delta'\chi\psi$. The β -clamp then binds the α subunit (polymerase), forming a “sliding clamp” that allows DNA polymerase to move along the template without easily being dislodged. This increases the processivity of DNA synthesis (i.e. increases the number of bases added before DNA polymerase leaves the template). On the leading strand, where DNA is synthesized in one continuous strand, loading of the clamp only needs to be done once on an undamaged DNA template. On the lagging strand, where DNA is synthesized in short 1000 bp fragments, β dimers must be loaded and unloaded many times. This is facilitated by the clamp loader being part of the replication machine, moving with the fork.

5. Completion of lagging strand. Once started, synthesis on the lagging strand will continue until the polymerase meets with the 5' terminus of a previously formed RNA portion of an Okazaki fragment. At this point, the polymerase will dissociate from the DNA.

6. Proofreading. Recalling the fidelity with which DNA is replicated in spite of the speed, leads to the question of how fidelity is maintained. Polymerases in general incorporate a relatively large number of incorrect bases while replicating. This, however, sets up a base pair mismatch situation that can be recognized as incorrect by the 3' $\rightarrow$ 5' exonuclease proofreading activity of the DNA polymerase holoenzyme. The component of DNA polymerase responsible for proofreading is DnaQ. This subunit allows polymerase to hesitate, excise the incorrect base, and then insert the correct base. However, even with this elegant proofreading system, some mistakes still occur. Chapter 3 covers how the cell deals with mistakes.

7. Replacing the primer. Once Pol III has replicated DNA up to the point where a preexisting RNA primer resides (on the lagging strand), the cell must remove that

RNA and replace it with DNA. RNase H cleaves the RNA primer and DNA Pol I uses its 5' → 3' polymerase activity to replace the cleaved RNA with DNA.

8. Repairing single-strand nicks on the lagging strand. When Pol I is finished, there remains a phosphodiester break(or nick) between the 3'-hydroxyl end of the last nucleotide synthesized by Pol I and the 5'-phosphoryl end of the adjacent DNA segment. DNA ligase will “seal” this nick by using NAD as a source of energy (Fig. 2-11).

Train Versus Factory Model of Replication. Two general models have been proposed for DNA replication. In one model, DNA polymerase moves along the DNA like a train on a track. In the second model, DNA polymerase is stationary (like a factory) in the cell and DNA is pulled through it. Support for the factory model came from visualizing that a fluorescently tagged polymerase in *Bacillus* remained localized at discrete locations at or near the midcell rather than being distributed randomly throughout the cell.

Initiation of DNA Replication

Replication of the bacterial chromosome proceeds bidirectionally from a fixed origin (*oriC*) located at 83.7 minutes on the standard *E. coli* linkage map (3881.5 kb) (Fig. 2-12). It takes about 40 minutes to completely replicate the *E. coli* genome. Yet, the generation time of *E. coli* can be as short as 20 minutes. To accomplish this feat, a single cell of *E. coli* will initiate a second and sometimes a third round of replication before the first round is complete. That way, after round one is finished and the cell divides, each daughter cell is already halfway through replicating its chromosomes.

In rapidly growing *E. coli* containing multiple origins of replication, all origins fire at precisely the same time in the cell cycle, and each origin only fires once per cell cycle. It is crucial to cell viability that initiation of chromosome replication is tightly controlled. This control is focused at the origin of replication (*oriC*), a 245 base pair sequence located at 83.68 minutes (3881.8 kb) on the circular *E. coli* map (Figs. 2-12 and 2-13). The region contains several 13-mer A + T-rich sequences that enable easy strand separation during initiation (the amount of energy required to separate A + T-rich DNA strands is considerably less than that required to separate G + C-rich DNA strands because A/T pairs are connected by two hydrogen bonds whereas G/C pairs are connected by three bonds). A number of potential RNA polymerase-binding sites have also been found in this region (see “RNA Synthesis” in this chapter). Transcription from some of these sites (e.g., *moiC*) helps separate the strands during assembly of the initiation complex.

Initiation of DNA synthesis is linked to the ratio of cell mass to the number of origins present in the cell. When a certain ratio is reached, the origins fire. As a result, slow-growing cells (one origin per cell) are smaller (less mass) than fast-growing cells (two or more origins per cell). The rule seems simple, but the molecular details of initiation are still being worked out today. Efforts to understand initiation started with the **replicon** theory proposed in 1963 which predicted that initiation could be controlled negatively and/or positively and that negative control might involve transient association of the DNA origin with the cell membrane. Amazingly, these predictions have proven true. A diagram of the origin is presented in Figure 2-13. The positive regulator of initiation is now known to be DnaA, 20 molecules of which must bind

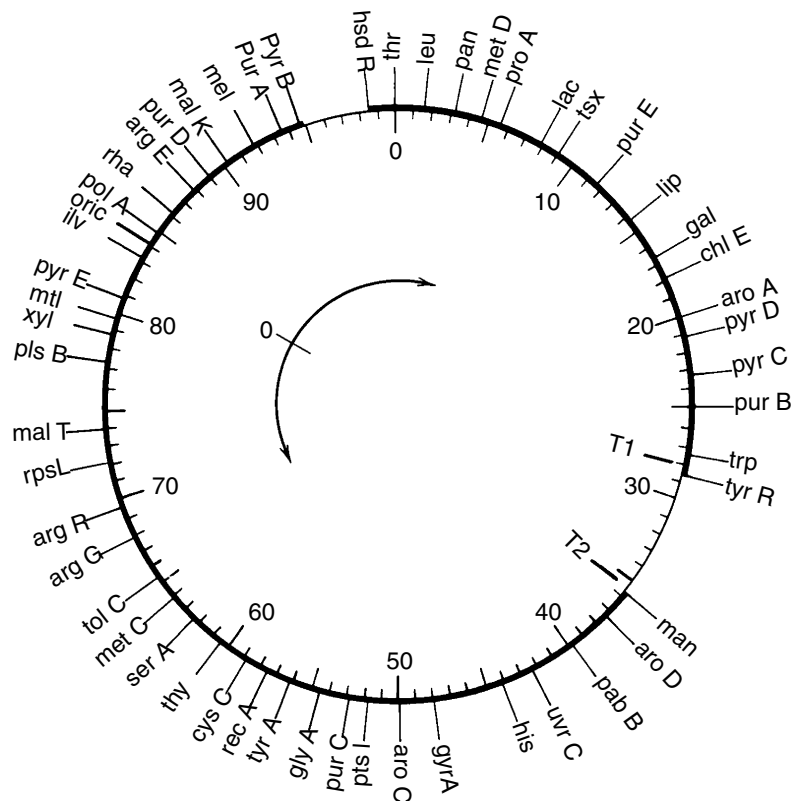


Fig. 2-12. Partial linkage and replication maps of *Escherichia coli*. Circular reference map of *E. coli* K-12. The large numbers refer to map positions in minutes, relative to the *thr* locus. From the complete linkage map, 52 loci were chosen on the basis of familiarity as longstanding landmarks of the *E. coli* K-12 genetic map.

to five repeated, 9 bp sequences called DnaA boxes located within the origin. DnaA binding will promote strand opening of the AT-rich 13-mers and facilitate loading of the DnaB helicase. DnaA protein levels remain relatively constant during the cell cycle, but the activity of DnaA and its accessibility to the origin are thought to be regulated (see below).

Preventing Premature Reinitiation. The negative regulator of initiation is a protein called SeqA that will sequester the *E. coli* chromosome origin to the membrane (actually the outer membrane). SeqA selectively binds to origins and inhibits reinitiation after replication has begun. This negative control mechanism hinges on when newly made origins become methylated at specific GATC sequences. The chromosome of *E. coli* contains many GATC palindromic sequences that are targets for an enzyme called deoxyadenosine methylase (Dam methylase). Discussed further in Chapter 3, Dam methylase adds a methyl group to the adenine in GATC sequences. However, immediately after a segment of chromosome is replicated, only the parental strand will contain GA^{Me}TC sequences. The newly synthesized strand remains unmethylated until it encounters Dam methylase. As a result, newly replicated DNA segments are

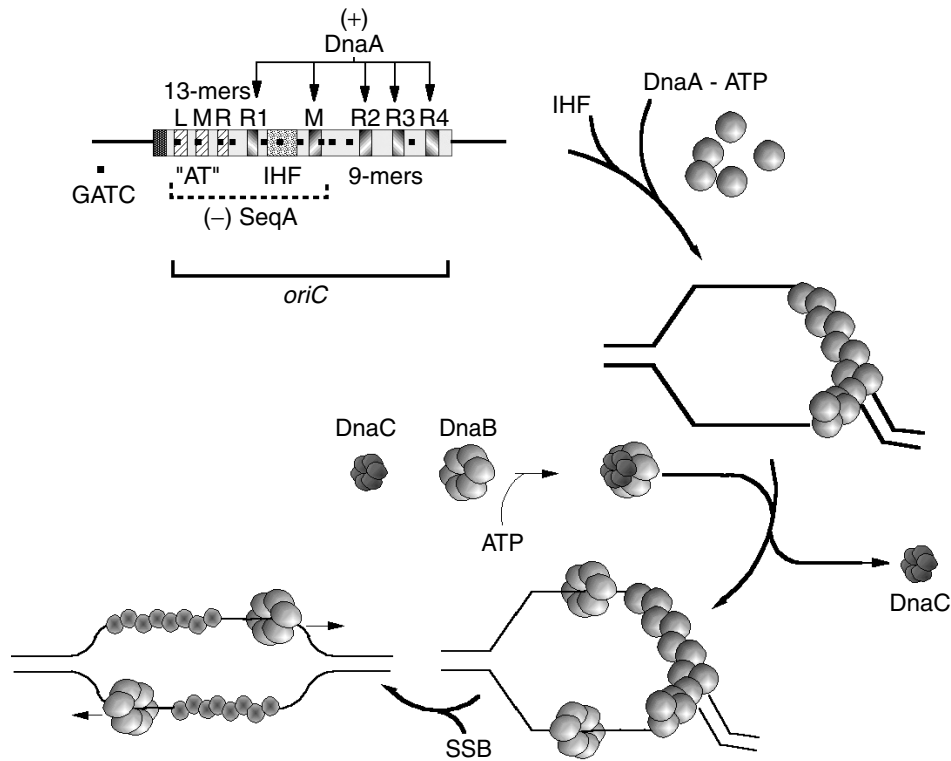


Fig. 2-13. Initiation of replication. The diagram depicts the area of the origin of replication. L, M, R, indicate left, middle, and right 13-mers; R1–4, 9-mer DnaA boxes. DnaA proteins bind to the DNA boxes to initiate replication. In collaboration with the histone-like protein HU and IHF, DnaA helps form an open complex. DnaC guides DnaB to the open complex, where it forms a sliding clamp. Its unwinding activity helps elongate the open complex and serves as a loading site for DNA polymerase and primosome components.

temporarily hemimethylated. It so happens that the SeqA protein has a high affinity for hemimethylated DNA. The *oriC* region contains 11 such GATC sites, so after the origin is replicated, SeqA will bind and sequester the hemimethylated daughter origins to the membrane, preventing more DnaA binding. As a result of its membrane association, methylation of *oriC* is delayed about 13 minutes after initiation.

Premature reinitiation is prevented not only by SeqA-dependent sequestration of the origin but through a timed reduction of DnaA activity. DnaA can bind either ATP or ADP but only the ATP form is active. As the replication machinery begins synthesizing DNA, the β subunit of DNA polymerase III stimulates ATP hydrolysis of DnaA–ATP, forming an inactive DnaA–ADP complex, thus decreasing the possibility of reinitiation. Regulation of initiation also depends on the titration of DnaA protein by a competing binding site called *data* located close to *oriC*. Once replicated, the two copies of *data* are predicted to bind DnaA molecules, thereby drastically reducing the concentration of free DnaA. Since *dat* does not contain any GATC sites, SeqA will not bind this area, leaving *dat* open to DnaA binding. In contrast, the *dnaA* promoter contains GATC sites that become hemimethylated after replication. As a consequence,

SeqA will bind and transiently repress *dnaA* transcription, further lowering the free DnaA concentration.

Initiation Hyperstructure. So what actually triggers initiation? The answer is complex but appears to involve increasing the level of active DnaA protein (DnaA–ATP) and the assembly of a so-called DnaA-dependent **initiation hyperstructure** at the cell membrane. In slow-growing cells, once replication stops, the β -clamp protein will no longer hydrolyze ATP bound to DnaA, and DnaA–ATP can build up. In rapidly growing cells, the increased copy number of *dnaA* genes relative to the number of actively replicating forks may lead to higher levels of DnaA and thus more DnaA–ATP, resulting in more frequent initiations. DnaA has been localized to the membrane, displaying an affinity for anionic phospholipids (e.g., cardiolipin) that are localized to cell poles and the septum. Interaction with these acidic phospholipids is also a requirement for DnaA activation. However, activation of DnaA alone is not sufficient for the initiation of DNA synthesis.

Immediately before initiation, the DNA-binding protein called IHF (integration host factor; see “ λ Phage” in Chapter 6) binds to a region in the origin and creates a bend in the DNA. The bend imposed by IHF brings DnaA box R1 in close proximity to the weaker DnaA-binding sites in the central part of *oriC*. This facilitates binding of multiple DnaA proteins to form the initiation complex. What might determine the timing of IHF binding is not understood.

Replication Hyperstructure. Distinctly different DnaA-dependent and SeqA-dependent giant hyperstructures at the origin have been proposed. The envisioned SeqA **replication hyperstructure** would bring both replication forks together and include DNA polymerases as well as enzymes responsible for making and delivering DNA precursors. This complex would confer efficiency to the replication process and assure bidirectional replication. SeqA may do this by recruiting specific genes (by binding to strategically located hemimethylated regions) to a localized membrane site. The origin is localized to the center of the cell just before replication, but after replication is complete the daughter origins move to the cell poles. SeqA protein, however, only moves halfway to the poles, meaning there must be a separation at some point. How the origin returns to midcell just prior to replication is not clear. The terminus follows the exact opposite pattern, starting out at the pole before initiation and migrating to midcell after initiation (see “Termination of DNA Replication and Chromosome Partitioning”).

The following list summarizes a proposed sequence of events leading to initiation:

1. Increased levels of DnaA–ATP. Origin moves to midcell.
2. Integration host factor binds to *oriC* and introduces bend.
3. DnaA–ATP binds to 9-mer DnaA boxes.
4. Open complex formation at 13-mer region, aided by transcription from *moiC*.
5. Formation of prepriming complex with DnaB helicase and DnaC loading protein.
6. ATP hydrolysis releases DnaC.
7. Helicase unwinds DNA and allows DnaG primase to enter and synthesize RNA primers.
8. DNA replication by DNA Pol III leads to hemimethylation of origin. β -clamp protein hydrolyses DnaA–ATP.

9. Sequestering of hemimethylated origin to membrane via SeqA.
10. Migration of origin toward poles.
11. Detachment of origin from SeqA membrane hyperstructure; methylation.
12. Slow increase in DnaA-ATP.

Termination of DNA Replication and Chromosome Partitioning

Once initiated, replication proceeds to the terminus (*terA*, *B*, *C*) between 30 and 35 minutes on the linkage map. The time required to complete one round of replication is about 40 minutes. There are two groups of *ter* sites arranged with opposite polarity that serve to inhibit the two converging replication forks. The T1 terminator (*terA*, *D*, *E*; at 28 minutes) permits clockwise-traveling replication forks to enter the terminus region but inhibits counterclockwise-traveling forks that might exit the region. T2 (*terF*, *B*, *C*; at 35 minutes) does the opposite, allowing counterclockwise forks to enter the region but inhibiting clockwise forks from exiting (Fig. 2-12). Another gene essential for termination encodes the **terminus utilization substance**, Tus. A monomer of Tus protein will bind to each *ter* site. The Tus-DNA complex can then act as a “conrahelicase,” inhibiting the DnaB helicase activity of an approaching replication fork. Since DnaB helicase drives replication fork movement, inhibiting the helicase will block further movement.

After termination, the two daughter chromosomes will form a linked concatemer due to the topological constraints inherent when separating the strands of a double-stranded helical circle. These linked molecules must be resolved through recombination if the cell is to divide (see Fig. 2-14). In *E. coli*, two tyrosine family site-specific recombinases, XerC and XerD, bind cooperatively at the 33 bp chromosomal site, *dif*, located at the replication terminus. XerC first recognizes *dif* and initiates a recombination (crossover) event between the two strands. The result is a structure, called a Holliday junction, which must be resolved (see Fig. 2-14 and “Recombination” in Chapter 3). A membrane protein, FtsK, located at the septum of a dividing cell, is hypothesized to modify the Holiday structure into a suitable substrate for the second recombinase, XerD. XerD acts on the Holiday structure, leading to completed recombination products and resolution of the concatemer.

Partitioning of the two chromosomes into separate daughter cells is essential for successful cell division. The process is not fully understood, but it appears that the SeqA protein pool, localized to the midcell, binds to newly hemimethylated DNA, then separates into two foci that migrate in opposite directions toward the poles. Type IV topoisomerase (ParC and ParE) is believed to decondense the nucleoid at the center of the cell by decreasing the supercoiling of the chromosome. The chromosome passes

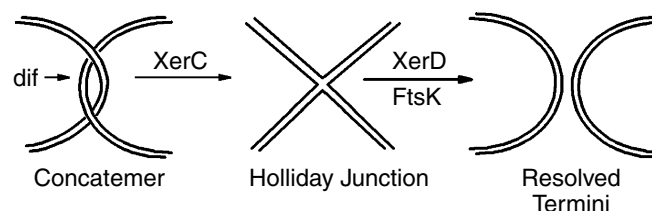


Fig. 2-14. Resolution of chromosome concatemers.

through a stationary replication machine and then is drawn to the cell poles by the action of the MukBKF complex, which is thought to supercoil DNA into a more condensed form. The FtsK protein may contribute to the segregation process as a DNA pump (see further discussion in Chapter 17).

RNA SYNTHESIS: TRANSCRIPTION

RNA Synthesis

Transcription is the process by which the information contained within genes is converted into RNA. The process is the same regardless of whether the gene encodes messenger RNA (mRNA), transfer RNA (tRNA), or ribosomal RNA (rRNA). The transcription of DNA to RNA requires a DNA-dependent RNA polymerase and proceeds in a manner similar to DNA synthesis but uses ribonucleic acid triphosphates (rNTP) rather than deoxyribonucleic acid triphosphates (dNTP).

RNA polymerase (RNAP) is an extremely complex machine that senses signals coming from numerous regulatory proteins as well as signals encoded in the DNA sequence. RNAP consists of four polypeptides: α , β , β' , and σ . Core polymerase consists of two α subunits plus one β and one β' subunit. A fifth subunit called omega is now believed to be involved in assembling the RNA polymerase complex. The core enzyme can bind to DNA at random sites and will synthesize random lengths of RNA. Holoenzyme, consisting of core enzyme plus a σ subunit, binds to DNA at specific sites called promoters and transcribes specific lengths of RNA. σ^{70} (70 kDa molecular weight) is considered the “housekeeping” σ , but there are several specialty σ factors that direct RNAP to specific promoters. Thus, the σ subunit plays an important role in promoter recognition by RNA polymerase.

The β subunit carries the catalytic site of RNA synthesis as well as the binding sites for substrates and products. The β' subunit plays a role in DNA template binding while the two α subunits assemble the two larger subunits into core enzymes ($\alpha_2\beta\beta'$). Rifampicin, a commonly used antibiotic, inhibits transcription by interfering with the β subunit of prokaryotic RNA polymerase. Figure 2-15 presents a view of RNA polymerase structure and important binding regions for each subunit. Electron microscopy and X-ray crystallography reveal the RNAP complex is shaped like a crab with a groove or trough that fits double-stranded helical DNA. Figure 2-16 represents an approximate density map of RNAP transcribing DNA.

Within a transcribing region, the coding strand refers to the DNA strand that is not transcribed but consists of the same sequence as the mRNA product (with a T substituting for U). The antisense or template strand is the strand that is transcribed (read) by RNA polymerase. Insofar as RNA polymerase binds to a promoter region, then progressively moves along the template strand from the region encoding the N terminus to the carboxy terminus, the terms *upstream* and *downstream* are used to describe regions relative to the direction of RNA polymerase movement. Thus, the promoter is upstream from the structural gene. It should be noted that RNA polymerase moves along the DNA template strand in the $3' \rightarrow 5'$ direction while synthesizing RNA $5' \rightarrow 3'$ (Fig. 2-17).

As with DNA replication, transcription can be described in three main steps: initiation, elongation, and termination. Initiation involves the binding of polymerase to the promoter with the formation of a stable RNA polymerase–DNA initiation

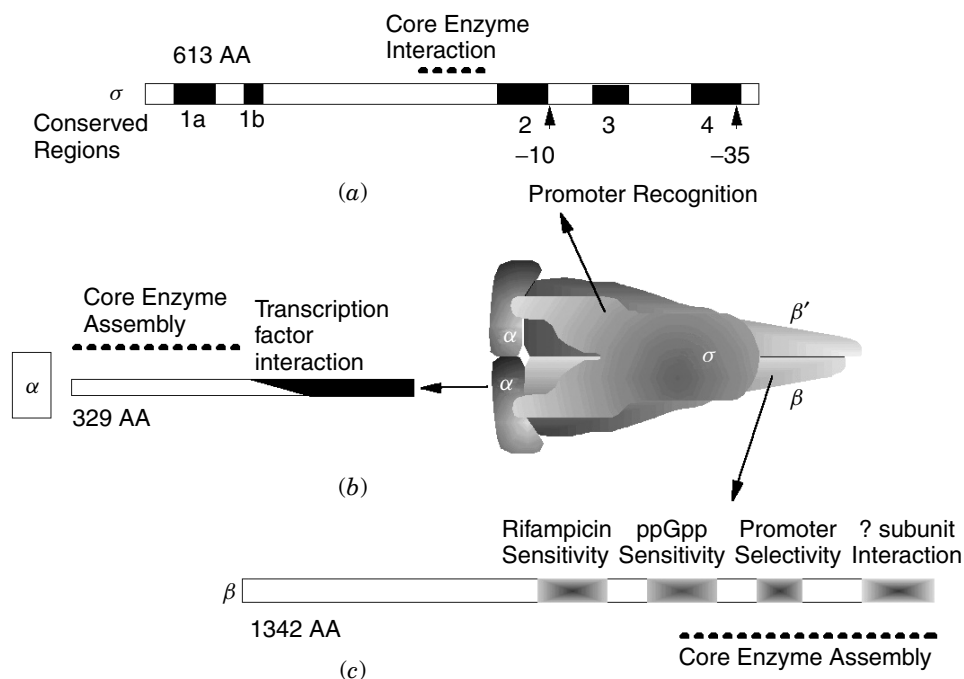
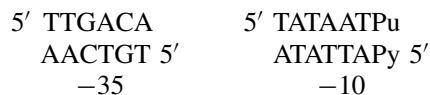


Fig. 2-15. Functional map of RNA polymerase subunits. The molecular model of RNA polymerase holoenzyme is shown in the middle. Areas of each subunit involved with various interactions between subunits, other protein factors, effector molecules, and promoter regions are indicated.

complex and the catalysis of the first $3' \rightarrow 5'$ internucleotide bond. Elongation is the translocation of RNA polymerase along the DNA template with the concomitant elongation of the nascent RNA chain. Termination, obviously, involves the dissociation of the complex. The overall process of transcription is illustrated in Figure 2-17.

Initiation. The site referred to as the promoter includes two DNA sequences that are conserved in most promoters. In general, highly conserved sequences are referred to as consensus sequences. The promoter consensus sequences are centered at -10 and -35 base pairs from the transcriptional start point (designated $+1$) and have been implicated in normal promoter function. The consensus -10 and -35 sequences for the σ^{70} housekeeping σ factor are



The -10 sequence is referred to as Pribnow's box while the -35 region is called the recognition site. It should be noted that different σ factors recognize different -10 and -35 sequences. One question that must be addressed, however, is how RNA polymerase can recognize these sequences when it would appear that direct interaction between RNA polymerase and DNA is precluded by the double-stranded nature of

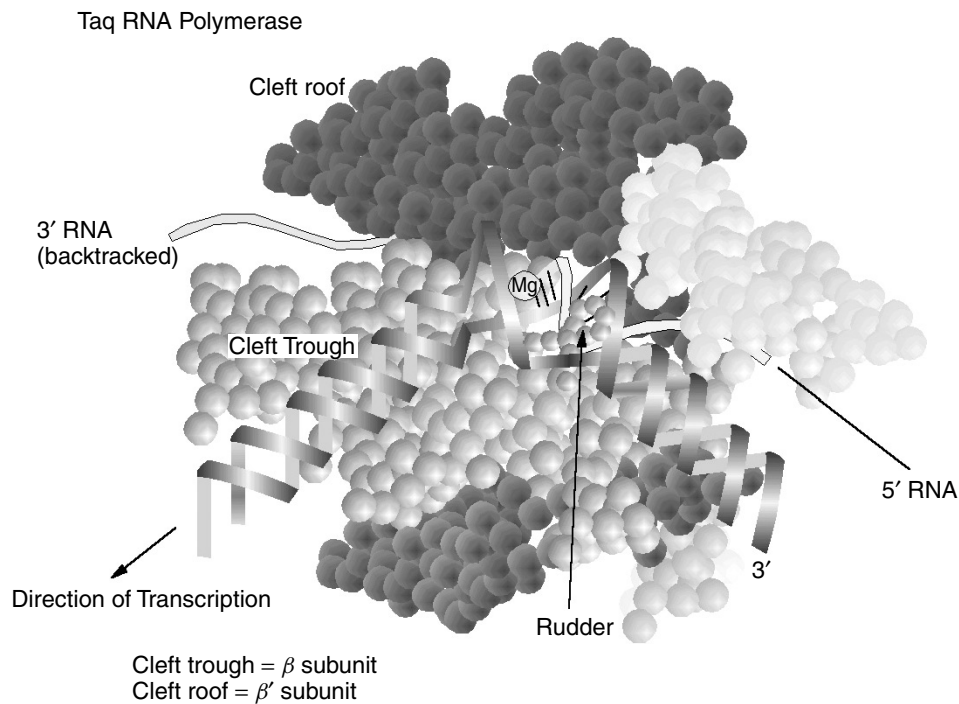


Fig. 2-16. Electron density map of Taq RNA polymerase. The cleft in which DNA fits is formed between the β subunit (cleft roof) and the β' subunit (cleft trough). Note the Mg^{2+} at the active site and the rudder that helps separate the strands. The position of a backtracked RNA is also shown.

DNA. Association is possible for any protein–nucleic acid interaction through base-specific groups that can be recognized in the major or minor grooves. RNA polymerase can interact with groups in the major groove and recognize the proper sequence upstream (–35 region) from the Pribnow box, then form a stable complex (closed complex) by moving laterally to the –10 region. σ factors are involved in recognizing both regions.

DNA Supercoiling and Transcription. The superhelical nature of the chromosome also plays a role in promoter function. In general, a negatively supercoiled chromosome is a better transcriptional template than a relaxed chromosome. Presumably, the torsional stress imposed by supercoiling makes certain areas of DNA easier to separate by RNA polymerase (i.e., lowers the melting temperature). However, it appears that supercoiling affects the expression of some genes more than others. There is some thought that the cell, through the use of topoisomerases Top I and DNA gyrase, can affect supercoiling in localized areas of the chromosome, facilitating the transcription of some genes while impeding the transcription of others.

Alternate Sigma Factors. Obviously, σ -70 plays an important role in normal transcription initiation. Alternate σ factors (e.g., σ -32, 32 kDa molecular weight) can change the promoter recognition specificity of core enzyme. Thus, σ -32 factor

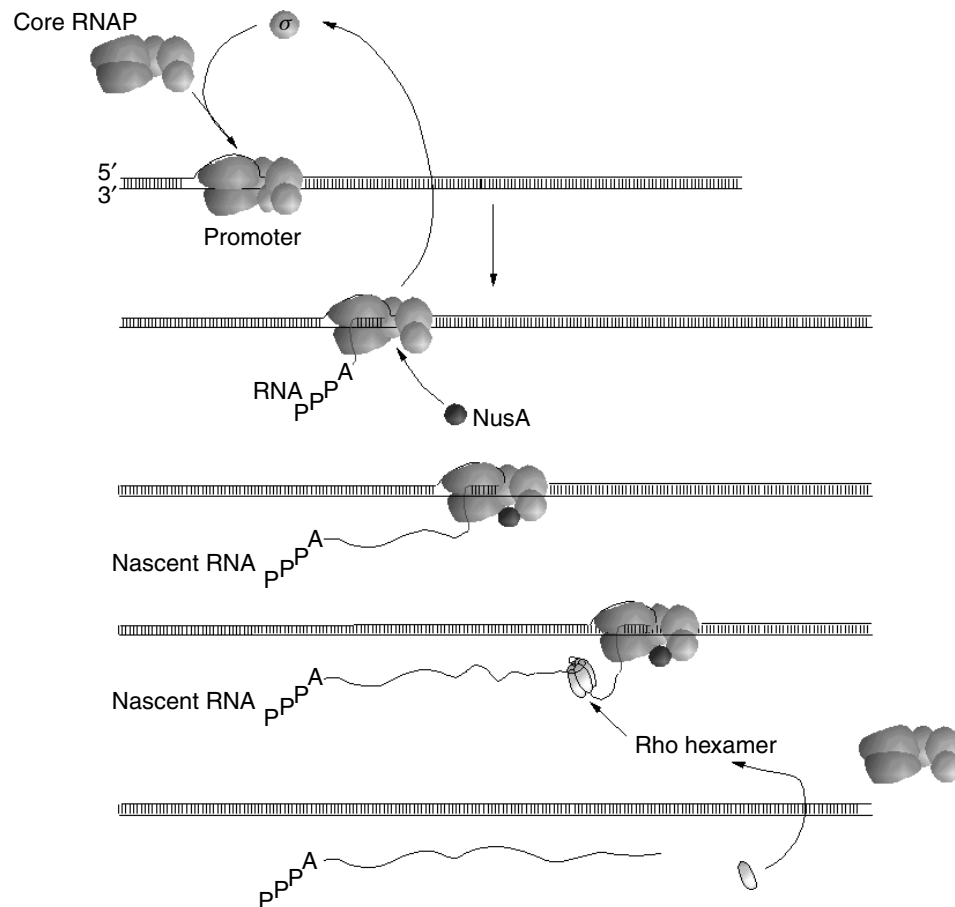


Fig. 2-17. Overview of the transcription process. Sigma factor (σ) directs core polymerase to the promoter region of a gene. After transcription of several nucleotides, σ dissociates and is replaced by NusA. When transcription is complete, RNA polymerase encounters transcriptional stop signals in the mRNA. One type of signal recruits ρ -factor protein to the RNA. The mRNA wraps around a ρ hexamer, which enables ρ to interact with RNA polymerase and trigger disassociation.

is utilized when cells undergo heat shock and is believed to be responsible for the production of proteins required to survive that stress (see Chapter 18). Other alternate σ factors are listed in Table 2-4 and in the discussion on sporulation in Chapter 19. Among the various sigmas there are highly conserved regions designated 1 to 4 as shown in Figure 2-15. The C-terminal part of region 2 (2.3, 2.4, and 2.5) is primarily involved in the use of the -10 region. Important strategies to control gene expression include (1) anti- σ factor proteins to inhibit the activity of a specific σ until needed and (2) regulated proteolysis of the σ factor. These mechanisms are discussed later.

DNA UP Element and Protein-Protein Interactions with RNA Polymerase. Promoter recognition by bacterial RNAP also involves interactions between the C-terminal domain of the RNAP α subunit and DNA sequences upstream of the core promoter.

TABLE 2-4. RNA Polymerase Subunits and Other Transcription Factors of *Escherichia coli*

Subunit	Structural Gene	Map Position	Molecular Weight ^a	Function
α	<i>rpoA</i>	74.1	36,000	Initiation
β	<i>rpoB</i>	90.1	150,000 ^b	Initiation, elongation, termination
β'	<i>rpoC</i>	90.1	160,000	Initiation
$\sigma^{-70}(\text{F}^D)$	<i>rpoD</i>	69.2	83,000	Initiation
$\sigma^{-32}(\text{F}^H)$	<i>rpoH</i>	77.5	32,000	Initiation, heat-shock response
$\sigma^{-28}(\text{F}^F)$	<i>fliA</i>	43.1	28,000	Flagellar genes, chemotaxis
$\sigma^{-24}(\text{F}^E)$	<i>rpoE</i>	58.4	24,000	Extreme heat shock
$\sigma^{-42}(\text{F}^S)$	<i>rpoS</i>	67.1	42,000	Stationary phase
$\sigma^{-54}(\text{F}^N)$	<i>rpoN</i>	72.0	54,000	Nitrogen
Omega	<i>rpoZ</i>	82.3	10,000	RNAP assembly
CRP	<i>crp</i>	75.1	22,500	Initiation enhancement
Rho (ρ)	<i>rho</i>	85.4	50,000	Termination
NusA	<i>nusA</i>	71.4	69,000	Pausing, transcription termination
NusB	<i>nusB</i>	9.4	14,500	Pausing, transcription termination
NusG	<i>nusG</i>	90.0	20,000	Pausing, transcription termination
GreA	<i>greA</i>	71.7	17,000	3' $\rightarrow$ 5' RNA hydrolysis proofreading
GreB	<i>greB</i>	76.2	8,384	3' $\rightarrow$ 5' RNA hydrolysis, proofreading

^aDaltons.^bZn²⁺ metalloenzyme.

Thus, bacterial promoters consist not only of σ -binding sites but α -binding sites as well. The α -binding sites have been named **UP elements** (for upstream). As noted in Figure 2-15, the N-terminal portion of α associates with the other subunits of RNAP. The C-terminal domain (CTD) is linked to the N-terminal domain by a flexible linker. As such, the CTD can interact with other transcriptional activator proteins and with DNA UP sequences. Part of the role of some regulatory proteins is to increase curvature of DNA to allow CTD access to the UP elements.

Many regulatory factors bound to DNA also directly contact RNA polymerase at the promoter region. These proteins can be divided into two classes. Class I transcriptional activators bind upstream from the promoter (e.g., CRP, AraC, Fnr, OmpR). Class II transcription factors overlap the promoter region. Many of these regulators contact the C-terminal end of the α subunit (Fig. 2-15) and activate transcription by engineering interactions with UP elements. Many of these factors are discussed in Chapter 5.

Conversion from a Closed to an Open Complex. The closed promoter complex, in which RNA polymerase-bound DNA remains unmelted, must be converted to an open promoter complex. This involves the unwinding of about one helix turn from the middle of Pribnow's box to just beyond the initiation site (the transcription bubble). Two aromatic amino acid residues within region 2 of σ^{70} are critical to unwinding. Note in Figure 2-16 that there is also a "rudder" structure in β' that separates the strands once unwound. The open complex then allows tight binding of RNA polymerase with the subsequent initiation of RNA synthesis. The first triphosphate in an mRNA chain is usually a purine. Initiation ends after the first internucleotide bond is formed (5'pppPupN).

Elongation. After the formation of transcripts eight to nine nucleotides in length, σ factor is abruptly released from the complex, suggesting that RNA polymerase undergoes a conformational change that causes a decrease in the affinity of σ for the RNA polymerase–DNA–nascent RNA complex (ternary complex). The released σ factor can be reused by a free core polymerase for a new initiation (Fig. 2-17).

Once elongation begins, transcription proceeds at a rate of between 30 and 60 nucleotides per second (at 37 °C). The elongation reaction as a whole includes the following steps: (1) nucleotide triphosphate binding; (2) bond formation between the bound nucleotide and the 3'-OH of the nascent RNA chain; (3) pyrophosphate release; and (4) translocation of polymerase along the DNA template. An essential feature of the β' catalytic site is a bound Mg^{++} ion (Fig. 2-16). Movement of RNA polymerase, by necessity, involves the melting of the DNA template ahead of the transcription bubble (ca. 17 nucleotides) as well as reformation of the template behind the bubble. Part of the β subunit forms a “flap” thought to hold the DNA.

The rate of elongation along a template is not uniform. Regions in a template where elongation rates are very slow are called **pausing sites**. Pausing sequences in general are GC-rich regions possessing dyad symmetry that form hairpin loops in the RNA (an RNA–RNA stem-loop structure). The base of a typical pause site stem is located approximately 11 base pairs from the active center of RNAP (an Mg^{++} -binding site) and is thought to cause the β subunit flap to open. The RNA–DNA hybrid would then move away slightly from the active site, slowing elongation.

The **NusA** and **NusG** proteins are believed to associate with core polymerase after σ disassociates and modulate RNAP elongation rates. The Nus interaction with the ternary complex is nonprogressive, involving rapid association and disassociation. These proteins enhance pausing at some sites and may be involved with Rho-dependent transcription termination (see below). Recent evidence suggests that NusA stabilizes the pause hairpin–flap interaction, thereby lengthening the pause period.

The transcription process is not without its flaws. Sometimes RNAP inappropriately backtracks along a nascent message, leaving the 3' end of the message protruding. This backtracking obviously disengages the 3' end of RNA from the catalytic site. In this event, transcription is arrested. Two proteins, **GreA** and **GreB**, associated with RNAP, trigger an endonucleolytic activity at the catalytic center that cleaves and removes the 3' end of the nascent message. This releases RNA polymerase, allowing transcription to proceed.

It should be noted at this point in the discussion that the RNA transcript is normally longer than the structural gene. Most messages contain a promoter-proximal region of mRNA called the **leader sequence** (or untranslated region) that exists between the promoter and the translation start codon. The leader sequence carries information for ribosome binding (see “Shine-Dalgarno Sequence” in Initiation of Polypeptide Synthesis) and in some cases is important in determining whether RNA polymerase will proceed into the structural gene(s) proper (see “Attenuation Controls” in Chapter 5).

Termination. Transcriptional termination includes the following events: (1) cessation of elongation, (2) release of the transcript from the ternary complex, and (3) dissociation of polymerase from the template. There are two classes of termination: ρ -independent and ρ -dependent transcription termination. The ρ -independent termination signal includes a GC-rich region with dyad symmetry, allowing formation of an RNA stem-loop structure about 20 bases upstream of the 3'-OH terminus and a stretch

of four to eight consecutive uridine residues. The transcript usually ends within or just distal to the uridine string. The RNA stem-loop structure causes RNA polymerase to pause and disrupts the 5' portion of the RNA–DNA hybrid helix. The remaining 3' portion of the RNA–DNA hybrid molecule includes the oligo(rU) sequence. The relative instability of rU–dA base pairs causes the 3' end of the hybrid helix to melt, releasing the transcript.

As the term implies, termination at ρ -dependent terminators requires the ρ -gene product. ρ -factor termination only occurs at strong pause sites apparently located at specific distances from a promoter region—that is, a strong pause site located closer to the promoter will not serve as a termination site. ρ -utilization (*rut*) sequences on RNA are GC rich with no secondary structure. ρ will assemble as a hexamer at *rut* sites (Fig. 2-17) after which the single-stranded mRNA is thought to wind around the outside of ρ -factor, facilitated by ρ -dependent ATP hydrolysis (Fig. 2-17). This process serves to bring ρ in contact with RNA polymerase. The extended dwell time of RNA polymerase elongation complexes at pause sites allows the ρ protein translocating along the nascent RNA to catch up with the transcription complex. The association of ρ to core enzyme may occur via NusG protein. Subsequently, activation of a ρ RNA–DNA helicase activity causes the mRNA chain and core RNA polymerase to disassociate from the DNA. Free core RNA polymerase can then interact with a σ factor and be used to initiate a new round of transcription.

RNA Turnover

Cellular RNA, thus transcribed, can be classed into two major groups relative to their decay rates in vivo: stable and unstable RNA. Stable RNA consists primarily of ribosomal RNA (rRNA) and transfer RNA (tRNA) while unstable RNA is mRNA. In *E. coli*, 70 to 80% of all RNA is rRNA, 15 to 25% is tRNA, and 3 to 5% is mRNA. Factors that contribute to stability include the association of rRNA in ribonucleoprotein complexes (ribosomes) and the extensive secondary structure exhibited by both rRNA and tRNA. Both of these structural considerations serve to protect stable RNA at the 5' terminus from ribonucleases.

The average mRNA (about 1200 nucleotides) has a **half-life** (time required to reduce the mRNA population by one-half) of approximately 40 seconds at 37°C. This number represents an average as each mRNA species has a unique degradation rate. Degradation occurs overall in the 5' → 3' direction; however, all exoribonucleases present in the cell only degrade in the 3' → 5' direction. One theory that attempts to resolve this paradox is that an initial, random endonucleolytic event occurs in the mRNA near the 5' end that would allow for exonucleolytic degradation of the 5' end. Since no new ribosomes can bind to mRNA that has lost its 5' end, the remaining message is progressively exposed as the already bound ribosomes move toward the 3' end. Thus, additional endonucleolytic events followed by 3' → 5' exonucleolytic degradation would be possible. Most RNA degradation appears to occur via a multicomponent **degradosome complex** involving polynucleotide phosphorylase (*pnp*), RNase E (*rnaE*), an ATP-dependent RNA helicase (RhlB), a DnaK chaperone (see Chapter 18, Heat Shock response), and the enzyme enolase.

Poly A Tails. A feature originally thought to be unique to eukaryotic cells is the addition of poly(A) tails to the 3' ends of many of their mRNAs. It is now evident

that poly(A) tails also occur in bacterial mRNA. At least two poly(A) polymerases have been identified in *E. coli*: Pap I (*pcnB*) accounts for 80% of poly(A) and polynucleotide phosphorylase (Pnp) accounts for the remainder. It has been proposed that polyadenylation provides the basis for the binding of the degradosome to mRNAs. Thus, Pnp serves not only as a 3' exonuclease but will add poly(A) tails to messages facilitating their degradation.

Message stability can be enhanced by the presence of stem-loop structures present at the end or beginning of the mRNA. Stem loops at the 3' end of the message will inhibit 3'–5' exonuclease processivity while the 5' loops may interfere with endonuclease association. However, Pnp-dependent polyadenylation of mRNAs containing 3' stem-loop structures is suggested to increase the susceptibility of those messages to degradation.

RNA Processing

All stable RNA species and a few mRNAs of *E. coli* must be processed prior to their use. For example, each of the seven rRNA transcription units is transcribed into a single message as follows:

5' leader–16S rRNA–spacer–23S rRNA–5S rRNA–trailer-3'

The spacer always contains some tRNA gene. Obviously, this long multigene (polycistronic) precursor RNA molecule must be processed to form mature 16S rRNA, 23S rRNA, and 5S rRNA species. Another example of processing involves the tRNA genes, which are frequently clustered and cotranscribed in multimers of up to seven identical or different tRNAs. These multimers must be processed to individual tRNAs. Along with the required cutting and trimming, the initial transcripts of rRNA and tRNA lack the modified nucleosides of the mature species. Thus, processing of stable RNAs (i.e., tRNA and rRNA) includes the modification of bases in the initial transcript.

There are four basic types of processing. The first involves the precise separation of polycistronic transcripts into monocistronic precursor tRNAs. Second, there must be a mechanism by which the mature 5' and 3' termini are recognized followed by the removal of extraneous nucleotides. The third type of processing involves the addition of terminal residues to RNAs lacking them (e.g., the 3' CCA end of some bacteriophage tRNAs such as T4 phage). Finally, the appropriate modification of base or ribose moieties of nucleosides in the RNA chain must be accomplished. Obviously, not every RNA molecule is subject to all four of these processes. The following constitutes a brief description of several enzymes involved with RNA processing in *E. coli*:

RNase P, a Ribozyme. This endonucleolytic enzyme is required for the maturation of tRNA (Fig. 2-18). RNase P removes a 41-base-long fragment from the 5'-side of a wide variety of tRNA precursor molecules. The enzyme appears to recognize tRNA secondary and tertiary configurations rather than a specific nucleotide sequence. There is an absolute requirement for the presence of the CCA-terminal sequence within the precursor (see below). The subunit structure of this enzyme is unusual in that there is a polypeptide component (C5) encoded by the *rnpA* locus and an RNA component (M1/M2) encoded by *rnpB*. The RNA component

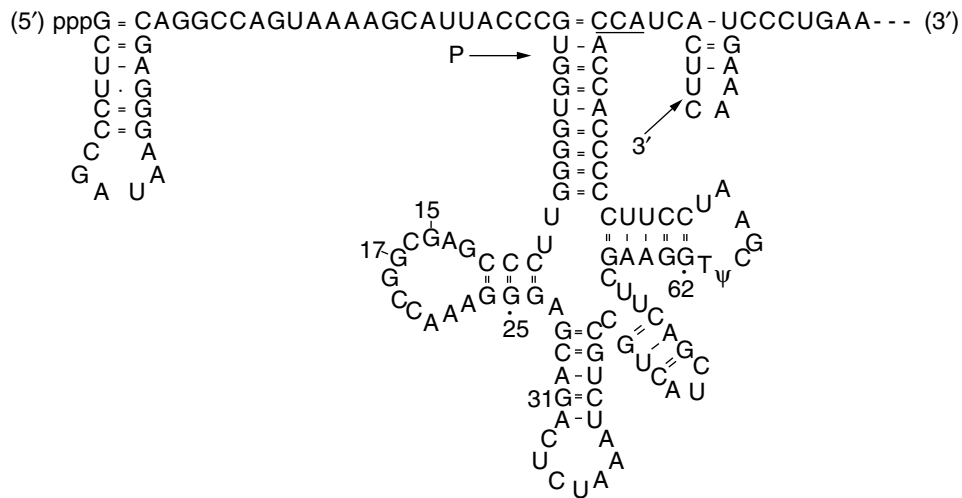


Fig. 2-18. tRNA precursor structure. The sequence shown is that of the tRNA^{tyr}₁ transcript. Transcriptions in vivo proceed through a second tRNA^{tyr}₁ gene. Sites of in vivo endonuclease cleavage are shown on the 5' side of the mature tRNA (RNaseP) and near the 3' end. The mature 3' terminus is underlined. Numbered nucleotide positions are those at which base changes result in reduced RNase P cleavage in vivo.

is integral to the mechanism by which RNase P cleaves tRNA precursor. It is, in fact, an example of a catalytic RNA. Hence, RNaseP is described as a ribozyme.

RNase II. Encoded by *rnB*, it is one of the major 3' → 5' exonucleases in *E. coli*.

RNase III. Encoded by the *rnC* locus, this enzyme contains two identical polypeptide units of 25,000 molecular weight with no RNA component. RNase III cleaves dsRNA by making closely spaced ss breaks (ca. every 15 bases). Other than recognizing perfect double-stranded RNA stems, there does not appear to be a unique recognition sequence for this enzyme.

RNase D. Maturation of tRNA precursors requires the removal of extra nucleotides distal to the CCA sequence located within the precursor tRNA destined to become the mature 3'-OH end of the tRNA molecule. The CCA-OH sequence is required for amino acid acceptor activity on all tRNAs. RNase D (*rnd*) is a monomeric protein with a molecular weight of 38,000. It is a nonprocessive 3' exonuclease that releases mononucleotide 5' phosphates from the RNA substrate.

RNase E. The *rne* locus is the structural gene for RNase E (molecular weight 70,000). The enzyme cleaves p5s precursor rRNA from larger transcripts. The enzyme first cleaves between the ss region of 5s precursor and the double-stranded region. The second cleavage occurs within the double-stranded area (Fig. 2-19). In addition to ribosomal RNA processing, RNase E is part of the RNA degradosome noted above.

RNase H. Encoded by *rnH*, this enzyme degrades the RNA strand of a DNA-RNA hybrid molecule.

RNase R. A major 3' → 5' exoribonuclease, (*rnR* or *vacB*), mutants defective in both *rnR* and *pnp* are inviable.

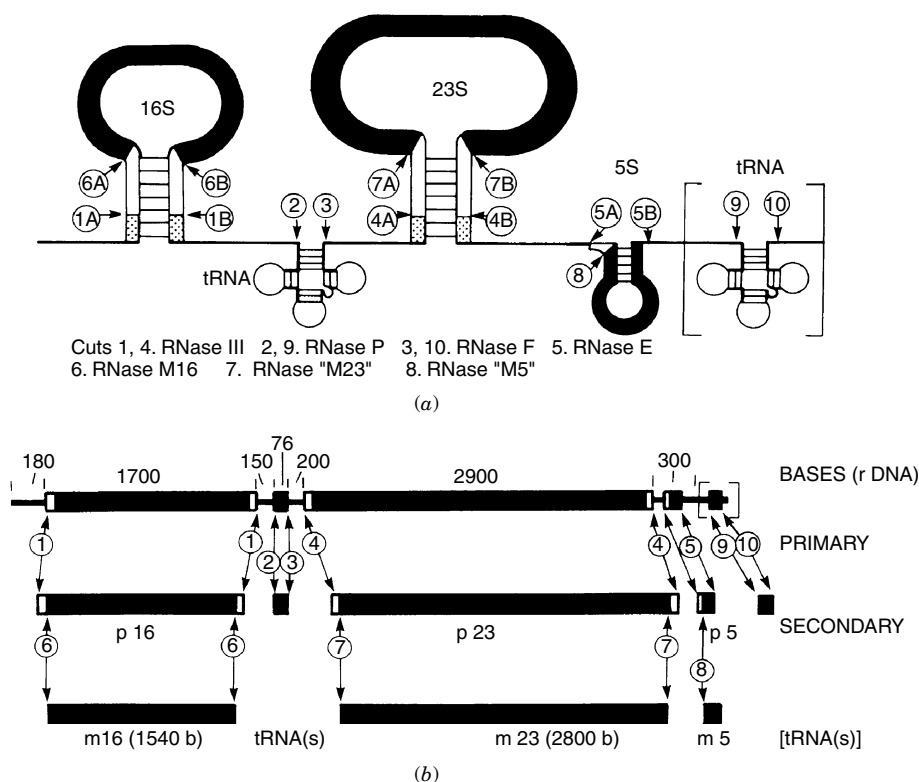


Fig. 2-19. Structure and processing of rRNA transcripts. (a) Structure and cleavage sites of the rRNA primary transcript (not to scale). Distal (trailer) tRNAs are bracketed because not all rDNAs contain them. Transcripts may contain one or two spacer tRNAs, and, one or two trailer tRNAs or none. Arrows indicate endonucleolytic cleavage sites. Each cutting event is given a separate number, referring to the enzyme involved; A and B indicate that two (or more) separate cuts may be required. Thick, solid segments represent mature rRNA sequences; thick, open segments represent precursor-specific sequences removed during secondary processing steps; stippled segments are sequences found only in p16b and p23b of RNase III⁻ cells; and thin lines (except for tRNAs) represent nonconserved sequences discarded during primary processing. Enzymes are discussed in the text. (b) Processing in wild-type strains. The first line shows the transcriptional map of a representative rDNA unit, drawn approximately to scale. Distances in bases are between vertical bars above the map. The primary and secondary cuts, numbered as in (a), are shown above the products they generate. Open and solid segments are as in (a).

tRNA nucleotidyltransferase. This enzyme can repair prokaryotic tRNAs in which the CCA sequence is missing. The product of the *cca* locus (67 minutes), nucleotidyltransferase, will sequentially add a 3'-CCA-terminal sequence to these tRNAs. Although it is probably not essential, mutants with a reduced level of this enzyme grow at a rate slower than normal.

Modifying enzymes. These enzymes chemically modify nucleotides present within tRNA and rRNA precursor transcripts and include methylases, pseudouridyllating enzymes, thiolases, and others. There are several examples that illustrate the importance of these modified bases. For example, mutants resistant to the

antibiotic kasugamycin have lost an enzyme that specifically methylates two adenine residues to form dimethyl adenine in the sequence A-A-C-C-U-G near the 3' end of 16S rRNA. Lack of this modification diminishes the affinity of kasugamycin for 30S ribosomal subunits.

In addition, there are several modifications in tRNA that are important. Modifications in the third base of the anticodon loop can contribute to the “wobble” of anticodon–codon recognition (see “Transfer RNA”).

Intervening sequences. Some organisms, such as *Salmonella enterica*, do not contain intact 23S rRNA in their ribosomes. The 23S rRNA locus of *S. enterica* contains an extra 90 base pair sequence that is not present in the *E. coli* homolog. This sequence, called an intervening sequence (IVS), is excised from the original transcript much as an intron is excised from within the mRNA of most eukaryotic genes. However, unlike intron processing, the remaining pieces of the *S. enterica* 23S rRNA are not spliced together. Since the rRNA does not encode a protein, there is no need to splice the fragments. The pieces function normally in all aspects of ribosomal assembly and protein synthesis. The rRNA superstructure is unaffected because the fragments remain associated through secondary and tertiary contacts. RNase III is the enzyme responsible for processing the IVS as proven by observing normal processing of *S. enterica* 23S rRNA in *rnaE*⁺ but not *rnaE*[−] mutants of *E. coli*. The origins of IVSs are unknown. They may be the “footprints” of transposable elements inserted and then excised from the 23S rRNA genes (see Chapter 3). A summary of RNA processing in *E. coli* is shown in Figure 2-20.

PROTEIN SYNTHESIS: TRANSLATION

During translation, the information coded in mRNA is specifically read and used to form a polypeptide molecule with a specific function. A given nucleotide sequence in mRNA codes for a given amino acid. Since there are 20 naturally occurring amino acids, there must be at least three nucleotides in the code for each amino acid (triplet code). Since there are only four bases, a doublet code would account for only 4² or 16 amino acids. With a triplet code, 4³ or 64 combinations are possible. Since there are only 20 amino acids. There is usually more than one triplet codon that can code for a given amino acid (see Table 2-5). For this reason, the code is termed **degenerate**. On the other hand, a given triplet codon cannot code for any other amino acid. For example, although the triplets UUU or UUC can both code for the amino acid phenylalanine, they cannot code for any other amino acid, indicating that the code is **nonoverlapping**. Of considerable significance is the fact that three codons, UAG, UAA, and UGA do not usually code for any amino acid, and tRNA molecules with the corresponding anticodons are uncommon. [There are exceptions in mitochondria and selenocystine-containing proteins (see below).] These codons (**nonsense** codons) serve as termination signals, stopping the translation process. The code is commaless; that is, there is no punctuation. If a single base is deleted or if another base is added in the sequence, then the entire sequence of triplets will be altered from that point on (**frameshift**).

Although the code is considered to be universal, there are some important differences in detail that are considered later. At this point it is sufficient to note just a few specific

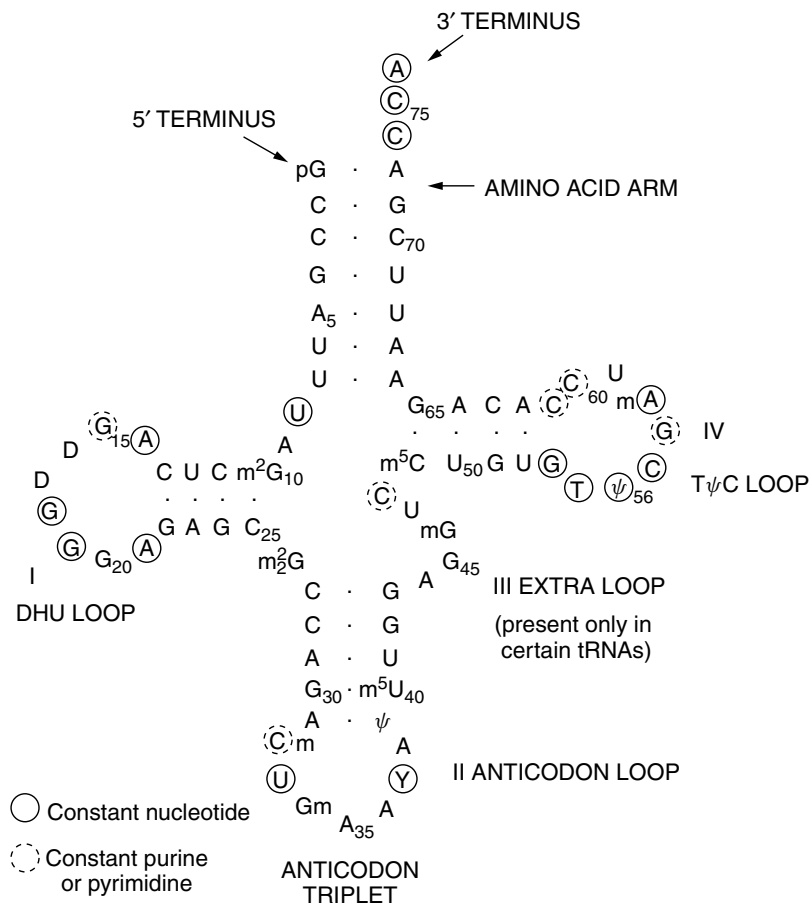


Fig. 2-20. Nucleotide sequence of yeast phenylalanyl tRNA. The structure is shown in the extended cloverleaf configuration. Loop I is termed the DHU loop since it contains dihydroxyuridine. The anticodon loop (II) contains the triplet bases which recognize the triplet code words in mRNA. Loop III varies in size and may even be absent from certain tRNAs. Loop IV is termed the TψC loop, since it contains (ribo)thymidylate, pseudouridine (ψ), and cytosine. Circled bases are constant in all tRNAs, and dashed circles indicate positions that are occupied constantly by either purines or pyrimidines. However, eukaryotic initiator tRNAs do not have the same constant nucleotides. (Modified from Kim et al., 1974.)

aspects. Methionine, rather than n-formylmethionine, is the initiating amino acid in eukaryotic cells as well as the archebacteria. The termination process is similar in both types of organisms. The process of protein synthesis in mitochondria is more closely analogous to that in bacteria than it is to what occurs in the cytoplasm of eukaryotes. Those differences that do exist between eukaryotic and prokaryotic protein synthesis may form the basis for selective inhibition by antibiotics or toxins on one cell type in the presence of the other. For example, some toxins have been shown to preferentially inhibit protein synthesis in eukaryotes. Diphtheria toxin specifically inactivates the elongation factor 2 of eukaryotic cells while having no effect on the analogous factor in prokaryotic cells.

TABLE 2-5. Amino acid code^a

Second →	U	C	A	G	Third ↓
First ↓	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
U	Leu	Ser	Ochre	Opal	A
	Leu	Ser	Amber	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

^aFirst, second, and third refer to the order of the bases in the triplet sequence. Ochre, amber, and opal are termination codons.

Transfer RNA

The incorporation of amino acids into polynucleotide chains in a specified order involves the formation of an activated complex between an amino acid and a specific RNA molecule. This activation (charging) of specific transfer RNA (tRNA) molecules is accomplished by aminoacyl tRNA synthetases (aminoacyl tRNA ligases). Each amino acid has its specific synthetase and specific tRNA that can read the code in the mRNA and thereby direct the incorporation of the amino acid into the growing peptide chain. Each tRNA has the capacity to recognize a specific triplet code by virtue of a **codon recognition site** and a site that can recognize the specific tRNA synthetase (ligase) that adds the amino acid to the tRNA (**ligase recognition site**). Additional specificities are required: a specific **amino acid attachment site** and a **ribosome recognition site**. The amino acid attachment site is the 3' hydroxyl of a terminal adenine ribose. All tRNA molecules have a 3' terminus consisting of a cytosine–cytosine–adenine (CCA) nucleotide sequence and a 5'-terminal guanosine nucleotide as shown in Figure 2-20.

Transfer RNAs consist of approximately 80 nucleotides. A characteristic feature is the high content of unusual nucleotides (inosine, pseudouridine, and various methylated bases) that occupy specific positions. Figure 2-21 illustrates a number of these unusual bases. They are formed by modification of the structures by specific **modification enzymes** after the synthesis of the tRNA molecule has been completed (posttranscriptional modification). The secondary structure of tRNA appears as a

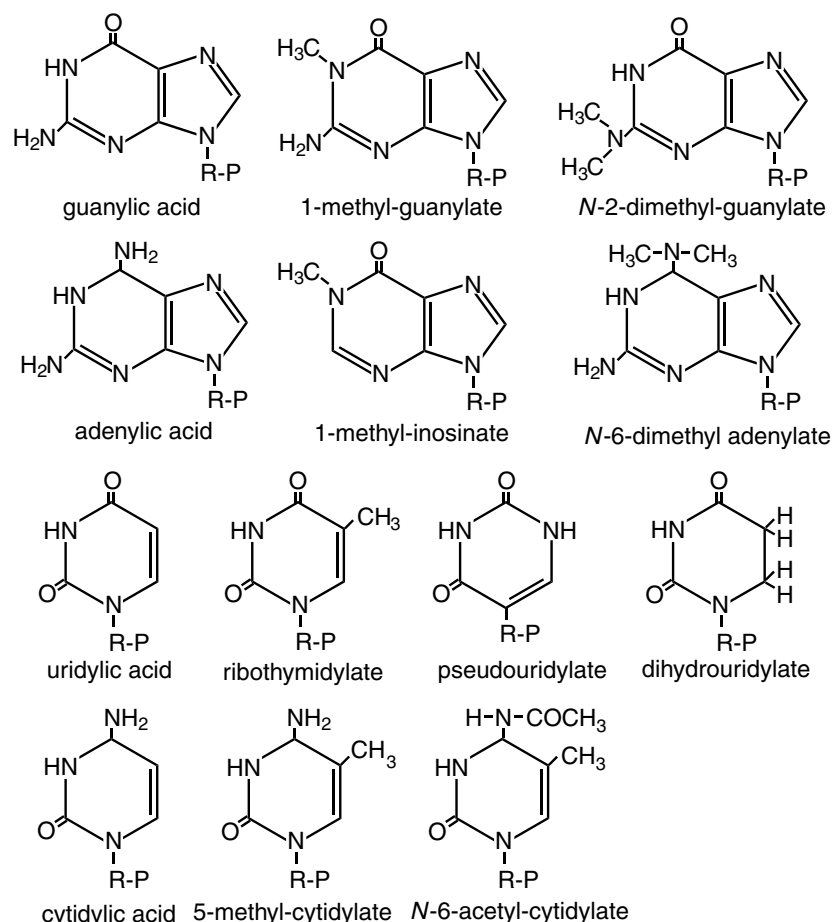


Fig. 2-21. Examples of some of the modified bases occurring in tRNAs.

cloverleaf with three loops containing the unusual bases. Loops result from the fact that some bases do not form double-stranded base-paired regions.

Loop I contains dihydroxyuridine (DHU) and is termed the DHU loop. Loop II contains the **anticodon**, a triplet of bases that recognizes the code in mRNA. Loop III varies in size, and, in those tRNAs in which it is large, it is double-stranded. Loop IV contains (ribo)thymidylate, pseudouridine, and cytosine (Fig. 2-21), a constant feature of all tRNAs that have been examined.

The activity of tRNA is dependent on the secondary and tertiary structures. Yeast phenylalanyl-tRNA was the first tRNA in which the complex tertiary structure was elucidated (Fig. 2-22). The 3'-terminal CCA required in all tRNAs is present upon transcription but requires processing (see "RNA Processing"). While it was generally considered that the DHU loop serves as the recognition site for tRNA ligase and loop III as the ribosome recognition site, it is now clear that the unusual tertiary structure of many sites in tRNA is involved in recognition. In the anticodon loop (loop II), each tRNA has a specific anticodon that directs the attachment of the proper amino acid as specified by the codon in the mRNA. Recognition is dependent on base pairing as in

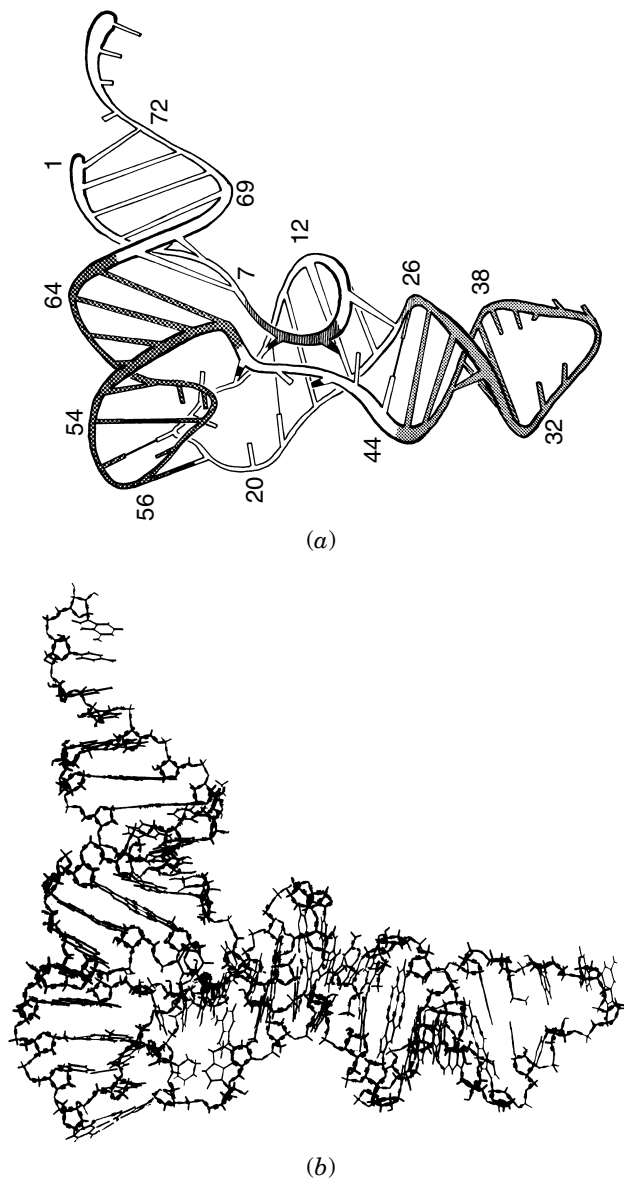


Fig. 2-22. Molecular and schematic models of yeast phenylalanyl tRNA. (a) Photograph of a molecular model built with Kendrew wire models at a scale of 2 cm per angstrom. The molecule is oriented with the anticodon loop at the bottom, the acceptor stem at the upper right, and the T ψ C loop at the upper left. The view is approximately perpendicular to the molecular plane. (b) Schematic model drawn from the coordinates of the molecular model shown in (a). The ribose phosphate backbone is drawn as a continuous cylinder with bars to indicate the hydrogen-bonded base pairs. The positions of single bases are indicated by rods, which are intentionally shortened. The T ψ C arm is heavily stippled, and the anticodon arm is marked by vertical lines. The black segments of the backbone include residues 8 and 9 as well as 26. Tertiary structure interactions are illustrated by black rods. The numbers indicate various nucleotides in the sequence as shown in Figure 2-20. (Modified from Kim et al., 1974.)

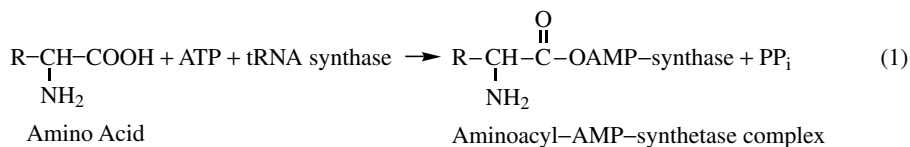
DNA. For example:

tRNA anticodon:	3'AAA	AUG	UUU	GGG	UGA
mRNA codon:	5'UUU	UAC	AAA	CCC	ACU
Amino acid added:	phe	tyr	lys	pro	thr

Since a number of amino acids have more than one codon, one question that must be addressed is how many tRNAs are required. There are, in a number of instances, isoaccepting species of tRNA—that is, different tRNAs that are charged by the same amino acid. This accommodates, in part, the degeneracy of the code. However, when there is more than one codon for a single amino acid, it is usually only the last nucleotide of the triplet that differs. The tRNA anticodon often contains one of the unusual nucleotides noted in Figure 2-21. These odd bases have “sloppy” base-pairing properties that permit them to bind with several different bases. This capability of recognizing more than one base is termed the “wobble” in codon recognition.

Charging of tRNA

The transfer of an amino acid to tRNA occurs via a two-step process. The amino acid must first be activated by ATP to form an aminoacyl–AMP complex with tRNA synthetase. The aminoacyl–AMP complex is then transferred to tRNA:



Aminoacyl–AMP–synthetase complex + tRNA



Each amino acid has a specific aminoacyl–tRNA synthetase (charging enzyme) and a specific tRNA. Although there may be more than one species of tRNA for a specific amino acid, there is only one charging enzyme for each amino acid. The synthetase must have at least two recognition properties. It must be able to differentiate one amino acid from another and it must also be able to recognize a specific tRNA. These recognition properties are necessary to assure that the specific amino acid is charged on the proper tRNA molecule. Recognition of the codon on mRNA by the specific anticodon on the tRNA is required for the incorporation of an amino acid into the proper position in the polypeptide sequence.

Ribosome Structure and Synthesis

Ribosomes are the protein-synthesizing machines of the cell. They conduct the remarkable task of choosing which amino acids must be added to a growing peptide chain by reading successive mRNA codons. The complex problem of converting information (mRNA) into product (protein) is reflected in the elaborate structure of the ribosome. The prokaryotic ribosome is composed of two subunits that together form

the 70S ribosome. The smaller subunit is 30S (900,000 mwt), containing 21 r-proteins (designated S1 through S21 in *E. coli* for small subunit proteins) and one 16S rRNA. The larger subunit is 50S (1.6 million mwt), containing 31 r-proteins (designated L1 through L34 for large subunit proteins) and one copy each of two rRNA species, 23S rRNA and 5S rRNA. Thus, there are 52 different proteins (r-proteins) and three distinct RNAs (rRNA) that make up an intact ribosome.

The 50S subunit contains **peptidyl transferase** activity that forms peptide bonds in a growing protein chain. The 30S subunit contains the mRNA **decoding site** where complimentary interactions between the mRNA codon and tRNA anticodon ensure translational fidelity. The ribosome also has three tRNA-binding sites formed at the interface of the two subunits: the aminoacyl-tRNA-binding or decoding site (A site), the peptidyl-binding site (P site), and the exit (E) site. The A site accepts all incoming charged tRNAs while the P site contains the previous tRNA attached to the nascent polypeptide (peptidyl-tRNA). The E site is discussed later.

X-ray crystallographic and cryomicroscopic analyses of ribosomal subunits from several organisms have given us an amazingly detailed view of the ribosome and provided unique insights into the catalysis of peptide bond formation. One of the more stunning discoveries is that the ribosome is, in fact, a **ribozyme**. The previous assumption was that rRNA simply existed as a framework upon which the catalytic proteins imbued with the real work of peptide bond formation were assembled. The fact is that the peptidyl transferase active center in the 50S subunit is devoid of any protein within an 18 Å radius. That, coupled with the finding that the part of 23S rRNA located within the active center is highly conserved across species, led to the conclusion that RNA (and not protein) is responsible for catalysis. Consequently, it is protein that helps shape rRNA into its proper form to catalyze peptide bond formation, not the other way around.

The r-protein genes of *E. coli* are scattered among 19 different operons (Table 2-6; see also “Translational Repression” in Chapter 5), but each ribosomal protein is only represented by one gene. In striking contrast, the rRNA genes exhibit a considerable amount of redundancy. There are seven rRNA operons each arranged as

5' leader–16s rRNA–spacer–23s rRNA–5s rRNA–trailer–3'

(see “RNA Processing”). The redundancy may be a protective strategy designed to reduce the potential danger associated with mutations in a single rRNA gene. Note that the various rRNA operons cluster near the origin of replication and that transcription of a given operon occurs in the same direction as replication. This is thought to minimize collisions between the two polymerases.

Figure 2-23 illustrates the association of the two subunits and their relationships with some translational factors and mRNA. The ribosome is constructed by sequentially adding subunit r-proteins to the processed rRNA (Fig. 2-24). It is a highly organized process involving a series of protein–RNA and protein–protein interactions. For example, protein S7 is a primary binding protein that organizes the folding of the 3' major domain of 16S rRNA and enables subsequent binding of other ribosomal proteins to form the head of the 30S. Protein S8, encoded by *rpsH*, also plays a critical role in both the assembly of the 30S ribosomal subunit and in the translational regulation of ribosomal proteins encoded by the *spc* operon (see below and Chapter 5). During 30S assembly, S8 binds to the central domain of the 16S rRNA and interacts cooperatively

TABLE 2-6. Genetic Nomenclature for Components of the Translation Apparatus

Gene Symbol	Gene Product
<i>Ribosome Components</i>	
<i>rplA</i> to <i>rplY</i>	50S ribosomal subunit proteins L1 to L25
<i>rpmA</i> to <i>rpmG</i>	50S ribosomal subunit proteins L27 to L33
<i>rpsA</i> to <i>rpsU</i>	30S ribosomal subunit proteins S1 to S21
<i>rrnA</i> to <i>rrnH</i>	rRNA polycistronic operons
<i>rrf</i>	5S rRNA (encoded within <i>rrn</i> operons)
<i>rrl</i>	23S rRNA (encoded within <i>rrn</i> operons)
<i>rrs</i>	16S rRNA (encoded within <i>rrn</i> operons)
<i>Accessory Factors</i>	
<i>cca</i>	tRNA nucleotidyl transferase
<i>fusA</i>	Translation factor EF-G
<i>prmA</i> , <i>prmB</i>	Methylation of ribosomal proteins
<i>rimB</i> to <i>rimM</i>	Maturation of ribosome
<i>rna</i> to <i>rnp</i>	Ribonuclease (I to P)
<i>trmA</i> to <i>trmD</i>	tRNA methyltransferase
<i>infA</i> to <i>infC</i>	Initiation factors IF-1 to IF-3
<i>tufA</i> , <i>tufB</i>	Translation factor EF-T _U
<i>rrf</i>	Ribosome recycling factor
<i>Transfer RNAs and Their Charging^a</i>	
<i>alaT</i> , <i>alaU</i>	Alanine tRNA _{1B}
<i>valT</i>	Valine tRNA ₁
<i>alaS</i>	Alanyl-tRNA synthetase
<i>valS</i>	Valyl-tRNA synthetase
etc.	

Source: From Berlyn. 1998. *Microbiol. Mol. Biol. Rev.* **62**:814.

^aGenerally, capitals of T and further in the alphabet indicate an RNA gene, whereas S and earlier letters are used for synthetases.

with several other small subunit proteins to form a well-defined ribonucleoprotein neighborhood. A similar strategy is employed for 50S subunit assembly using large subunit proteins.

Because rRNA molecules are innately capable of folding into a variety of different nonfunctional secondary structures, the sequential addition of ribosomal proteins direct the RNA into the proper folding pathway. As noted earlier, 16S and 23S rRNAs contain about 30 posttranscriptionally modified nucleotides. All of the modified nucleotides in 16S rRNA form a group at the mRNA decoding site surrounding the anticodon stem loops of tRNA-binding sites (A and P sites; see below). The modified bases in 23S rRNA cluster around the peptidyl transferase center of the 50S subunit. It has been proposed that these modifications fine-tune interactions between tRNA and rRNA. To see more detail of rRNA secondary structure visit <http://www-rna.uia.ac.be/secmodel/> or <http://geta.life.uiuc.edu/~nikos/Ribosome/rproteins.html>. [Internet addresses should not normally be hyphenated. How can this be handled?]

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Fig. 2-23. (a) Diagrammatic representation of the ribosome exit and translational domains. (b) Structure of the 70S ribosome and its functional center. *Top:* The tRNA molecules span the space between the two subunits. The channel in the 50S subunit through which the growing peptide protrudes is shown in dashed lines. *Bottom:* The 30S (left) and 50S (right) subunits have been opened up to give a better view of the three binding sites for tRNA; the A, P, and E sites. The 30S subunit shows the approximate location of the site where the mRNA codons are read by the tRNA anticodons. The 50S subunit has the tRNA sites shown from the opposite direction. The acceptor ends of the A and P sites are close to each other in the peptidyl transfer site, which is close to the exit channel located behind a ridge in the 50S subunit. The binding site for EF-G and EF-T_U is located on the right-hand protuberance of the 50S subunit. From L. Ijas, A. 1999. Function is structure. *Science* **285**:2077–2078.

Growth Rate–Dependent Control of Ribosome Synthesis. Bacteria such as *E. coli* can grow exponentially at a wide range of growth rates depending on the carbohydrate source and how hard it has to work to make things such as amino acids. The ability to grow at different rates, of course, has a consequence in terms of protein synthesis. Basically, the faster the organism grows (the shorter the generation time), the faster it must make proteins. Since the peptide elongation rate per ribosome is constant, faster protein synthesis requires more ribosomes per cell. As a result, de novo synthesis of ribosomes must be linked to the bacterial growth rate. The cell accomplishes this by linking rRNA synthesis to ribosomal protein synthesis.

The first step in the process is growth-rate control of rRNA synthesis. The rRNA operons have two promoters, P1 and P2, with growth rate control focused on P1.

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Fig. 2-24. Model for maturation of ribosomal RNAs in *Escherichia coli*. (a) The events occur in the following general sequence: (1) 16S, 23S, 5S genes are transcribed sequentially. (2) Key ribosomal proteins associate with 16S and 23S precursors before release from DNA. Transcript is cleaved into separate RNAs before transcription is complete. (3) 16S precursor is 10 to 20% methylated; 23S precursor is 60% methylated. First precursor particles sediment at 22S and 30S and are probably slightly folded. Sequential addition of other proteins gives intermediate precursor particles, sedimenting at 26S and 43S; these are folded into more compact structures. Final set of proteins added to give mature subunits. (b) *E. coli* 16S rRNA secondary structure. The binding site for ribosomal protein S8 is shown. From Zimmermann et al. 2000. How ribosomal proteins and rRNA recognize one another. In: *The Ribosome: Structure, function antibiotics and cellular interactions*. Ed. R. A. Gurratt et al. ASM Press, Washington, DC. pp 93–104.

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(b)

Fig. 2-24. (continued)

Transcription from P1 requires a high nucleotide triphosphate concentration (ATP or GTP). If growth slows, the high concentration of ribosomes will overtranslate relative to need and will consume more GTP (see below) than can be replenished at the slower growth rate. GTP levels then drop, causing decreased transcription of rRNA operons. With less rRNA scaffolding to bind, the concentration of free ribosomal proteins increases. How does this lower r-protein synthesis? Most ribosomal protein operons are subject to translational repression (see Chapter 5) in which a ribosomal protein binds to its own mRNA, decreases translation, and lowers its own production. This control is distinct from ppGpp-dependent stringent control discussed in Chapter 18. The end result is less rRNA, less ribosomal protein, and less ribosomes per cell.

Evolutionary Consideration. Comparative analysis of the nucleotide sequences of ribosomal RNA has provided a means of direct measurement of genealogical relationships. Ribosomal RNAs, particularly 16S rRNA, serve as phylogenetic markers for prokaryotic organisms. Detailed technical and statistical considerations have revealed that an oligonucleotide of six bases is unlikely to recur more than once in

16S rRNA. When the 16S rRNAs from different organisms include the same six bases, this reflects true homology.

By comparing the occurrence of these six-base sequences in various organisms, it is possible to compile a “dictionary” of them for a given organism. The data can then be analyzed to compare the degree of relatedness among organisms. This type of analysis from a large number of organisms has shown that certain classical criteria (e.g., gram stain) used in bacterial classification distinguish valid phylogenetic relationships, whereas other criteria (e.g., morphology) do not. Interestingly, this type of analysis has added weight to the theory that mitochondria in eukaryotic organisms evolved from prokaryotes. The nucleotide sequence of 16S rRNA from mitochondria relates more closely to bacteria than to the cytoplasmic ribosomes of eukaryotes.

Initiation of Polypeptide Synthesis

The processes of initiation and elongation of the polypeptide chain involve several specific and complex mechanisms (Figs. 2-25 and 2-26). The first stage in initiation finds the 30S and 50S subunits separated from each other with the 30S subunit complexed with two initiation factors (IF-1, mwt 9000, and IF-3, mwt 22,000). In the absence of mRNA, IF-1 and IF-3 function to prevent the association of the 30S and 50S ribosomal subunits. The next events involve the association of the mRNA and initiator tRNA to the 30S subunit (Fig. 2-25.2).

IF-3 binds to both the 30S subunit and to mRNA. As such, IF-3 helps bring mRNA to the ribosome. The 30S-[IF-3][IF-1] complex binds to mRNA around the initiation codon (in order of preference, AUG, GUG, UUG, CUG, AUA, or AUU). IF-3 inhibits initiation on any codon other than the three canonical initiation codons, AUG, GUG, or UUG. How does IF-3 recognize an initiation codon? Every mRNA molecule also includes within the untranslated region a ribosome-binding site for each polypeptide in a polycistronic message. The consensus sequence (5')AGGAGGU(3'), called the **Shine-Dalgarno (S-D) sequence**, is important in the binding of mRNA to the 30S complex. The S-D sequence has been shown to base pair to a region located at the 3' end of 16S rRNA. This pairing will position the initiating AUG codon to bind the anticodon of the appropriate initiator tRNA and presumably allows unambiguous distinction between initiation and internal met codons. IF-3 also binds to the ribosome in this area.

Next, cytosolic initiation factor 2 (IF-2, mwt 115,000) complexes with GTP and the initiator tRNA, N-formylmethionyl-tRNA (fMet-tRNA). In this form, IF-2 directs the binding of fMet-tRNA directly to the P site within the 30S subunit. Thus, initiator tRNA bypasses the A site where all other aminoacyl-tRNAs first bind the ribosome (Fig. 2-25.3). Binding of N-formylMet-tRNA then permits the association of the 30S and 50S ribosomal subunits (Fig. 2-25.4). Removal of IF-3, which occurs at the same time, is essential, since its presence prevents the association of the two ribosome subunits. The initiation complex, at this point, consists of an association of mRNA with the 30S ribosomal subunit, IF-1, IF-2-GTP, and fMet-tRNA. Union of the 30S initiation complex with the 50S ribosomal subunit causes the immediate hydrolysis of the bound GTP to GDP. The union itself does not require this hydrolysis, since it can be accomplished in the presence of an analog of GTP, 5'-guanylyl (methylene diphosphonate), GDPCP. The hydrolysis of GTP is required to eject IF-1 and IF-2 from the ribosome, which is required before proceeding to the elongation stage of translation (Fig. 2-25.5).

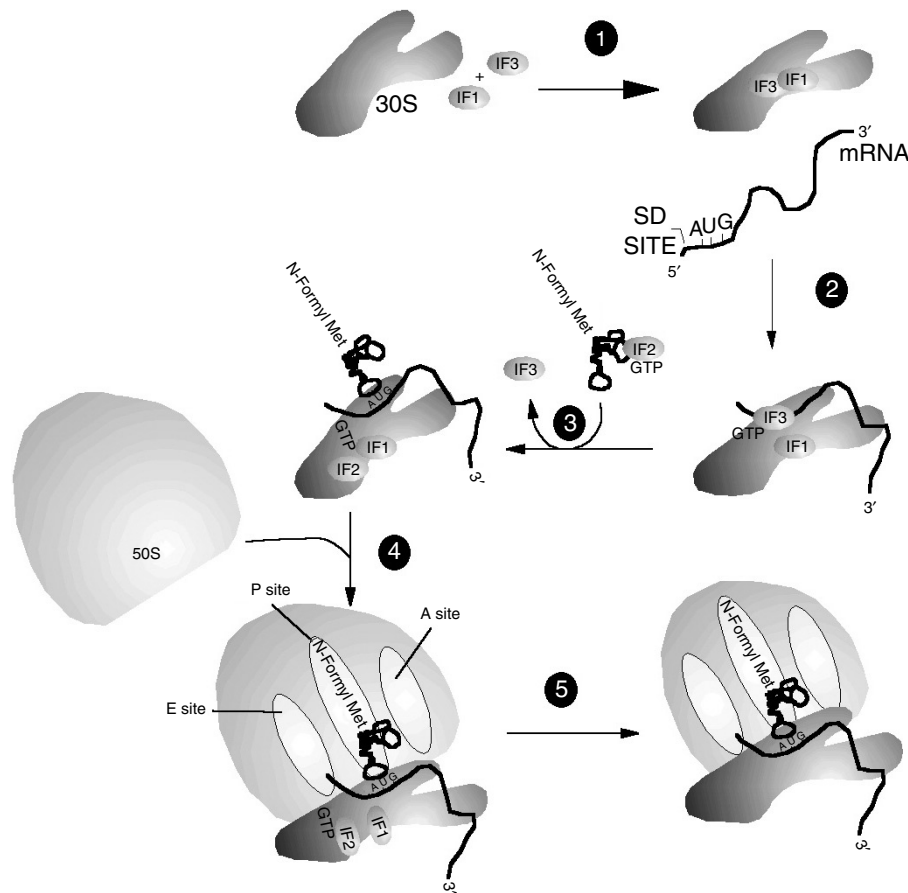


Fig. 2-25. Sequence of events involved in the initiation of polypeptide synthesis.

Elongation

The addition of amino acids to the growing polypeptide chain occurs at approximately 16 residues per second (at 37°C) in the order specified by the mRNA code. At the completion of initiation, the 70S ribosome contains fMet-tRNA in the P site while the A site is unoccupied and free to accept the next aminoacyl-tRNA directed by the triplet base code in the mRNA. Elongation factor T is involved in shuttling the proper aa-tRNA into the A site (Fig. 2-26). EF-T has been shown to consist of two subunits. One subunit (EF-T_U, mwt 44,000) is unstable while the other (EF-T_S, mwt 30,000) is stable. EF-T_U is the most abundant protein in *E. coli*, comprising 5 to 10% of the total cellular protein. GTP binds to EF-T and causes its dissociation into EF-T_U-GTP and EF-T_S (Fig. 2-26.1). EF-T_U-GTP complexes with all aminoacyl-tRNAs except the initiator tRNA. The resulting ternary complex (GTP-EF-T_U-aminoacyl-tRNA) is a required intermediate in the binding of aminoacyl-tRNA to the ribosome. EF-T_S plays no role in this process. How ternary aminoacyl-tRNA complexes initially find the A site is due in part to EF-T_U and 5S rRNA molecules recognizing sequences in the TψC loop (loop IV) of tRNA (Fig. 2-20).

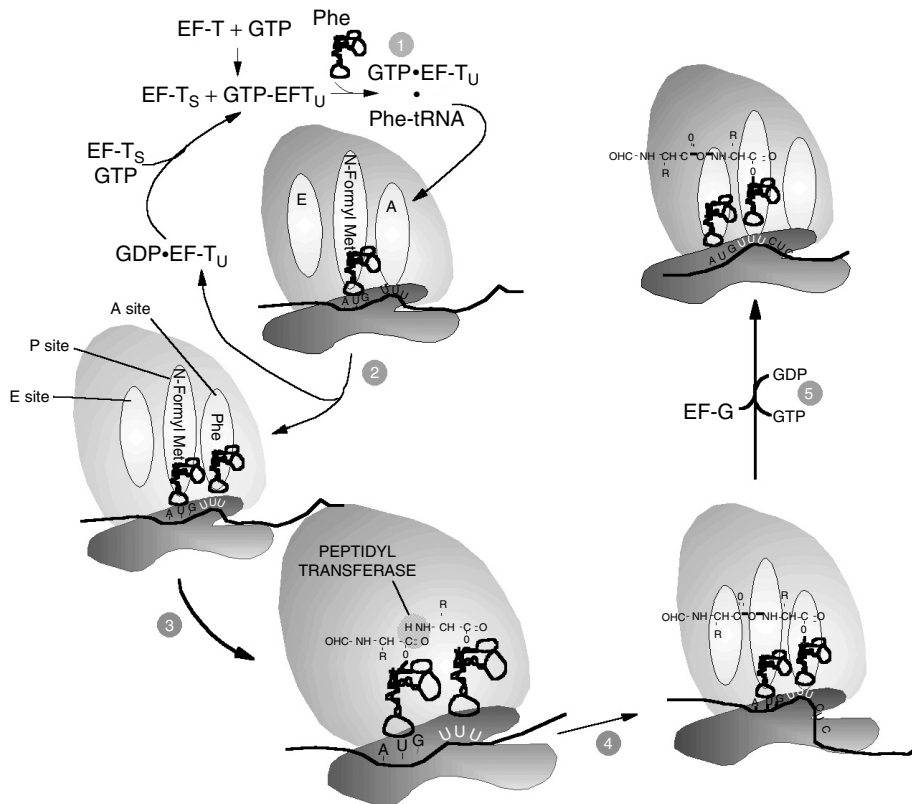


Fig. 2-26. Sequence of events involved in the continuation of peptide chain formation.

Although the correct selection of aminoacyl-tRNA-EF-T_U-GTP at the A site depends on the codon:anticodon match, the binding energy of this interaction is not sufficient to account for the exquisite selectivity of this process. The ribosome must sense and amplify the tRNA fit. Information regarding fit is probably conveyed between the 50S and 30S subunits by alterations in tRNA structure and by a relay of conformational changes through 16S rRNA to 23S rRNA. Residues 1492 and 1493 of 16S rRNA are in a position within the ribosome to monitor codon:anticodon fit via hydrogen bonding. If the fit is not perfect, the tRNA complex is rejected. One supportive piece of evidence for this model is that the antibiotic paromomycin alters the structure of 16S rRNA to mimic conditions of perfect codon:anticodon fit so that incorrect aminoacyl-tRNAs may be selected.

Once aminoacyl-tRNA has bound to the A site, GTP is hydrolyzed and a complex of EF-T_U-GDP is released from the ribosome. One GTP is hydrolyzed per aminoacyl-tRNA bound as shown in Figure 2-26.2. This GTP hydrolysis is not essential for aminoacyl-tRNA binding, since nonhydrolyzable GDPCP can substitute. Although the aminoacyl-tRNA can bind to the A site in the absence of GTP hydrolysis, EF-T_U cannot be released. Therefore, one result of GTP hydrolysis is the release of EF-T_U from the ribosome. After release of EF-T_U-GDP, EF-T_S will catalyze a nucleotide exchange that yields active EF-T_U-GTP. GTP hydrolysis, however, may serve another

critical function at this stage. Following GTP hydrolysis, EF-T_U undergoes an unusually large conformation change predicted to move the amino acid-carrying CCA end of the tRNA from the codon:anticodon testing site to the peptidyl transferase site (Fig. 2-26.3).

Peptide Bond Formation

Once the incoming aminoacyl-tRNA has bound to the A site, peptide bond formation occurs (see Fig. 2-26.3 and 2-26.4). The peptide bond is formed between the amino group of the incoming amino acid and the C terminal of the elongating polypeptide bound to the tRNA at the P site. As already noted, the activity responsible for peptide bond formation, the peptidyl transferase center, is not a protein but is a ribozyme comprised of a segment of the 23S rRNA in the 50s ribosomal subunit. No soluble factors appear to be necessary.

Translocation

After the formation of the peptide bond, the growing peptide chain becomes bound to the tRNA that was carrying the incoming amino acid (see Fig. 2-26.4) and occupies the A site. This peptidyl tRNA must now be moved from the A site to the P site to allow the next EF-T_U-aminoacyl-tRNA to enter the A site. In addition, the mRNA must be moved by one codon so that appropriate codon-anticodon recognition takes place (decoding). Elongation factor G (EF-G, mwt 80,000), in which G = GTPase, and GTP hydrolysis are required for these processes (Fig. 2-26.5). EF-G binds to the same site as EF-T_U; hence the requirement noted above for T_U recycling. Once bound, EF-G catalyzes the hydrolysis of another GTP. Therefore, during elongation two molecules of GTP are hydrolyzed per peptide bond. One hydrolytic step is EF-T dependent and the other is EF-G dependent. Binding of EFG combined with GTP hydrolysis triggers a rotation of the 30S subunit relative to the 50S subunit (Fig. 2-27) and a widening of the

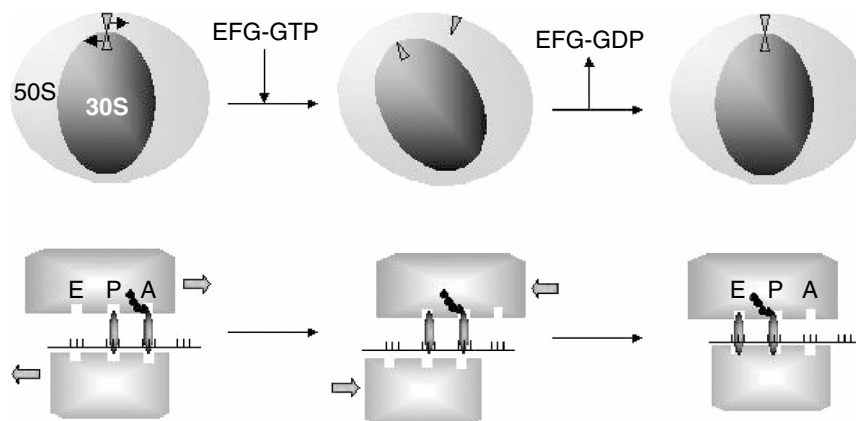


Fig. 2-27. Ratchet movement of ribosome along mRNA. *Top:* Top view illustrating rotation of 30S and 50S subunits relative to each other. *Bottom:* Side view illustrating how the rotation moves uncharged and nascent peptide-containing tRNA molecules from the P and A sites (respectively) to the E and P sites.

mRNA channel. This rotation moves the tRNAs in the A and P sites to the P and E sites, respectively. Thus, EF-G has been suggested as the “motor” for translocation, with the energy for this process coming from GTP hydrolysis. To complete one translocation event, EF-G must be released from the ribosome. Unless EF-G is released, the 30S subunit will not fully ratchet back to its original position relative to the 50S subunit and elongation cannot continue.

At the end of a translocation cycle, the deacylated tRNA resides in the E site, the peptidyl-tRNA sits in the P site, and the A site is open to accept the next incoming aa-tRNA ternary complex. How, then, is the deacylated t-RNA expelled from the E site? The next incoming charged tRNA binds to the unoccupied A site, which causes a conformational change in the ribosome that reduces affinity of the E site for the deacylated tRNA, allowing it to leave.

Termination

When translocation brings one of the termination codons (UAA, UGA, or UAG) into the A site, the ribosome does not bind an aminoacyl-tRNA-EF-T_U-GTP complex. Instead it binds one of two peptide release factor proteins—RF-1, mwt 44,000, or RF-2, mwt 47,000—which then activate peptidyl transferase, hydrolyzing the bond joining the polypeptide to the tRNA at the P site. Specific recognition of the stop codon by release factors involves protein side chains mimicking nucleic acid base–base interaction. The result is release of the completed polypeptide.

Finally, ribosome disassembly and recycling must occur. RF-3 is known to catalyze the disassociation of RF-1 or RF-2 from ribosomes after polypeptide release. The next step is the disassociation of 50S subunits from the 70S posttermination complex catalyzed by the binding of ribosome release factor (RRF) and EF-G to the A site. This process also requires GTP hydrolysis. Subsequent removal of deacylated tRNA from the resulting 30 S : mRNA : tRNA posttermination complex is then needed to permit rapid 30S subunit recycling. This step requires IF-3 binding. However, some 30S subunits are not ejected from mRNA but are allowed to slide onto the next cistron in a polycistronic message. This has been shown to occur when the beginning of a downstream ORF lies within 3 or 4 nucleotides of the translational stop of the upstream ORF. This is an example of **translational coupling** in which the successful translation of one ORF is required to achieve translation of the neighboring ORF. A very different form of translational coupling is when the translation of one gene in a message relieves secondary structure that sequesters a downstream ribosome-binding site. If the secondary structure includes the 3' end of the preceding gene, ribosome movement through that region will disrupt stem base pairing and expose the previously hidden ribosome-binding site.

Macromolecular Mimicry. It is interesting to note that the three extraribosomal translation factors possessing GTPase activity all interact with L7/L12. Cryomicroscopic structure analyses have revealed that the overall shape of the EF-T_U ternary complex is surprisingly similar to the shape of EF-G : GDP, with part of the tRNA structure mimicked by portions of EF-G (Fig. 2-28). In addition, sequence comparisons of EF-T_U, EF-G, IF-2, and RF-3, based on the known structures of EF-T_U and EF-G, indicate they all contain similar domains, called 1 and 2, with strikingly similar folds. It is implied that they all bind to the ribosome in similar modes, and that the GTPase-activating center on the ribosome is the same for all of them. This macromolecular

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Fig. 2-28. Molecular mimicry, structural comparison of elongation factors. On the left is the ternary complex of Phe-tRNA : EF-T_U : GDPNP and on the right is EF-G : GDP. Nucleotides bound to domain 1 are shown in ball-and-stick models, and an Mg²⁺ ion in EF-T_U is shown as a gray ball. The anticodon of tRNA is at the bottom of the ternary complex, and Phe attached to the terminal ribose is seen in black between domains 1 and 2 of EF-T_U. From: Nissen et al. *EMBO J.* **19**:489 (2001).

mimicry appears important in maintaining the proper order of events by forcing one factor to leave the ribosome before another can bind. However, there are other potential roles for structural mimicry in translation beyond simple site sharing. For example, the part of IF-2 that resembles tRNA may mimic a tRNA bound to the A site. This would ensure the exclusive binding of initiator tRNA to the P site. Another possibility is that by binding to the A site of the 30S subunit via its macromolecular mimicry of tRNA, EF-G could prevent the peptidyl-tRNA from rebinding to the A site, thereby controlling the correct progress of the elongation cycle.

Posttranslational Processing

The free polypeptide, *in vivo*, undergoes one of two posttranslational processing steps: (1) the formyl group of the N-terminal fMet may be removed by the enzyme methionine deformylase, or (2) the entire formylmethionine residue may be hydrolyzed by a formylmethionine-specific peptidase (methionyl amino peptidase or MAP). There is apparently some discrimination involved in channeling different polypeptides through

these two alternative steps, since not all N-terminal methionines are removed. The discriminating factor appears to be the side-chain length of the penultimate amino acid. The longer the side chain, the less likely MAP will remove the methionine. Other processing can occur such as acetylation (e.g., acetylation of L12 to give L7) or adenylation (see “Regulation of Nitrogen Assimilation and Nitrogen Fixation” in Chapter 5).

When Nonsense Makes Sense

There are several examples of proteins such as formate dehydrogenase (FdhF) that contain a modified form of cysteine called selenocysteine. The gene for formate dehydrogenase contains an in-frame TGA (UGA in message) codon that encodes the incorporation of selenocysteine instead of terminating translation as a nonsense codon. The requirements for this process include a specific tRNA^{SEC} (*selC*) containing a UCA anticodon, a unique translation factor (*selB*) with a function analogous to elongation factor T_U (see “Translocation”), and a specific mRNA sequence (SECIS) downstream of the internal UGA codon in the *fdh* gene that signals selenocysteine incorporation. This region forms a loop that could, in part, shield the UGA from release factor. SelB is a GTPase that binds specifically to selenocysteyl-tRNA^{SEL} and to the SECIS sequence. These interactions direct selenocysteyl-tRNA to the ribosome only when the SECIS sequence is present in the ribosome A site.

Coupled Transcription and Translation

Prokaryotes differ significantly from eukaryotes in that prokaryotic ribosomes can engage mRNA before it has been completely transcribed (see Figs. 1-1 and 1-8 in Chapter 1). This coupled transcription-translation affords bacteria novel mechanisms for regulation (see “Attenuation Controls” in Chapter 5). Because the transcription process is faster than translation, the cell must protect itself from situations in which a long region of unprotected mRNA might stretch between the ribosome and RNA polymerase. To avoid that situation, most mRNAs contain RNA polymerase pause sites (discussed earlier) that allow the ribosome to keep pace with polymerase, thereby protecting the message from endonucleolytic attack.

PROTEIN FOLDING AND CHAPERONES

A continuing challenge for biochemists is to unravel the processes required to convert the one-dimensional information encoded within genes (i.e., an amino acid sequence) into three-dimensional protein structures that contain biochemical activity. Historically, protein biogenesis was thought to involve only spontaneous folding of polypeptide domains. We now realize that the process is more complex than previously envisioned. Most, if not all, proteins in the living cell require assistance to fold properly. This assistance comes from proteins that are not final components of the assembled product. These “foldases” are called chaperones (or chaperonins, depending on their structure).

The proposed function of chaperone proteins is to assist polypeptides to self-assemble by inhibiting alternative assembling pathways that produce nonfunctional structures. During protein synthesis, for example, the amino-terminal region of each

polypeptide is made before the carboxy-terminal region. The chance of incorrect folding of a nascent polypeptide is reduced through interaction with chaperones. Another process in which chaperones can be invaluable is protein secretion or translocation. Proteins that cross membranes do so in an unfolded or partially folded state. Often they are synthesized by cytosolic ribosomes and must be prevented from folding into a translocation-incompetent state.

Cells also take advantage of chaperonins when faced with an environmental stress that will denature proteins into inactive forms that aggregate. To protect against such stresses, cells accumulate proteins that prevent the production of these aggregates or unscramble the aggregates so that they can correctly reassemble (see “Thermal Stress and the Heat Shock Response” in Chapter 18).

Key members of the major chaperone pathway are GroEL(60KDa), GroES(10KDa), DnaK(70KDa), and trigger factor (*tig*) of *E. coli*. They are all abundant proteins that increase in amount after stresses such as heat shock and therefore are called heat shock proteins (HSPs). Their importance to life in general is evident in that they have been preserved throughout evolution. Representatives of the major chaperone families have been found in all species examined. DnaK-like chaperone proteins (called HSP70s) are found in prokaryotes and most compartments of eukaryotes. HSP60s, also called chaperonins, are found in bacteria, mitochondria, and chloroplasts.

How do chaperones identify proteins in need of folding? The specifics are not clear, but it is apparent that most chaperones recognize exposed hydrophobic surfaces on target proteins. These regions of a protein should be buried within the protein structure and held away from water. Surface exposure indicates improper folding.

Folding Stages

There are several stages in the folding pathway of newly synthesized proteins (Fig. 2-29). The ribosome-associated **trigger factor** is the initial chaperone followed by the DnaK and the GroEL/GroES systems. Trigger factor is a 48 kDa protein with an amino terminus that binds ribosomes, a central domain with peptidyl proline isomerase activity, and a carboxy-terminal domain whose function is not defined. As a peptide is synthesized, it first encounters trigger factor on the ribosome. This chaperone is supposed to help the nascent peptide orient proline residues. The peptidyl proline isomerase activity mediates cis-trans conversions of peptidyl proline bonds. Because of their structure, proline residues introduce critical turns or bends in the secondary structure of a protein. Thus, rotating the bond between the COOH end of proline and the NH₂ end of the adjacent amino acid dramatically alters protein structure.

If the protein comes off the ribosome still incompletely folded, DnaK intervenes. DnaK is monomeric with two domains. Its N-terminal portion contains ATPase activity and binds the cochaperone **GrpE**. The C-terminal domain binds polypeptide substrates and the cochaperone **DnaJ** (HSP40). DnaK binds to surface-exposed hydrophobic areas on the target peptide and DnaJ binds DnaK to modulate DnaK ATPase activity. After hydrolysis of ATP, DnaK binds and releases small hydrophobic regions of misfolded proteins, allowing the protein another chance to fold. Release of the peptide often requires an ADP/ATP exchange mediated by GrpE.

The third, and basically last, chance for a peptide to fold properly involves the HSP60 GroEL/HSP10 GroES chaperonin system. Hsp60s assemble into large double-ring structures whose central cavity allows proteins up to 60–65 kDa to fold in a

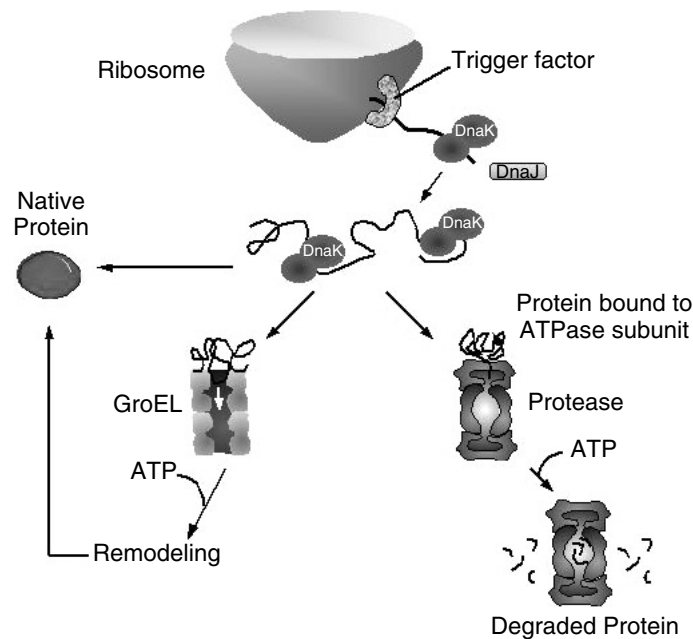


Fig. 2-29. Protein folding versus proteolysis pathways.

protected environment. Reversible association of the small capping protein GroES controls entrance to the chamber. Cycles of ATP binding and hydrolysis produce conformational changes within the chamber that enable binding and release of partially folded substrates.

There is some redundancy between the function of DnaK and trigger factor, since cells remain viable if one or the other remains active. GroEL, however, is essential for viability.

Protein Folding and Chaperone Mechanisms Outside the Cytoplasm

Proteins that move into the periplasm have special folding considerations. Generally, periplasmic proteins are secreted in their unfolded forms because attempts to export folded proteins clog most transport systems. Consequently, soon after entering the periplasmic space, the exported protein must be folded. Since the periplasm is an oxidizing environment whereas the cytoplasm is a very reducing environment, sulfhydryl groups in periplasmic proteins are subject to oxidation and formation of disulfide bonds. To prevent this, there are a number of periplasmic disulfide bond reductases (Dsb) that convert disulfide bonds into sulfhydryl groups and keep them that way. In addition, there are several peptidyl prolyl isomerases (Skp, SurA, FkpA, PpiA, PpiD) that, as noted earlier, rotate prolines within a peptide to achieve a proper bend. As opposed to the broad substrate chaperones noted above, there are other chaperones dedicated to a single purpose. For example, the FimC chaperone of *E. coli* protects the FimH pilin component until the FimH subunits are assembled into a mature pilus.

Quality Control

After initial folding and assembly, proteins may suffer damage as a consequence of environmental stress (e.g., acid pH). These damaged proteins go through a quality-control process that either targets the protein for refolding or degradation (Fig. 2-29). Chaperones and proteases both recognize exposed hydrophobic regions on unfolded peptides. Damaged proteins go through a kind of triage process where they first interact with chaperones that attempt to refold the protein. If the protein is released from the chaperone without being fully renatured, it remains in the pool of nonnative proteins and can either be rebound by a chaperone for another attempt at remodeling or will bind a protease that will target it for degradation. One or the other outcome for each damaged protein is critical if the cell is to survive the stress.

Future research on molecular chaperones will focus on how chaperones work and the structural basis by which a given chaperone recognizes and binds to some features present in a wide variety of unrelated proteins but which is accessible only in the early stages of assembly.

PROTEIN TRAFFICKING

The bacterial inner membrane, periplasmic space, and outer membrane all contain proteins not found in the cytoplasm. There are other proteins exported completely out of the cell into the surrounding medium or into eukaryotic host cells. All of these proteins are first synthesized in the cytoplasm but somehow find their way out. How does this happen?

Proteins destined to be integral membrane proteins are tagged with a very hydrophobic N-terminal **signal sequence** that anchors the protein to the membrane. N-terminal signal sequences range from 15 to 30 amino acids and include 11 hydrophobic amino acids preceded by a short stretch of hydrophilic residues. In contrast to integral membrane proteins, proteins predestined for the periplasm or outer membrane have their signal sequences cleaved following export. The roles of the signal sequence are to mediate binding of nascent polypeptides to the membrane and confer a conformation on the precursor that renders it soluble in the membrane. Signal sequences alone, however, are not enough to export proteins. There are actually seven general mechanisms of protein export that manage the remarkable feat of inserting proteins into membranes and passing hydrophilic proteins through hydrophobic membrane barriers. It is important to note that each system traffics different sets of exported proteins.

Insertion of Integral Membrane Proteins and Export of Periplasmic Proteins

Trigger Factor Versus Signal Recognition Particle. The process begins on the ribosome where either trigger factor (see “Protein Folding and Chaperones”) or SRP (signal recognition particle) can bind nascent proteins (Fig. 2-30). These two factors help direct proteins toward different transport systems. Recall that trigger factor is associated with the ribosome. When trigger factor binds to a nascent protein emerging from the ribosome, it will prevent SRP from binding and allow the protein to be completely synthesized into the cytoplasm. Presecreted proteins that bind trigger factor generally have signal sequences targeting them for secretion rather than for

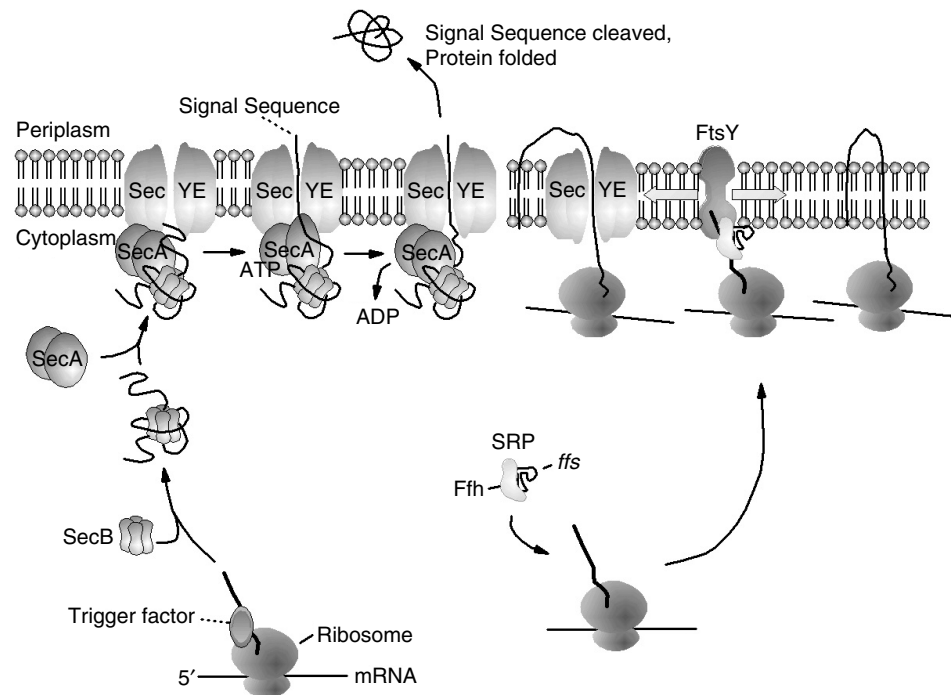


Fig. 2-30. General secretion pathways (Sec dependent, FtsY dependent). SRP, signal recognition particle.

membrane insertion (see “Sec-Dependent Protein Translocation”). On the other hand, SRP, composed of a 54 kDa protein called Ffh (with GTPase activity) and a 4.5S RNA molecule encoded by *ffs*, interacts with integral inner membrane precursor proteins (not presecretion proteins) by binding to their very hydrophobic, noncleaved signal anchor sequences (see SRP database http://carrier.gnf.org/dbs/html/gkd023_gml.html). SRP delivers these precursor proteins, still being translated, to a membrane-bound receptor called FtsY. Secretion from this point occurs while the protein is still being translated (**cotranslational export**). Binding of Ffh to FtsY, a protein–protein interaction actually mediated by *ffs* RNA, greatly enhances the FtsY GTPase activity. Once bound to FtsY, translation of the nascent protein resumes, and, depending on the signal sequence, the protein, as it is translated, has two possible fates. Some FtsY-bound nascent proteins are delivered to the SecYE translocon for insertion into the membrane (see “Sec-Dependent Protein Translocation”). Other FtsY-bound nascent proteins can be cotranslationally inserted directly into the cytoplasmic membrane in a Sec-independent manner.

Sec-Dependent Protein Translocation. The central feature of this system is the SecYE translocon located in the cytoplasmic membrane (Fig. 2-30). Two protein delivery systems converge at this translocon, one of which delivers proteins that will be secreted to the periplasm while the other escorts proteins destined for inner membrane insertion. Signal recognition particle FtsY interacts with SecY, as already noted, in a process that will insert proteins into the membrane. In contrast, proteins with cleavable signal sequences (thus secreted into the periplasm rather than inserted

into the membrane) interact with trigger factor on the ribosome and are completely translated into the cytoplasm, where they bind to a chaperone called SecB. SecB is a piloting protein that delivers the fully translated presecretory protein in an unfolded state to the SecA ATPase associated with the SecYE translocon. Upon binding of ATP and the preprotein, SecA deeply inserts into the membrane at the SecYE translocon. Part of SecA actually inserts into the SecYE channel.

Subsequent ATP hydrolysis leads to disinsertion of SecA after which SecA can rebinding ATP, precursor protein, and SecY, then reinsert into the SecYE channel, promoting the translocation of the next 20 to 30 amino acids of the preprotein. Following ATP-dependent insertion of the precursor protein into the translocon, proton motive force alone can energize translocation of the remaining polypeptide chain (see Chapter 9). Other components of the system have been identified (SecDF YajC), but their functions are unclear and their roles are dispensable. It should be noted that SecYE channel serves a dual purpose in mediating translocation of secretory proteins to the periplasm (SecA dependent) and the insertion of membrane proteins.

During or immediately following translocation, a **signal peptidase** (LepB) will cleave the signal sequence of the exported protein if it is destined for the periplasm, outer membrane, or external environment. Anchor signal sequences of integral cytoplasmic membrane proteins are usually not cleaved by LepB. Table 2-7 summarizes many of the genes involved with the basic protein translocation system.

Twin Arginine Translocation (TAT). Proteins such as TorA (molybdoenzyme trimethylamine N-oxide reductase, a component of an anaerobic respiratory chain) contain a twin arginine motif in their N-terminal signal sequence (RRXFXK). These proteins are secreted in a Sec-independent fashion using the twin arginine translocase comprised of integral membrane proteins TatA, B, C, and E. In contrast to the Sec translocase, which is ATP driven (see above), transport via TAT is dependent on proton motive force (see Chapter 9). This is an example of a Sec-independent protein export system that also does not rely heavily on SRP. The Tat system is unique because it can transport prefolded, active proteins across the inner membrane; the Sec-dependent system only transports unfolded proteins. The function of the Tat proteins, especially how they contribute to the formation of a pore and whether they are involved in recognizing Tat signal sequences, is currently not known.

Traveling to the Outer Membrane. Outer membrane proteins such as OmpA or the porins OmpC and F are first deposited in the periplasm and then make their way to the outer membrane. How they do this remains a matter of debate. There are basically two schools of thought. The first is that these proteins weave their way, perhaps associated with some form of pilot protein, through the periplasm and peptidoglycan to the outer membrane. The second model is that they do not really travel through the periplasm but through zones of adhesion between the inner and outer membranes called Bayer junctions. However, the existence of Bayer junctions has been hotly contested. Regardless of the path, OMPs must contain information that direct them to the outer membrane. The nature of that information has remained elusive. The studies seem to suggest that the overall structure of OMPs is responsible for their targeting to the outer membrane. Whether other factors exist remains unknown.

TABLE 2-7. Components of the *E. coli* Cytoplasmic Membrane Secretory Protein Translocation System and Their Presence in Other Bacteria

Group and Characteristics	Name	Size (kDa)	Location	Presence in Other Bacteria
Secretory chaperonins (pilot protein ^a)	SecB	18	Cytoplasm	Widespread in members of Enterobacteriaceae
General chaperones	DnaK	69	Cytoplasm	Universal
	GroEL	62	Cytoplasm	Universal
	GroES	11	Cytoplasm	Universal
Secretory ATPase (pilot protein)	SecA	102	Cytoplasm, ribosome, peripheral cytoplasmic membrane	<i>Bacillus subtilis</i> , widespread in members of Enterobacteriaceae
Translocase	SecD	67	Cytoplasmic membrane	<i>Brucella abortis</i> , archaea
	SecE	14	Cytoplasmic membrane	<i>B. subtilis</i>
	SecF	39	Cytoplasmic membrane	<i>B. subtilis</i> , archaea
	SecY	48	Cytoplasmic membrane	<i>B. subtilis</i> ^b
Signal peptidases	LepB ^c	36	Cytoplasmic membrane	<i>Salmonella enterica</i> , <i>Pseudomonas fluorescens</i> , <i>B. subtilis</i> (probably widespread)
	LspA ^d	18	Lipoprotein signal peptidase; cytoplasmic membrane	<i>Pseudomonas fluorescens</i> , <i>Enterobacter aerogenes</i> (probably widespread)
	Ppp ^e	ca. 25	Cytoplasmic membrane	<i>Pseudomonas aeruginosa</i> (PilD/XcpA), <i>Vibrio cholerae</i> (TcpJ), <i>K. oxytoca</i> (PulO), <i>B. subtilis</i> (ComC)
Others ^h	4.5S RNA (ffs)		Cytoplasm	<i>B. subtilis</i> (6S), <i>Thermus thermophilus</i> , <i>P. aeruginosa</i> , <i>Halobacterium halobium</i> (7S), <i>Mycoplasma pneumoniae</i>
	Ffh	48	Cytoplasm	<i>Streptococcus mutans</i> , <i>Neisseria gonorrhea</i> , others
	FtsY	54	Cytoplasmic membrane	<i>Sulfolobus solfataricus</i> , <i>B. subtilis</i> , Archaeobacteria, others

^aSee text.^bGenes coding for proteins with >20% overall sequence identity to SecY have been cloned and sequenced from a variety of different sources, ranging from a plastid genome to *Saccharomyces cerevisiae*. Only in *S. cerevisiae* and *B. subtilis* has any direct participation in protein traffic been demonstrated.^cAlso called leader peptidase or signal peptidase I.^dAlso called prolipoprotein signal peptidase, lipoprotein signal peptidase, or signal peptidase II.^ePrepilin; the gene-based designation varies according to the organism from which it is derived and its function.

Secretion of Proteins Across the Outer Membrane

Type I Secretion: ABC Transporters. This is the simplest of the protein secretion systems. It consists of just three protein components: an outer membrane channel, an inner membrane ABC (ATP-binding cassette) transporter, and a periplasmic protein attached to the inner membrane. This system generates no periplasmic intermediates because there is a continuous channel from the cytoplasmic side of the inner membrane to the extracellular side of the outer membrane. The inner membrane and periplasmic components of type I secretion systems are substrate specific, but the outer membrane component, TolC, couples to many different systems to transport a variety of substrates ranging from the very large *E. coli* HlyA hemolysin (110 kDa) to small organic molecules (multidrug efflux). Figure 2-31a illustrates the story with the protein HlyA (*E. coli* hemolysin). The inner membrane transporter (HlyB₂) contains an ATP-binding cassette (the ABC transporter). The periplasmic component is HlyD. Energy is required at two stages in the secretion process. Proton motive force is required early, probably for subunit assembly, and ATP hydrolysis is necessary to secrete HlyA.

TolC contains a β -barrel channel that passes through the outer membrane and a long α -helical tunnel protruding into the periplasm that forms a periplasmic bridge with the inner membrane HlyB transporter. The α -helical channel opens as much as 30 Å once it contacts the inner membrane and periplasmic components of the system. After HlyA has been transported, the three-component system disassembles.

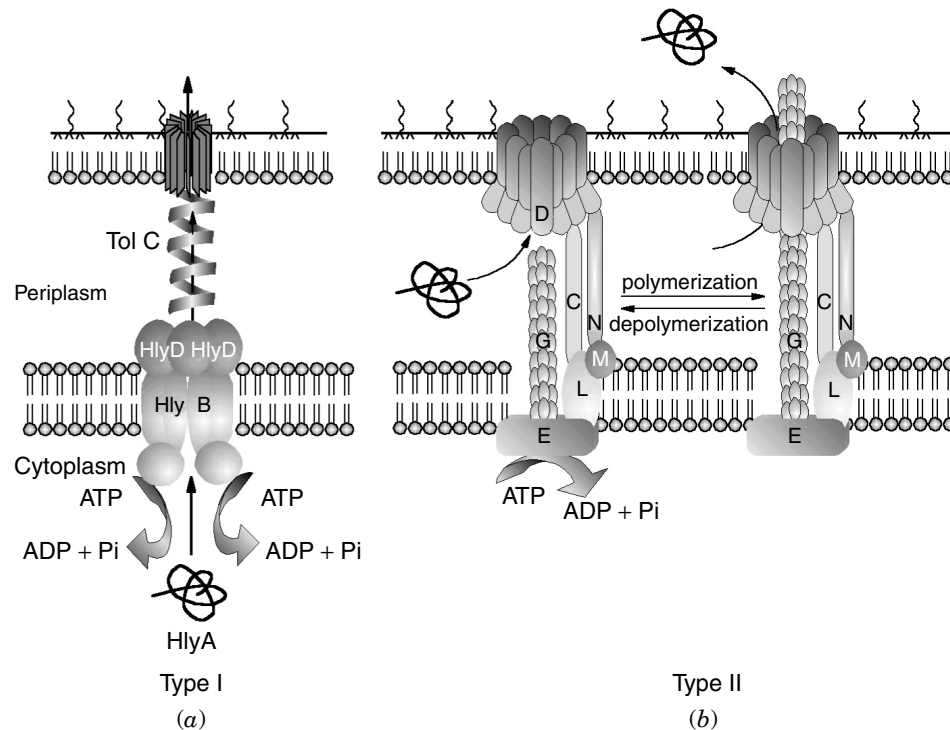


Fig. 2-31. Types I and II secretion systems.

Type II Secretion (Similar to Pilus Assembly). This system is also referred to as the main terminal branch of the general secretory pathway (the Sec-dependent system). Type II secretion differs from other systems in that the substrates consist of folded rather than unfolded proteins (Fig. 2-31). Each system is comprised of at least 12 different components that span the periplasmic space. Type II systems assemble a pilus-like structure, thus sharing many features with type IV pilus biogenesis systems, and are thought to extrude proteins via a piston-like motion of the pilus. Targets for type II secretion have N-terminal signal sequences that allow Sec-dependent translocation across the inner membrane (see above). This is followed by removal of the signal peptide, then folding and release of the mature peptide in the periplasmic space. Type II systems are very specific, and are able to distinguish proteins to be secreted from resident periplasmic proteins. It is not known, however, how type II systems recognize their substrates.

Type III Secretion (Similar to Flagella Assembly). Type III systems are syringe-like structures designed to inject proteins from the cytoplasm of a bacterial cell directly into a eukaryotic cell (Fig. 2-32). They are commonly used by pathogens to deliver toxins and pharmacologically active proteins into the host cell (see Chapter 20). As such, these systems are activated upon contact with the eukaryotic cell. The key distinguishing feature of this class is that the components bear remarkable homology to proteins involved in flagellar assembly. Pathogens that harbor these systems include *Yersinia* species, *Salmonella enterica* serovar Typhimurium, *Shigella flexneri*, Enterohemorrhagic *E. coli*, *Bordetella pertussis*, and *Chlamydia trachomatis*. Proteins secreted by this system are typically escorted to the system by specific chaperone proteins.

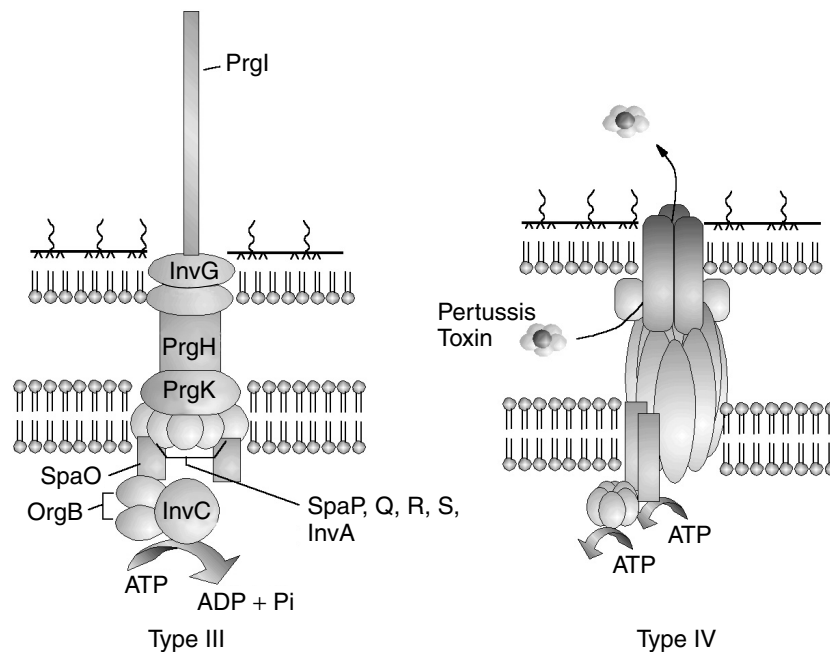


Fig. 2-32. Types III and IV secretion systems.

Type IV Secretion (Related to Conjugation Systems). Conjugation is a highly promiscuous DNA exchange mechanism in which DNA from a donor cell is moved into a recipient cell. The exchange mechanism requires cell–cell contact and an elaborate membrane protein superstructure (see “Conjugation” in Chapter 3). Several pathogens use secretion systems whose subunits are evolutionarily related to those conjugation systems for delivering effector proteins to eukaryotic cells. For example, *Agrobacterium tumefaciens*, a plant pathogen that induces tumorous growth of infected tissues, uses such a mechanism to transfer oncogenic T-DNA and several effector proteins to cell plant nuclei (Fig. 2-32). *Helicobacter pylori*, a causative agent of gastric ulcers, uses this type of system to deliver the 145 kDa CagA protein to mammalian cells.

Autotransporters. Certain extracellular proteins such as gonococcal IgA protease mediate their own transport across the outer membrane (see Chapter 20). Like the type II system, these proteins use the Sec-dependent system to move across the inner membrane followed by cleavage of an N-terminal signal sequence. Once in the periplasm, these proteins can bind to the outer membrane and form their own pores. Autoproteolytic cleavage at the carboxy-terminal end of the surface-bound protein releases IgA into the medium. Other type IV secreted proteins include the vacuolating toxin from *Helicobacter pylori* and the EspC protein produced by enteropathogenic *E. coli*.

Two-Partner Secretion. A collection of large virulence exoproteins produced in *Bordetella pertussis*, *Serratia marcescens*, and *Proteus mirabilis* are secreted by two-partner secretion (TPS) pathways. For each exoprotein there is a specific channel-forming β -barrel transporter protein (TspB). Thus, a noteworthy feature of this secretion pathway is its lack of versatility. Computer-assisted scans of sequenced genomes have identified over 25 potential TspB proteins. Initial transport of the exoprotein across the inner membrane is Sec dependent. The secretion domain of the exoprotein then enters the outer membrane TspB channel and is exported.

PROTEIN DEGRADATION

Degradation of Abnormal Proteins

Bacteria contain a variety of cytoplasmic, periplasmic, and membrane-bound proteases that are critical to the cell (Table 2-8). Under what circumstances would bacteria have to degrade proteins they worked so hard to make in the first place? One class of targets are nonfunctional proteins produced as the result of mutations or misfolding. These abnormal proteins are degraded much more rapidly than their normal counterparts. Stress in the form of high temperature, low pH, or other environmental extremes can also cause partial or complete unfolding of proteins. In order to clear the cell of debris, the levels of several proteases increase in response to conditions such as high temperature (see “Thermal Stress and the Heat Shock Response” in Chapter 18).

Another class of protease targets are timing proteins that regulate complex regulatory pathways. Their functions are required for limited periods of time within a growth cycle. Bacteria with developmental pathways provide striking examples of this class. In addition, proteases often target “sentry sigmas” that direct the expression of stress response genes only in times of trouble (see “Regulated Proteolysis”). These proteins

TABLE 2-8. Proteases and Peptidases of *Escherichia coli* and *Salmonella typhimurium*

	Location	Gene (Map Position)	Nature	Mwt × Subunit	Inhibitors
<i>Proteases</i>					
Do (HtrA or DegP)	Cytosol	<i>degP</i> (3.9)	Serine protease	52,000 × 10	D
Re	Cytosol + periplasm		Serine protease	82,000	D, E, O, TPCK
Mi			Serine protease	110,000	D, E, O
Fa	Cytosol		Serine endoprotease	110,000	D, E, O, TPCK
So	Cytosol		Serine protease	77,000 × 2	D, TPCK
La (Lon)	Cytosol	<i>lon</i> (10)	ATP-dependent serine protease	87,000 × 4	D, E, NEM
Ti(ClpAP)	Cytosol	<i>clpA</i> , <i>clpP</i>	ATP-dependent serine protease	(84,000 × 6) + (21,500 × 14)	
ClpXP	Cytosol	<i>clpX</i> , <i>clpP</i>	ATP-dependent serine protease	(46,000 × 6) + (21,500 × 14)	
ClpYQ		<i>hslU</i> , <i>V</i> (88.8 m)	ATP dependent, proteasome-like	49,000 + 19,000	
HflA	Membrane	<i>hflK</i> , <i>C</i> , <i>X</i> (94.8 m)		45,000, 37,000, 48,000	
HflB (FtsH)		<i>hflB</i> (71.6 m)	ATP-dependent protease	70,000	
II	Cytoplasm	<i>ptrB</i> (41.5 m)	Tryp-like protease	79,000	D, TLCK
III (Pi)	Periplasm	<i>ptrA</i> (63.7 m)	Metalloprotease	107,000	E, O
IV	Inner membrane	<i>sppA</i> (39.8 m)	Signal peptide peptidase	67,000	D
V	Membrane	—	—	—	D
VI	Membrane	—	Serine protease	43,000	
OmpT(VII)	Outer membrane	<i>ompT</i> (12.5 m)	Tryp-like protease	36,000	
	Ci	Cytoplasm	Metalloprotease	54,000	
DegQ	Periplasm	<i>hhoA</i> (72.8 m)	Serine protease	47,000	

<i>Peptidases</i>						
Dipeptidases						
PepD	Cytoplasm	<i>pepD</i> (5.5 m)	Broad specific	53,000		
PepQ	Cytoplasm	<i>pepQ</i> (86.8 m)	X-Pro	50,000		
PepE	Cytoplasm	<i>pepE</i> (91.0 m)	Asp-X	24,000		
Aminotripeptidase						
PepT	Cytoplasm	<i>pepT</i> (25.5 m)	Tripeptides	45,000		
Signal Peptidases						
I	Inner membrane	<i>lep</i> (55 m)	Precursors of secreted proteins	36,000		
II	Inner membrane	<i>lsp</i> (0.5 m)	Prolipoproteins	18,000		—
Amino Peptidases						
PepN	Cytoplasm	<i>pepN</i> (21.3 m)	Aminoexopeptidase, broad	99,000		Amino acids
PepA (PepI)	Cytoplasm	<i>pepA</i> (96.6 m)	Broad	55,000 × 6		E, Zn ²⁺
PepP	Cytoplasm	<i>pepP</i> (65.8 m)	Proline aminopeptidase II, X-Pro-Y	50,000 × 4		E
PepM	Cytoplasm	<i>map</i> (4.1 m)	N-terminal methionine aminopeptidase, Met-X-Y	29,000		E
C-terminal Peptidases						
Dipeptidylcarboxypeptidase	Periplasm	<i>dcp</i> (35 m)	Ac-Ala3	77,000		C

Abbreviations: C, Captopril; D, Diisopropyl fluorophosphate; E, EDTA; O, o-phenanthroline; TPCK, N-tosyl-phenylalanine chloromethyl ketone; NEM, N-ethylmaleimide; Tryp, trypsin.

are constantly synthesized and degraded in the absence of an environmental threat. However, a sudden encounter with stress immediately stops degradation of the sentry regulator, allowing it to immediately direct synthesis of protective proteins. This seemingly wasteful strategy actually allows a very rapid response to potentially lethal threats and helps ensure cell survival, a consideration of paramount importance to the population.

A final set of targets has been called “proteins without partners.” These are individual components of multiprotein complexes. As a complex, these components are usually quite stable, but when the stoichiometry is disrupted by mutation or overproduction of one subunit, rapid degradation of the uncomplexed protein may occur.

Abnormal proteins are progressively degraded into smaller and smaller fragments by a series of proteases starting with ATP-dependent endoproteases such as Lon (see below). The polypeptide products (mwt > 1500) are then degraded to smaller and smaller peptides by endoproteases, tripeptidases, and dipeptidases until they have been reduced to individual amino acids. Table 2-8 lists many of the proteases identified in *E. coli* and *Salmonella*.

Energy-Dependent Proteases

Many proteases in the cell require ATP to function. The purpose of ATP hydrolysis is to help make cytoplasmic proteases selective for specific substrates and to contribute to the unfolding of substrates in a manner that will facilitate their degradation. For example, the **Lon protease** (also called protease LA) is a cytoplasmic ATP-dependent protease that plays a primary role in the degradation of many abnormal proteins as well as unstable regulatory proteins such as the cell division inhibitor Sula (see SOS inducible repair in Chapter 3) and the positive regulator of capsule synthesis RcsA. It is a homotetrameric protein with each subunit carrying an active site serine (**serine protease**). In the presence of ATP, Lon protease will degrade intracellular proteins to acid-soluble peptides usually greater than 1500 daltons. The fact that the Lon protease only recognizes and acts upon unfolded proteins partially explains how intracellular proteases can exist free in the cytosol without destroying essential cell proteins. In addition, ATP hydrolysis results in repeated autoinactivation after each proteolytic event. The ADP remains bound to inactive Lon protease until a new protein substrate induces its release. This phenomenon will help ensure against indiscriminate proteolysis in vivo. The Lon protease also plays a role in the SOS response to DNA damage (see Chapter 3) and other regulatory systems (see Chapter 5).

The ClpA, ClpP, and ClpX proteins comprise two mix-and-match proteases in which the ATPase and protease active sites reside in different subunits. ClpP, the protease subunit, can combine either with ClpA or ClpX ATPase subunits to form active cytoplasmic protease **ClpAP** (protease Ti) or **ClpXP**. These proteases are composed of two seven-member rings of ClpP arranged back to back, forming a proteolytic barrel (see inset Fig. 2-33). This structure is capped, top and bottom, with six-member rings of ClpX or ClpA. The ATPase subunits appear to be the gatekeepers for substrate translocation into the proteolytic chamber. The role of the ClpXP protease in cell physiology is discussed below.

Degradation of Nascent Proteins Permanently Stalled on Ribosomes. A ribosome that translates a damaged or prematurely truncated mRNA molecule will

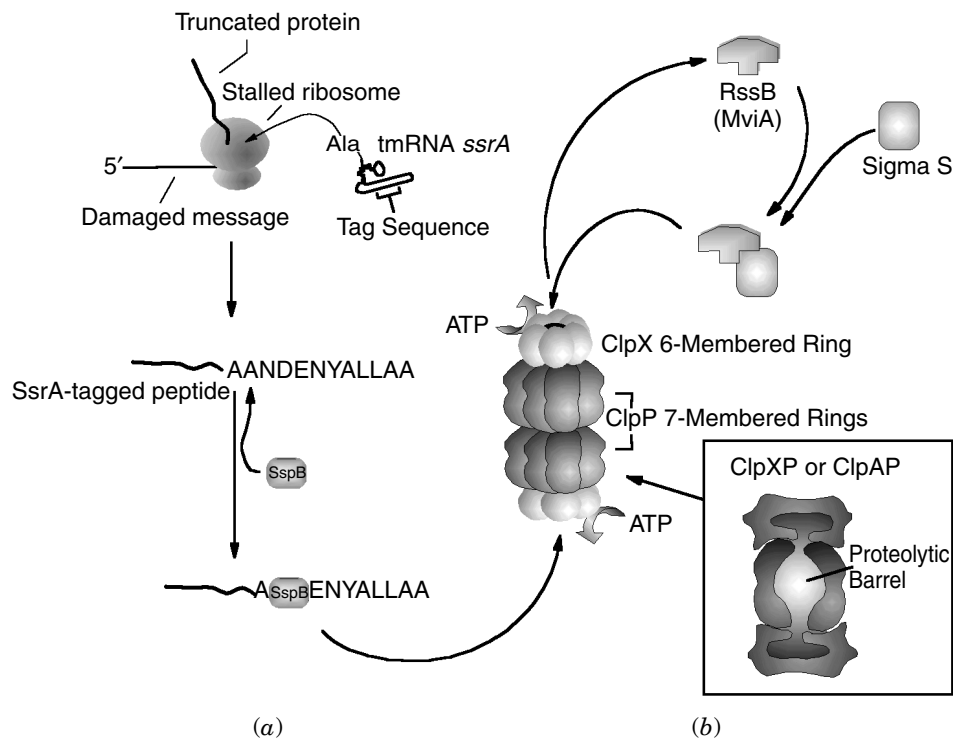


Fig. 2-33. Targeted protein degradation to the ClpXP proteasome. (a) SsrA tagging system. (b) RssB(MviA) recognition of σ^S . Inset shows a cutaway of the ClpXP protease revealing the internal proteolytic barrel.

often find itself at the end of the damaged message with no stop codon in sight. Without a stop codon the ribosome cannot release the message. *E. coli* and other prokaryotes have designed a clever strategy that circumvents this problem. A small RNA molecule called *smpA* directs a new type of RNA molecule to the stalled ribosome. This new RNA is called tmRNA because it has the properties of tRNA and mRNA. The tmRNA encoded by the *ssrA* gene helps the ribosome disengage from the message and targets the faulty protein for degradation (Fig. 2-33). The tRNA part of *ssrA* is charged with alanine, finds the ribosome A site, and triggers peptidyl transferase activity, thus adding alanine to the stuck nascent peptide. The mRNA part of *ssrA* can then be translated by the ribosome. This places a specific amino acid tag (AANDENYALLAA) on the end of the truncated protein. This amino acid tag is recognized by another protein called SspB, which presents the doomed protein to the ClpXP protease for degradation.

Degradation of Normal Proteins. The level of a particular protein in the cell is determined not only by its rate of synthesis but also by its rate of degradation. Degradation of a normal protein, however, is not a stochastic or random process. The cell goes to considerable lengths to regulate which proteins are degraded and when. During starvation for a carbon or nitrogen source, for example, the overall rate of protein degradation increases several-fold. This degradation of preexisting cell proteins can provide a source of amino acids for continued protein synthesis or for a source

of energy. The enhanced proteolysis occurs coordinately with the cessation of stable RNA synthesis and the accumulation of the signal molecule guanosine tetraphosphate (see “Stringent Control” in Chapter 18).

Another role of proteolytic pathways is to confer short half-lives on proteins whose concentrations must vary with time or alterations in cellular state. This contributes to the overall metabolic economy of the cell, since unbridled persistence of a regulatory protein will not allow target genes to react quickly to changing conditions. How is it that some proteins resist degradation whereas others are degraded rapidly? Certain amino acid sequences, conformational determinants, or chemically modified protein structures confer degradation signals, called **degrons**, on proteins. One example is the N-degron, which is the N-terminal residue of a protein. Manifested as the N-end rule, the in vivo half-life of a protein was found to be related to the identity of its N-terminal residue. Several gene products of *E. coli* have been associated with enforcing the N-end rule. For example, the *aat* product is required to degrade proteins with N-terminal Arg or Lys. Mutations in *clpA* will stabilize proteins with N-terminal Phe, Leu, Trp, Thr, Arg, or Lys (see Table 2-8).

Regulated Proteolysis. A recurring theme emerging from proteolysis research is that key regulatory proteins are often presented to a given protease by a specific adaptor or chaperone protein. For example, the sigma factor σS (important to survival under many environmental stress conditions) is rapidly degraded in logarithmically growing cells by the ATPase protease ClpXP. However, the ClpX chaperone does not directly recognize σS . An adaptor protein called RssB (MviA in *Salmonella*) binds to σS and ClpX, presenting σS in a form that ClpX can then feed to the actual protease ClpP (Fig. 2-33). Upon encountering stress, the degradation of σS slows by virtue of some unknown effect on RssB or other component of the RssB/ClpX/ClpP proteosome. Decreased degradation allows σS to accumulate and trigger expression of genes required to survive the stress. This strategy allows the cell to respond quickly to potentially lethal environments by cutting out the need to synthesize the regulator in the first place.

There are many other proteases listed in Table 2-8 that we have not discussed. However, the physiological roles of several of them are explained within other sections of this book.

ANTIBIOTICS THAT AFFECT NUCLEIC ACID AND PROTEIN SYNTHESIS

Much of our knowledge concerning the synthesis of macromolecules was gleaned from studies designed to elucidate the modes of action of several antimicrobial agents. The following section describes some of these agents and their mechanisms of action.

Agents Affecting DNA Metabolism

Intercalating Agents. Many rigidly planar, polycyclic molecules interact with double-stranded DNA by inserting, or intercalating, between adjacent stacked base pairs of the double helix (see Fig. 2-34). To permit this intercalation, there must be a preliminary unwinding of the double helix, providing space between the bases into

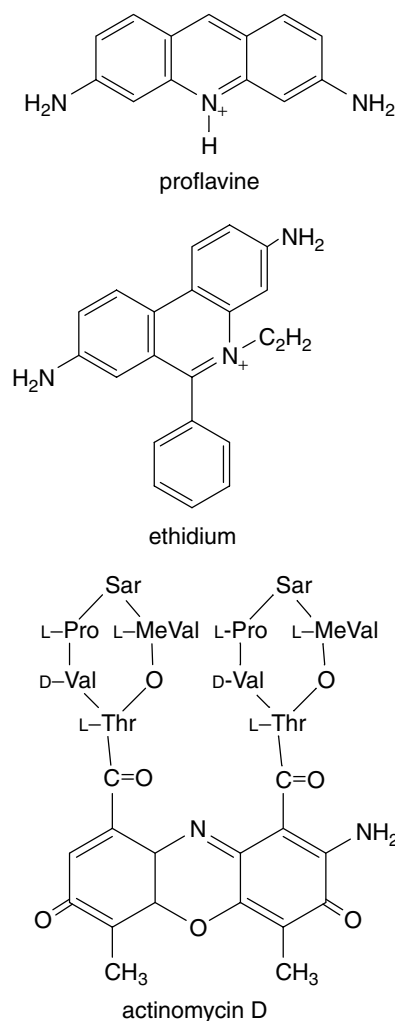


Fig. 2-34. Three agents that intercalate with DNA. Actinomycin D is sometimes called dactinomycin. Thr, threonine; Val, valine; Pro, proline; Sar, sarcosine; MeVal, N-methylvaline.

which the drug can move. Hydrogen bonding remains undisturbed; however, some distortion of the phosphodiester backbone occurs.

Examples of intercalating agents are proflavine, ethidium, and actinomycin D (Fig. 2-34). Treatment of cells harboring extrachromosomal elements called plasmids (Chapter 8) by proflavine or ethidium bromide will often lead to the disappearance of the plasmids from the cells, which is apparently due to the preferential inhibition of plasmid replication.

Actinomycin D inhibits DNA replication but inhibits transcription at lower drug concentrations. DNA must have certain features to interact with actinomycin D: (1) the DNA must contain guanine; (2) the DNA must be a double-stranded helical structure; and (3) the sugar moiety must be deoxyribose. Double-stranded RNA does not interact with this drug.

Mitomycins. The mitomycins are a group of antitumor and antibacterial agents. Mitomycin C (Fig. 2-35) is one of the more active and more widely studied members of this group. It causes cross-linking between complementary strands of DNA but produces a number of toxic side effects in mammalian host tissues, making it impractical as a chemotherapeutic agent against bacterial infections. Several lines of evidence suggest that mitomycin C and related derivatives (e.g., porfiromycin, the ziridine N-methyl analog of mitomycin) form cross-links by bifunctional alkylation of guanine residues in opposing DNA strands. By binding to sites on each of the two complementary DNA strands, replication of a segment of the DNA in the immediate vicinity of the cross-link fails to occur and a loop of DNA is formed. This inhibition of replication appears to be an antimicrobial mode of action of mitomycin C.

DNA Gyrase Inhibitors. Nalidixic acid has been widely used in the treatment of urinary tract infections caused by *E. coli*, *Klebsiella pneumoniae*, *Enterobacter aerogenes*, *Proteus*, and other gram-negative bacilli. Oxolinic acid is a chemically related derivative of nalidixic acid (Fig. 2-35) that has been found to be more potent than nalidixic acid in its antimicrobial action. Both compounds have an identical mode of action. The primary effect of nalidixic acid in intact, living bacteria is an immediate cessation of DNA synthesis. DNA gyrase is the target protein for the action of nalidixic and oxolinic acids. The *gyrA* or “swivelase” component of DNA gyrase is sensitive

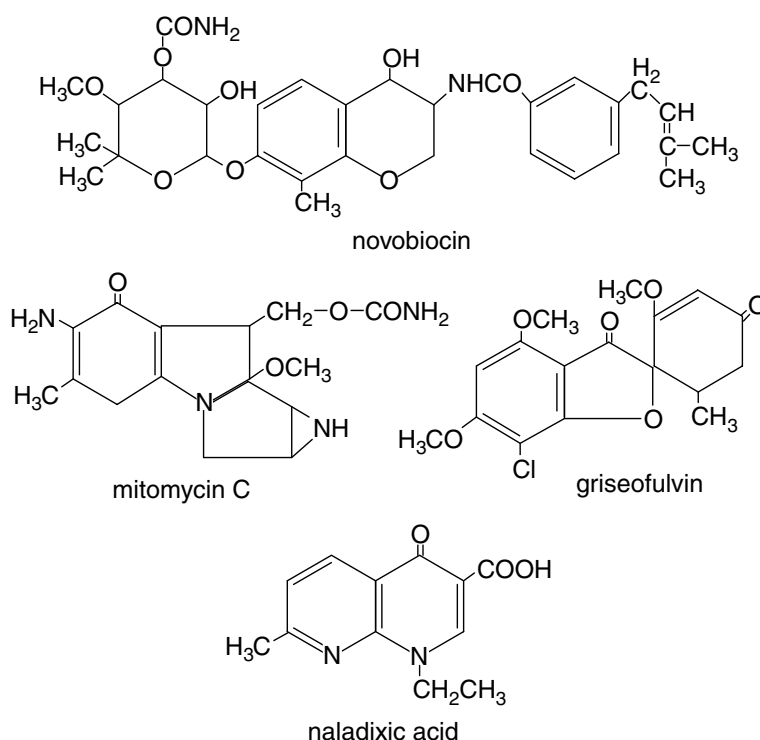


Fig. 2-35. Structures of antimicrobial agents affecting the metabolism, structure, or function of DNA.

to the action of nalidixic and oxolinic acids. DNA gyrase isolated from nalidixic acid-resistant mutants is resistant to the action of nalidixic and oxolinic acids but not to novobiocin or the related compound, coumermycin. Conversely, novobiocin-resistant mutants yield a DNA gyrase that is sensitive to nalidixic acid. Novobiocin binds to the B subunit of DNA gyrase (*gyrB*) and inhibits its ATPase activity.

The **quinolone** antibiotics (e.g., ciprofloxacin) are structurally related to nalidixic acid based on the presence of a quinolone ring. The structural modifications in the quinolones increase their potency and utility against a broad spectrum of microorganisms, both gram positive and gram negative. They are also useful clinically because mutants resistant to nalidixic acid remain quite sensitive to the quinolones.

Agents Affecting Transcription

Rifamycin/Rifampin. The rifampicin or rifamycin group of agents and the streptovaricins are chemically similar (Fig. 2-36). Rifampin and streptovaricin are semisynthetic compounds, while the rifamycins are naturally occurring antibiotics. All of these agents affect the initiation of transcription by specific inhibition of bacterial RNA polymerase (see Fig. 2-15). The RNA polymerase of eukaryotic cells is not affected. Rifamycin forms a tight, one-to-one complex with the β subunit of RNA polymerase (*rpoB*). Rifamycin-resistant variants of *E. coli* contain mutations in *rpoB* that reduce affinity of the protein for the chemical. The drug acts to inhibit initiation of transcription by inhibiting formation of the second phosphodiester bond. Once transcription is allowed to proceed past the third nucleotide (i.e., to the elongation stage), rifamycin is no longer inhibitory.

Streptolydigin. Streptolydigin is similar to rifamycin in that it inhibits transcription by binding to the β subunit of RNA polymerase. However, this antibiotic will inhibit

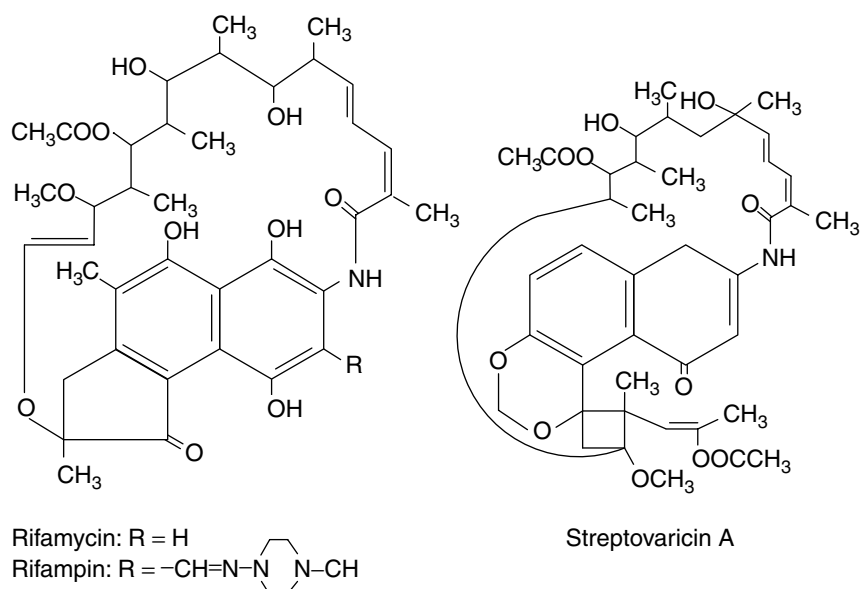


Fig. 2-36. Comparison of the structures of rifamycins and streptovaricins.

chain elongation as well as the initiation process in vitro. In vivo, streptolydigin appears to accelerate the termination of RNA chains. The binding of the drug to RNA polymerase may destabilize the transcription complex, permitting premature termination.

Agents Affecting Translation

Chloramphenicol. One of the simplest antibiotic structures (Fig. 2-37), chloramphenicol was originally isolated as a product of *Streptomyces venezuelae* but is now produced commercially by organic synthesis. Inhibition of peptidyl transferase activity by chloramphenicol is mediated by binding to the 50S subunit of bacterial ribosomes. Its bacteriostatic effect is readily reversed with removal of the drug. The drug binds to several ribosomal proteins including S6, L3, L6, L14, L16, L25, L26, and L27. Protein L16, which is located near the peptidyl transferase active center and forms part of the tRNA acceptor site, is preferentially labeled. This drug does not affect mammalian cells because they contain 60S ribosomal subunits. However, chloroplasts

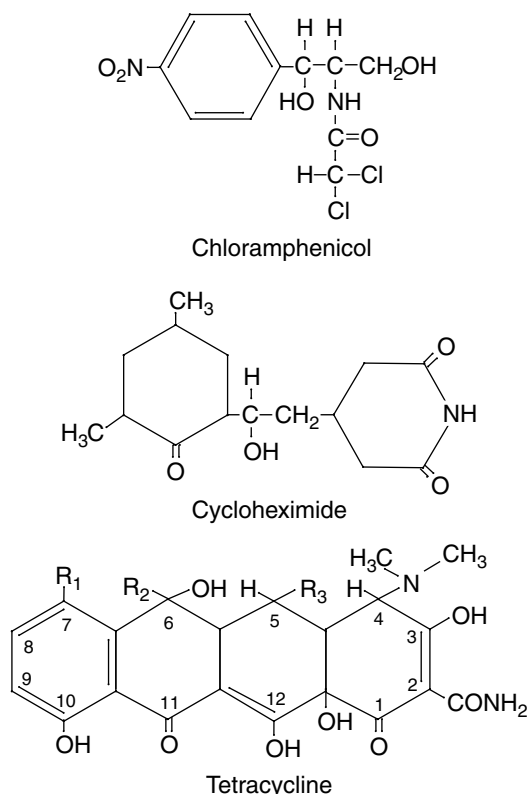


Fig. 2-37. Structures of chloramphenicol, cycloheximide, and tetracycline derivatives. Chloramphenicol bears the chemical name, D-threo-1-p-nitrophenyl-2-dichloroacetamido-1,3-propanediol. Cycloheximide is B-2-(3,5-dimethyl-2-oxocyclohexyl)-2-hydroxyethyl-glutarimide. In tetracycline (R_1 , H; R_2 , CH_3 ; R_3 , H). In chlortetracycline (R_1 , Cl; R_2 , CH_3 ; R_3 , H). In oxytetracycline (R_1 , H; R_2 , CH_3 ; R_3 , OH). In demethylchlor-tetracycline (R_1 , Cl; R_2 , H; R_3 , H.)

and the mitochondria of mammalian cells and lower eucaryotes contain 70S ribosomes comparable to those of prokaryotes. Thus, chloramphenicol also affects protein synthesis in these eukaryotic organelles. This finding explains the toxicity of chloramphenicol toward mammalian cells. Bone marrow toxicity, resulting in aplastic anemia, provides a strong contraindication for the routine use of chloramphenicol, particularly in situations where alternatives in the choice of a therapeutic agent are available.

Cycloheximide (Actidione). Cycloheximide (Fig. 2-37) inhibits protein synthesis in eukaryotic cells (yeasts, fungi, higher plants, and virtually all mammalian cells), but not in prokaryotic organisms (bacteria), by interfering with the activity of cytoplasmic ribosomes but not the ribosomes in mitochondria. The mode of action of cycloheximide resembles that of chloramphenicol. Cycloheximide binds to the 80S ribosomes of eukaryotic cells and prevents ribosome movement along the mRNA. The general effectiveness of cycloheximide on eukaryotic cells of all types accounts for the lack of applicability to the treatment of fungal infections, despite its high *in vitro* activity against a wide variety of yeasts and molds.

Tetracyclines. This group of antibiotics exhibits broad-spectrum activity (bacteriostatic) against both gram-positive and gram-negative bacteria as well as rickettsiae, chlamydiae, and mycoplasmas (Fig. 2-37). The mechanism of action of all the tetracyclines is based on their ability to prevent the binding of aminoacyl-tRNA to the A site on the 30S ribosome. Although the antibiotic can bind to many sites on both ribosomal subunits, the strongest binding occurs to S7, which helps define an area involved in contact between the two subunits.

Resistance to tetracycline is plasmid (R factor) mediated in many bacteria (e.g., *E. coli*, *S. aureus*). Plasmid-mediated tetracycline resistance is an inducible property that prevents the active accumulation of the drug in the bacterial cell by actively pumping the compound out of the cell. In the enteric bacteria, expression of tetracycline resistance following induction appears to result from the insertion of one or more proteins into the cell envelope. Similar proteins have been reported to be present in *S. aureus*.

Macrolides. This large group of antimicrobial agents includes angolamycin, carbomycin (magnamycin), chalcomycin, erythromycin, kujimycin, leucomycin, macrocin, megalomycin, and several others including oleandomycin and spiramycin. All contain a large lactone ring (an aglycone) containing from 12 to 22 carbon atoms with few or no double bonds and no nitrogen atoms. Erythromycin A, more widely recognized because of its extensive clinical application, is shown in Figure 2-38. The macrolide antibiotics are active mainly against gram-positive bacteria and have relatively limited activity against gram-negative bacteria. All the macrolides are inhibitors of protein synthesis in bacterial but not in eukaryotic systems, as a consequence of their interaction with the 50S ribosomal subunit. Erythromycin binds at or near the peptidyl transferase center by blocking peptide bond formation when the donor site is occupied by a peptidyl moiety of a certain length. Erythromycin has been shown to bind to protein L15 and to 23s rRNA with contacts at the central loop of domain V and residues in domain II, which form part of the peptidyltransferase center. It is proposed that macrolide antibiotics stimulate the disassociation of peptidyl-tRNA from the ribosome by triggering

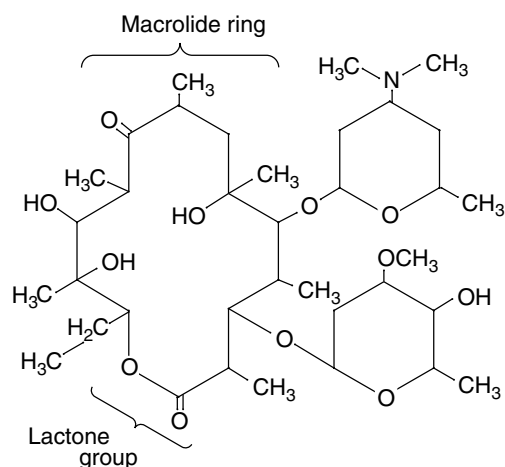


Fig. 2-38. Basic structure of macrolides. Structure of erythromycin A is shown.

an abortive translocation step. It is known that erythromycin is bacteriostatic in its action, and microbial resistance develops rather readily.

Resistance to erythromycin in *E. coli* depends on an alteration in L4 or L12 of the 50S ribosomal subunit, causing reduced affinity for the drug. In *S. aureus*, however, the 23S rRNA of the 50S subunit is modified by an inducible, plasmid-mediated ribosomal RNA methylase.

Aminoglycosides. This large group of antibiotics includes the streptomycins, gentamicins, sisomicin, amikacin, kanamycins A and B, neomycins, paromomycins, tobramycins, and spectinomycins. These agents are all chemically similar (Fig. 2-39). The ultimate mechanism of action is considered to be through binding to ribosomes, causing misreading of the amino acid code. In fact, at low concentrations, streptomycin can suppress some missense mutations by allowing misreading of the mutant codon. Streptomycin partially inhibits the binding of aminoacyl-tRNA and peptide synthesis. The effect of streptomycin on ribosome activity appears due to interactions with 16S rRNA.

Resistance (*rpsL* or *strA*) appears to develop through alteration of the S12 protein of the 30S ribosomal subunit in such a manner that the drug can no longer bind to the 16S rRNA. The drug clearly does not bind to S12. Streptomycin dependence has been explained on the basis of a mutationally altered site in S12 that functions normally only with the addition of streptomycin. Other aminoglycosides apparently are sufficiently dissimilar in their action to have little effect on these dependent mutants. Mutations of another type may render microorganisms resistant on the basis of altered permeability to streptomycin or other aminoglycosides.

Resistance to the aminoglycosides may also develop as a result of the ability to produce enzymes that inactivate various derivatives by the addition of phosphate groups (aminoglycoside phosphotransferases), the addition of nucleotide residues (aminoglycoside nucleotidyltransferases), or the acetylation of various positions (aminoglycoside acetyltransferases). These inactivating enzymes do not react equally with all aminoglycoside derivatives. For this reason, it may be possible to substitute one compound for

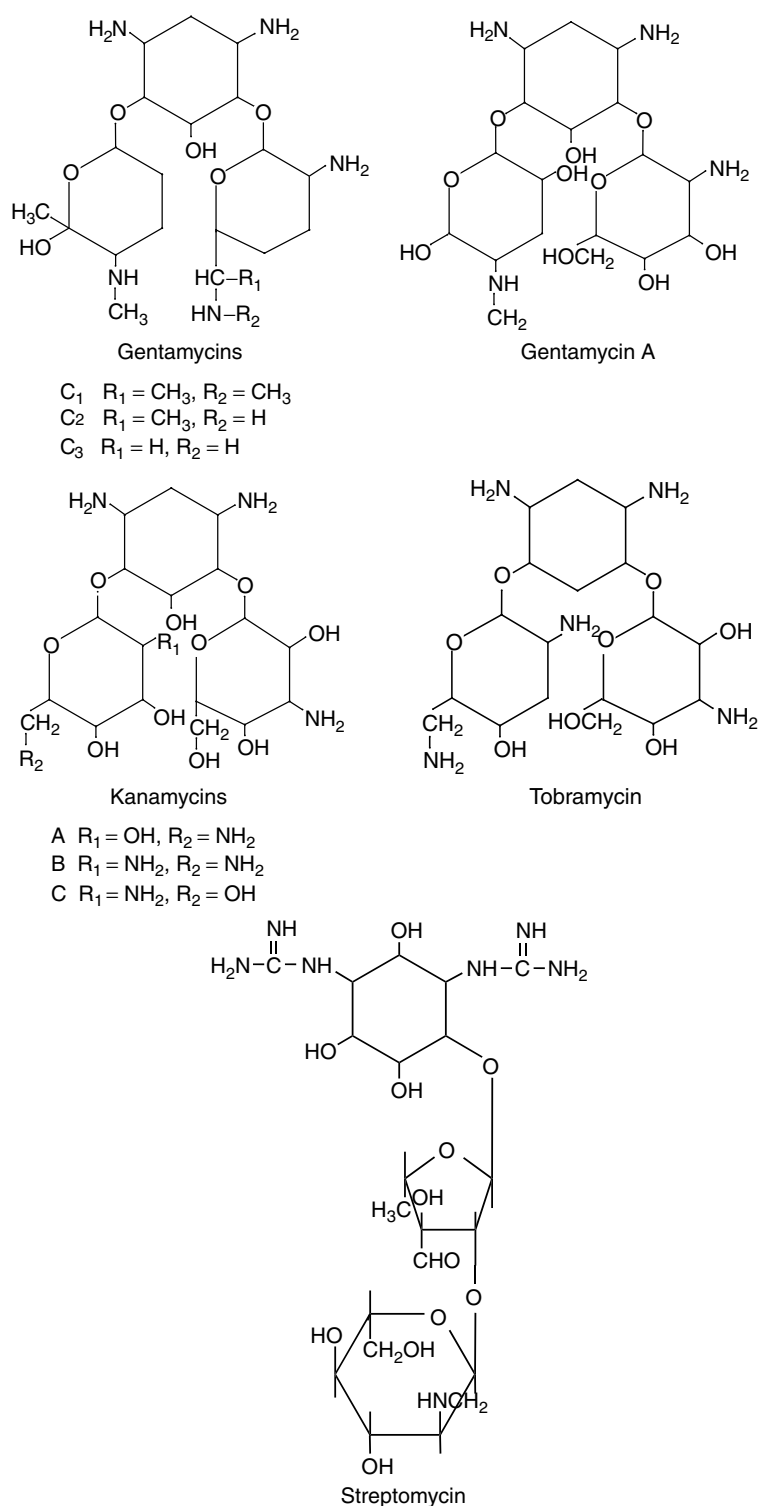


Fig. 2-39. Comparison of structures of aminoglycosides.

another. The aminoglycosides exhibit a broad spectrum of activity, being only slightly more effective against gram-negative bacteria than against gram-positive bacteria.

Kasugamycin is an aminoglycoside that acts on the 30S subunit of 70S ribosomes. It inhibits protein synthesis, but it does not induce misreading, nor can it cause phenotypic suppression, as does streptomycin. Kasugamycin-resistant mutants (*ksgA*) are unusual in that they have an altered rRNA instead of protein. The mutation actually affects the activity of an enzyme (S-adenosylmethionine-6-*N'*, *N'* adenosyl (rRNA dimethyltransferase) that specifically methylates two adenine residues to dimethyl adenine in the sequence AACCUG near the 3' end of 16S rRNA, the region that interacts with the Shine-Dalgarno region of mRNAs.

Lincomycin. Lincomycin and a chemically related derivative, clindamycin (Fig. 2-40), inhibit peptidyl transfer by binding to 23S rRNA in the 50S ribosomal subunit. These agents are effective against both gram-positive and gram-negative organisms. Clindamycin is especially effective against *Bacteroides* and other anaerobes because of its ability to penetrate into deeper tissues.

Fusidic Acid. Fusidic acid is a member of the steroidal antibiotics (Fig. 2-41). It inhibits the growth of gram-positive but not gram-negative bacteria. The failure to act upon gram-negative bacteria may be due to the inability of fusidic acid to penetrate these cells, since it will inhibit protein synthesis from gram-negative ribosomes in vitro. Addition of fusidic acid to 70S ribosomes in vitro prevents the translocation of peptidyl-tRNA from the acceptor to the donor site. This drug also inhibits the EFG-dependent cleavage of GTP to GDP. The data point to EFG as the target for fusidic acid. The drug forms a stable complex with EFG, GDP, and the ribosome that is unable to release EFG for subsequent rounds of translocation and GTP hydrolysis.

Puromycin. Puromycin inhibits protein biosynthesis by forming a peptide bond with the C-terminus of the growing polypeptide chain on the ribosome, thereby prematurely terminating the chain. A structural similarity between puromycin and the terminal 3' aminoacyl adenosine moiety of tRNA (Fig. 2-42) allows puromycin to bind to the ribosomal A site where it triggers peptidyl transferase activity, linking the amino group of puromycin to the carbonyl group of peptidyl-tRNA.

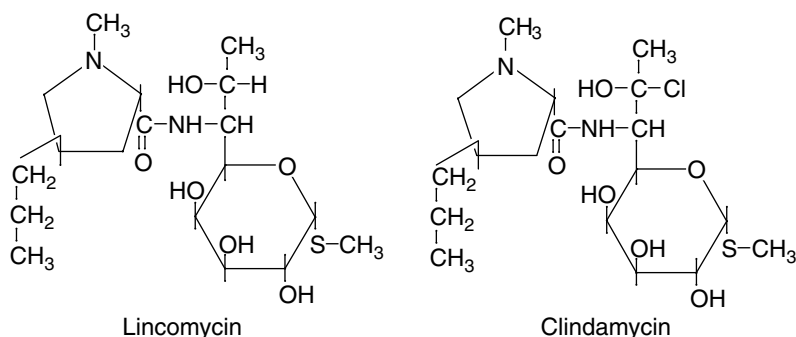


Fig. 2-40. Lincomycin and clindamycin structures.

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CHAPTER 3

BACTERIAL GENETICS: DNA EXCHANGE, RECOMBINATION, MUTAGENESIS, AND REPAIR

The plasticity of bacterial genomes is nothing short of amazing! The processes of DNA exchange, replication, and repair are major contributors to gene shuffling and critical players in the evolutionary history of present-day bacteria. The molecular footprints of ancient DNA exchanges between diverse genetic species are evident in most, if not all, of the bacteria for which complete DNA sequences are known. In addition to their evolutionary significance, these processes are crucial for bacterial survival in natural environments where chemical insults constantly threaten to damage DNA and destroy chromosome integrity. The goal of this chapter is to describe these processes, ultimately placing them in the context of evolution and survival.

TRANSFER OF GENETIC INFORMATION IN PROKARYOTES

Genetic recombination is the production of new combinations of genes derived from two different parental cells. Several processes have been described by which prokaryotic cells can exchange genetic information to yield recombinants. They are **conjugation**, **transformation**, and **transduction**. Conjugation requires physical contact between two cells of opposite mating type, resulting in the transfer of DNA from the donor cell to the recipient cell. The transfer of cell-free DNA into a recipient cell is called transformation, whereas transduction involves bacteriophage-mediated transfer of genetic information from a donor cell to a recipient cell.

In the following section, genetic exchange and DNA recombination processes are considered in detail. Often, only a small portion of the donor cell genome is transferred to the recipient cell. The recipient cell temporarily becomes a partial diploid (**merodiploid**) for the short region of homology between the transferred DNA and the DNA of the recipient chromosome. The ultimate fate of the donated DNA, whether it becomes incorporated into the recipient genome by recombination processes, degraded

by nucleases (host restriction), or maintained as a stable extrachromosomal fragment, occupies a considerable portion of the ensuing discussion.

PLASMIDS

Bacteria are sexually active organisms! This ability to transfer DNA through cell–cell contact was first discovered in the 1950s using *Escherichia coli*. Transfer of genetic information between different *E. coli* strains was found to depend on the presence in some cells (called donor cells) of a small “extra” chromosome called F (fertility) factor. F factor encodes the proteins necessary for the sexual process and is discussed in detail in this chapter. Subsequent to the discovery of F, many other extrachromosomal DNA elements, called plasmids, were discovered. All plasmids share some common features. They are generally double-stranded, closed circular DNA molecules capable of autonomous replication (i.e., independent of chromosomal replication). One exception to circularity is the linear plasmids found in the Lyme disease bacterium *Borrelia burgdorferi*.

Some plasmids, called episomes, commonly integrate into the bacterial chromosome. F factor is one example of an episome. The size of plasmids can range from 1 to 2000 kb. Compare this to the size of the *E. coli* bacterial chromosome, which is approximately 4639 kb. A plasmid that can mediate its own transfer to a new strain is called a **conjugative** plasmid, whereas one that cannot is referred to as **nonconjugative**. Plasmids that have no known identifiable function other than self-replication are often referred to as cryptic plasmids. However, they are cryptic only because we have not been clever enough to elucidate their true function.

It is important to examine and understand plasmids because of the enormous role they play in biotechnology and medicine. Plasmids are the workhorses of recombinant DNA technology, serving as the vehicles by which individual genes from diverse organisms can be maintained separate from their genomic origins. In addition, many plasmids harbor antibiotic resistance genes. Promiscuous transfer of these plasmids among diverse bacterial species has led to the rapid proliferation of antibiotic-resistant disease-causing microorganisms.

Partitioning

Why have plasmids been maintained throughout evolution? One reason is that they often provide a selective advantage to organisms that harbor them, such as toxins to kill off competing organisms (e.g. Colicins) or resistance genes to fend off medical antibiotics. On a more basic level, the reason plasmids persist in a population is due to their ability to partition. **Partitioning** assures that after replication each daughter cell gets a copy of the plasmid. For plasmids present in high copy numbers (50 to 100 copies per cell; e.g., pBR322), random diffusion may be enough to get at least one copy of the plasmid to each daughter cell. However, random segregation of low-copy-number plasmids (only 2 to 4 copies per cell) would likely mean that, following cell division, one of the daughter cells would not receive a plasmid. The plasmid would eventually be diluted from the population. Consequently, regulated partitioning mechanisms are essential for these plasmids. The mechanism used for partitioning differs depending on the plasmid.

Partitioning, especially of low-copy-number plasmids such as F factor and P1 (see below), usually involves genes called *par* organized on plasmids as gene

cassettes. Almost all plasmid-encoded *par* loci consist of three components: a cis-acting centromere-like site and two trans-acting proteins that are usually called Par A and B, which form a nucleoprotein complex at the centromere-like site. **Cis-acting** means that, for a plasmid to replicate, the DNA sequence in question must be part of that plasmid. It does not encode a diffusible product. In contrast, a **trans-acting** replication gene encodes a protein that does diffuse through the cytoplasm. Thus, the gene does not have to reside on the plasmid to help that plasmid replicate.

The upstream gene in a typical *par* operon encodes an ATPase essential to the DNA segregation process. The downstream gene encodes the protein that actually binds to the centromere. The organization of the partitioning loci in F factor and P1 are the same, but the genes are called *par* and *sop*, respectively. ParB/SopB proteins bind as dimers to the downstream *parS/sopS* centromere-like regions. ParA /SopA (the ATPases) then bind through protein–protein interactions with the B proteins.

Genetic fusions between the Par proteins and green fluorescent protein (GFP) from the jellyfish *Aequorea victoria* have been used to visualize the location and movement of plasmids in the cell during segregation. The data suggest that the plasmids are intrinsically located at midcell where they replicate. After replication, the daughter plasmids pair with ParA and B proteins at the *parC/sopC* site, then move from midcell to quarter-cell positions, ensuring each daughter cell receives a plasmid copy. This movement presumably requires interaction with a host-encoded mitotic-like apparatus. What comprises that apparatus is not known.

Incompatibility

Incompatibility is a property of plasmids that explains why two very similar yet distinct plasmids might not be maintained in the same cell. Two plasmids that share some regulatory aspect of their replication are said to be incompatible. For example, if two different plasmids produce similar repressors for replication initiation, then the repressor of one could regulate the replication of the other and vice versa. The choice as to which plasmid will actually replicate is random. So, in any given cell, copies of one plasmid type could outnumber copies of a second plasmid. Because of this incongruity, cell division could result in a daughter cell that only contains one of the two plasmid types. This is called **segregational incompatibility**.

Nonconjugative, Mobilizable Plasmids

Many plasmids that are nonconjugative by themselves nevertheless possess a system that will allow conjugal transfer when present in the same cell as a conjugative plasmid. The colicin E1 (ColE1) plasmid is an example of a nonconjugative, mobilizable plasmid. It has a *dnaA*-independent origin of replication but contains a site called *bom* (basis of mobility) that functions much like *oriT* of the F factor (see below). While some ColE1 genes are required for mobilization (*mob* genes), transfer will not occur unless some functions are provided by F factor. Not all conjugative plasmids will fill this role, however. Where members of the incompatibility groups IncF, IncI, or IncP conjugative plasmids will efficiently mobilize ColE1, members of IncW will do so inefficiently.

Resistance Plasmids

Many extrachromosomal elements have been recognized because of their ability to impart new genetic traits to their host cells. One important factor, resistance (R) factor, was first recognized by the fact that organisms in which it was present were resistant to a number of chemotherapeutic agents. A single R factor may carry traits for resistance to as many as seven or more chemotherapeutic or chemical agents. R factors harbored by organisms in the normal flora of human beings or animals may be transferred to pathogenic organisms, giving rise to the sudden appearance of multiple resistant strains.

There is evidence that R factors did not arise as a direct result of the widespread use of antibiotics. Individuals living in highly sequestered geographic areas who have not been subjected to antimicrobial agents possess normal bacterial flora harboring R factors. Also, organisms maintained in the lyophilized state from time periods prior to the widespread use of antimicrobial agents have been shown to carry R factors. Thus, the origin of these plasmids probably occurred in response to encounters with antimicrobials produced in the natural environment. Regardless of their origins, the fact that R factors are transferred from normal flora to pathogenic microorganisms under natural conditions (e.g., the gastrointestinal tract) generates considerable concern with respect to the widespread and indiscriminant use of antimicrobial agents that could foster the further spread of these plasmids.

Plasmids in Other Bacterial Genera

The association of resistance to chemotherapeutic agents and other genetic traits with extrachromosomal elements has been well established in the Enterobacteriaceae. It is also possible to demonstrate intergeneric transmission of plasmids among members of this closely related group of organisms. Plasmid-associated resistance factors have been well characterized in most bacteria (one notable exception is *Rickettsia*). For example, resistance of *Enterococcus hirae* (previously called *E. faecalis*) to erythromycin and lincomycin is plasmid associated. Self-transferable plasmids encoding hemolysins and bacteriocins as well as multiple antibiotic resistance have also been found in *E. hirae*. As is described in this chapter, transfer of plasmids among the streptococci appears to take place through conjugal mechanisms.

Plasmid Replication

There are three general replication mechanisms for circular plasmids: theta (θ) type, rolling circle, and strand displacement. **Theta (θ)-type replication** has been examined most extensively among circular plasmids of gram-negative bacteria. The mechanism involves melting of the parental strands at a plasmid origin, synthesis of a primer RNA, and then DNA synthesis by covalent extension of that primer. θ -type DNA synthesis can start from one or multiple origins, and replication can be either uni- or bidirectional, depending on the plasmid. Electron microscopy revealed these replication intermediates look like θ -shaped molecules. Initiation generally requires a plasmid-encoded Rep initiator protein that binds to a series of directly repeated DNA sequences within the origin called **iterons** (see below).

Rolling circle replication (RC) is unidirectional. The current model posits that replication initiates by the plasmid-encoded Rep protein, introducing a site-specific nick on one strand (the plus strand) of the DNA. The resulting 3'-OH end serves as

a primer for leading strand synthesis involving host replication proteins (DNA pol III, ssb, and a helicase). Elongation from the 3'-OH end accompanied by displacement of the parental strand continues until the replicome reaches the reconstituted origin at which point a DNA strand transfer reaction takes place to terminate leading strand replication. The result is a double-stranded DNA molecule comprised of the parental minus strand and the newly synthesized plus strand, and a single-stranded DNA intermediate corresponding to the parental plus strand. The generation of this single strand is the hallmark of plasmids replicating by the RC mechanism. Finally, the parental strand is converted into dsDNA by the host replication machinery initiating at a second origin physically distant from the RC origin.

The best-known examples of plasmids replicating by the **strand displacement** mechanism are the promiscuous plasmids of the IncQ family. Replication is promoted by the joint activity of three plasmid-encoded proteins, RepA, B, C, with, respectively, 5'-3' helicase, primase, and initiator activities. RepC binds to iteron sequences within the origin and probably with the RepA helicase, promoting exposure of two small palindromic sequences (*ssi*) located on opposite strands at the origin. DNA synthesis occurs with opposite polarities from these two sites. Stem-loop structures generated by the *ssi* sequences are probably required for assembly of the RepB primase, an event required to initiate replication.

Although we tend to think of plasmids as circular, there are notable examples of linear plasmids. These plasmids have two basic structures. The spirochete *Borrelia* contains **linear plasmids** with a covalently closed hairpin loop at each end. This group replicates via concatameric intermediates. Linear plasmids in the gram-positive filamentous *Streptomyces* have a covalently bound protein attached at each end. Representatives of the second group have a protein covalently bound at their 5' ends and replicate by a protein priming mechanism.

Plasmid Copy Number Control. Plasmids are selfish genetic elements. They must ensure that they are stably inherited despite constituting an energy burden on the host cell. In addition to partitioning systems, plasmids contain mechanisms that maintain a certain number of plasmid copies per cell. Too few copies can result in plasmid loss after division, whereas too many copies resulting from runaway replication will overburden the cell, leading to cell death. As noted earlier, plasmids can be characterized into separate groups based on the number of copies normally present in a cell. F factor is a low-copy-number plasmid whose replication is stringently controlled. ColE1, in contrast, is a high-copy-number plasmid that can be present at 50 to 100 copies per cell. Even though there is a wide difference in their copy number, both types of plasmids have mechanisms designed to limit and maintain copy number.

The ColE1 plasmid undergoes unidirectional replication (one replication fork) and can replicate in the absence of de novo protein synthesis. It does this by utilizing a variety of stable host replication proteins (e.g., Pol I, DNA-dependent RNA polymerase, Pol III, DnaB, DnaC, and DnaG) rather than making its own. Because replication initiation of the bacterial chromosome requires de novo protein synthesis, the addition of chloramphenicol to ColE1-containing cells results in the amplification of the plasmid copies relative to the chromosome. But even though the ColE1 replication system allows for a fairly high copy number per cell, the system is regulated so as not to exceed a given number (Fig. 3-1).

Replication of ColE1 actually begins with a transcriptional event that occurs about 500 bp upstream of the replication origin (*ori*). As the transcript progresses through *ori*,

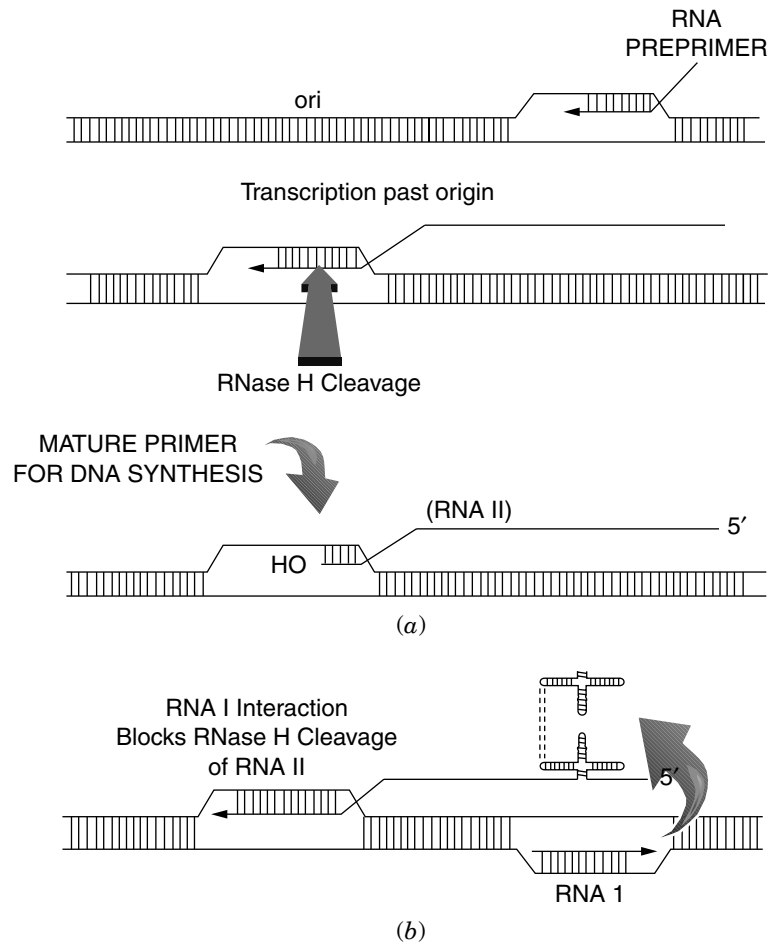


Fig. 3-1. Countertranscript replication control of plasmid ColE1.

an RNA–DNA hybrid molecule forms that is a suitable substrate for the bacterial host enzyme RNaseH. RNaseH cleaves the RNA primer, leaving a 3'-OH end suitable for DNA Pol III and replication. The processed RNA primer is called RNA II. Replication and, thus, copy number, is actually negatively controlled by an antisense RNA, a partial countertranscript of RNA II (100 nucleotide).

The countertranscript (RNA I) is transcribed near the start of the prepriming transcript (the RNA II precursor) but in the opposite direction (Fig. 3-1B). The RNA I and RNA II precursor molecules each contain a significant amount of secondary stem-loop structure, and because the two RNA molecules are complementary, the loop areas can base pair with the help of the plasmid Rom protein. In order for RNA II to prevent replication, it must achieve a concentration that will favor its interaction with the RNA II precursor. It is this aspect of the interaction that allows the plasmid to achieve 20 to 50 copies per cell before RNA I concentrations are high enough to prevent further replication. Following the initial molecular “kiss” of the loop areas, the molecules melt into each other, which extends the base pairing. Formation of the RNA–RNA

complex disrupts the RNA–DNA hybrid at *ori* and interferes with processing by RNaseH, thereby aborting replication initiation.

The RNA I–preprimer complex is now also susceptible to RNaseE (*rnaE*) digestion. RNaseE is part of a complex host RNA degradosome that includes polynucleotide phosphorylase, RNA helicase, and poly(A) polymerase (*pcnB*, as noted in Chapter 2, the addition of poly(A) tails to mRNAs may stimulate degradation). Thus, a lack of RNA I degradation decreases plasmid copy number because the resulting higher RNA I level reduces the number of possible replication events. So, under normal circumstances, when the plasmid copy number in a cell decreases, less RNA I is made and preexisting RNA I is degraded. This allows a greater number of initiation events, and the plasmid copy number will rise. Other plasmid systems (F factor, R1 plasmid) utilize a different type of countertranscript (antisense) control of copy number where the target is the mRNA for a rate-limiting initiator protein (see “Fertility Inhibition”).

Another type of plasmid copy number control is exemplified by **P1**, which is actually a bacteriophage DNA that exists either as a lytic phage or as a low-copy-number plasmid in its prophage state (Fig. 3-2). How does P1 maintain a low copy number as a plasmid? To understand this process you must first know that the P1-encoded RepA protein serves to initiate P1 plasmid replication. RepA protein must bind to sites within a region of the P1 genome called *oriR* in order to trigger P1 replication. Within *oriR* there are five copies of a 19 bp sequence, called an **iteron**, to which RepA can bind. Another nine iterons are found in a separate region called *incA*. Replication will only initiate if RepA binds to the iterons in *oriR*. The nine iterons present at *incA* were initially thought to titrate the level of available RepA to a point insufficient for binding to *oriR*, especially when multiple copies of P1 are present in the cell. The theory was that with multiple copies of P1 in the cell, the number of competing iterons would increase and thus decrease the probability that RepA would fully occupy the *oriR* iterons and initiate replication. However, artificially increasing RepA did not increase copy number.

It now appears that the monomeric forms of RepA that bind to iterons can subsequently bind through protein–protein interactions with other RepA monomer–iteron complexes, even those on sister plasmids. The result, called handcuffing, effectively

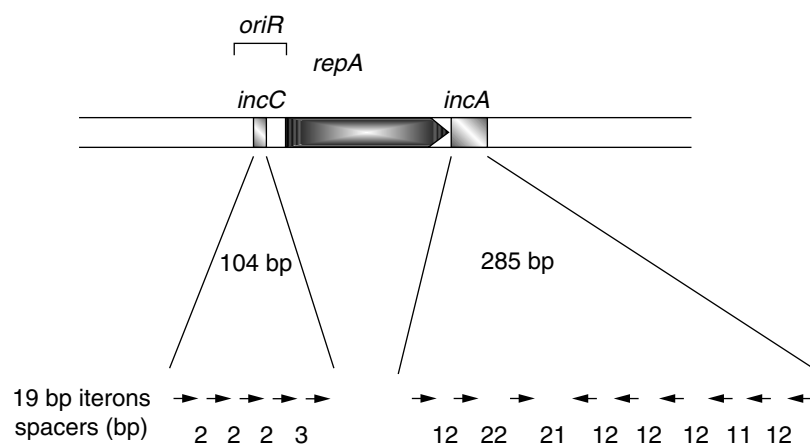


Fig. 3-2. Location of iterons used to maintain P1 copy number.

ties up the plasmids, preventing movement of the replication fork. As the cell volume increases, the RepA dimers disassociate and handcuffing decreases. Since RepA autorepresses its own expression, the increase in cell volume also derepresses *repA*. The resulting increase in RepA monomers leads to full occupancy of iterons in *oriR* and replication ensues, so after cell division, copy number in each daughter cell is maintained. F factor also utilizes an iteron system for copy number control.

Addiction Modules: Plasmid Maintenance by Host Killing: The *ccd* Genes

In addition to the partitioning systems already described, several plasmid systems utilize genes called addiction modules to prevent the development of a plasmid-less population. Low-copy-number plasmids such as F factor, as well as R1, R100, and P1, all have the remarkable ability to kill cells that have lost the plasmid. In F, two plasmid-associated genes are responsible: *ccdA* and *ccdB* (coupled cell division). The *ccdB* product (11.7 kDa) is an inhibitor of cell growth and division. Its activity is countered by the unstable *ccdA* product (8.7 kDa). If plasmid copy number falls below a certain threshold, the loss of CcdA through degradation will unleash the killing activity of CcdB. Affected cells form filaments, stop replicating DNA, and die. This phenomenon is referred to as **programmed cell death**. The population has the appearance of being addicted to the plasmid. The target of CcdB is DNA gyrase. CcdB toxin, in the absence of the CcdA antidote, binds to GyrA, thereby inhibiting DNA gyrase activity. Cell death may be due to DNA cleavage that results from CcdB trapping the DNA gyrase–DNA complex in a poststrand passage intermediate (see Chapter 2). Alternatively, the DNA gyrase–CcdA–DNA complex will inhibit movement of RNA polymerase.

In addition to proteic postsegregational killing systems, there are antisense RNA-regulated systems. The role of antidote is played by a small, unstable RNA transcript that binds to toxin-encoding RNA and inhibits its translation. The best-characterized system of this type is the *hok/sok* locus of plasmid R1. Translation of mRNA encoding the Hok (host-killing) protein is regulated by *sok* (suppressor of killing) antisense RNA. The unchecked production of Hok (52 amino acids) causes a collapse of the transmembrane potential (see Chapter 9).

In the orthodox view of bacteria as single-celled organisms, programmed cell death does not make adaptive sense. However, it is becoming increasingly clear that in nature bacteria seldom behave as isolated cells. They often take on characteristics normally associated with multicellular organisms. Biofilms are one obvious example. In this context, adaptive sense can be made for sacrificing an individual cell that fails to meet the community standards required for survival of the species.

CONJUGATION

Conjugation involves direct cell-to-cell contact to achieve DNA transfer (Fig. 3-3). For this process, certain types of extrachromosomal elements called plasmids are usually required. The prototype conjugative plasmid is the F, or fertility, factor of *E. coli*.

F Factor

F factor is an *E. coli* plasmid (100 kb) with genes coding for autonomous replication, sex pili formation, and conjugal transfer functions. In addition, there are several insertion sequences situated at various sites. F is considered an episome, since it replicates

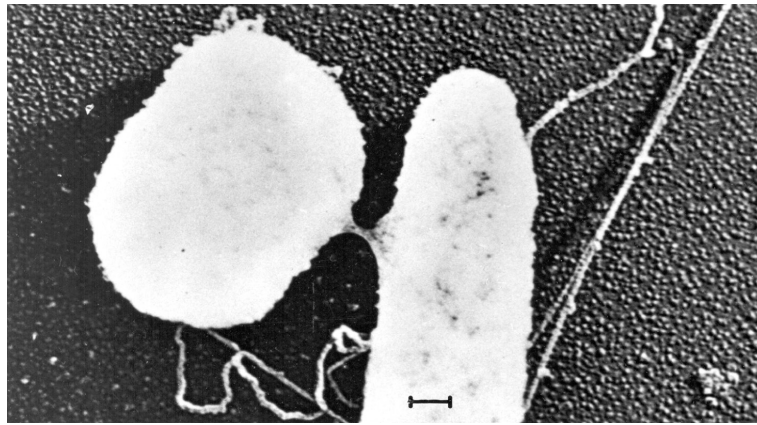


Fig. 3-3. Conjugation: cell-to-cell contact between bacterial cells. The long cell on the right is an Hfr donor cell of *E. coli*. It is attached by the specific F-pilus to a recipient (F^-) on the left. Bar equals 100 nm. (Source: From Anderson et al., 1957.)

either independently of the host chromosome or as part of the host genome. Cells containing an autonomous F are referred to as F^+ cells. Replication of the F factor in this situation requires host proteins but is independent of the *dnaA* gene product (see Chapter 2, initiation of DNA replication). There are only one to three copies of this plasmid per cell; thus, F is an example of a plasmid whose replication is stringently controlled.

Conjugal Transfer Process. A large portion of F-factor DNA is dedicated to the transfer process. Figure 3-4 presents a portion of the genetic map for F encompassing the *tra* region. It is useful to refer to this diagram during the following discussion of F conjugation.

In the Enterobacteriaceae, the presence of specific structural appendages (sex pili) on the cell surface is correlated with the ability of the cell to serve as a donor of genetic material. Of the 25 known transfer genes, 14 are involved with F-pilus formation (*traA*, *-L*, *-E*, *-K*, *-B*, *-V*, *-W*, *-C*, *-U*, *-F*, *-H*, *-G*, *Q*, *X*). *TraA* is the F-pilin protein subunit while all the other genes are involved in the complex process of assembling the sex pilus (see Chapter 7). Sex pilus assembly occurs in the membranous layers of the cell envelope. F-donor cells possess only one to three sex pili. The tip of the pilus is involved in stable mating pair formation (*TraN* and *TraG* involvement), interacting with the *OmpA* protein on the outer membrane of the recipient. Once initial contact between the donor pilus and the recipient is established, the pilus is thought to contract, bringing the cell surfaces of the donor and recipient cells into close proximity. This wall-to-wall contact causes a fusion of cell envelopes and a conjugation bridge that includes a pore formed by *TraD*. Mating mixtures of *E. coli* actually involve aggregates of from 2 to 20 cells each rather than only mating pairs.

Following mating-aggregate formation, a signal that a mating pair has formed is relayed from the *TraD* pore protein to *TraM*, a protein that binds the *TraD* pore, and ultimately to *TraY*, a site-specific DNA-binding protein that binds *oriT* (origin of conjugal transfer). The transfer of F^+ DNA initiates from *oriT* as opposed to *oriV*, the vegetative replication origin used for plasmid maintenance. *TraY* and integration

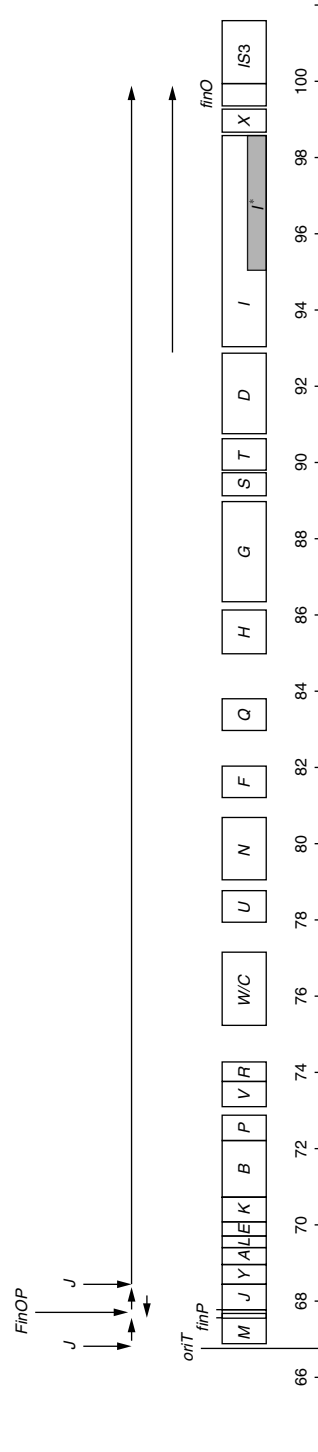


Fig. 3-4. Physical and genetic map of the transfer region of plasmid F. The numbers show kilobase coordinates and the horizontal lines above the genes represent transcripts. The direction of DNA transfer from *oriT* is such that the transfer region is transferred last. The *traM* and *traJ* promoter regions have been sequenced and the *traY-I* operon has been shown to have its own separate promoter. *finP* may be transcribed from the DNA strand opposite the long leader sequence of the *traJ* mRNA. Transcription from the promoters for *traM* and for the *traY* operon is dependent on the product of *traI*, which is in turn negatively regulated by the FinOP repressor. The *traI* and *traI** genes are transcribed constitutively from a second promoter at about 18% of the level from the *traJ*-induced *traY-I* operon promoter. Roles attributed to the genes are regulation, finP and traJ; pilus formation, *traA*, *-L*, *-E*, *-K*, *-B*, *-V*, *-W/C*, *-U*, *-F*, *-Q*, and *-G*; stabilization of mating pairs, *traN* and *traG*; conjugative DNA metabolism, *traM*, *-Y*, *-D*, *-I*, and *I**; surface exclusion, *traS* and *traT*.

host factor (IHF, an *E. coli* DNA-binding protein) change the architecture of *oriT* to a structure suitable for binding the bifunctional TraI endonuclease/helicase. The DNA transfer machinery, known as a **transferosome** or **relaxosome**, is then assembled at this origin. The plasmid-encoded TraI endonuclease/helicase nicks the F plasmid at *oriT*, and the 5' end of the DNA is attached to the protein (Figs. 3-5 and 3-6). The

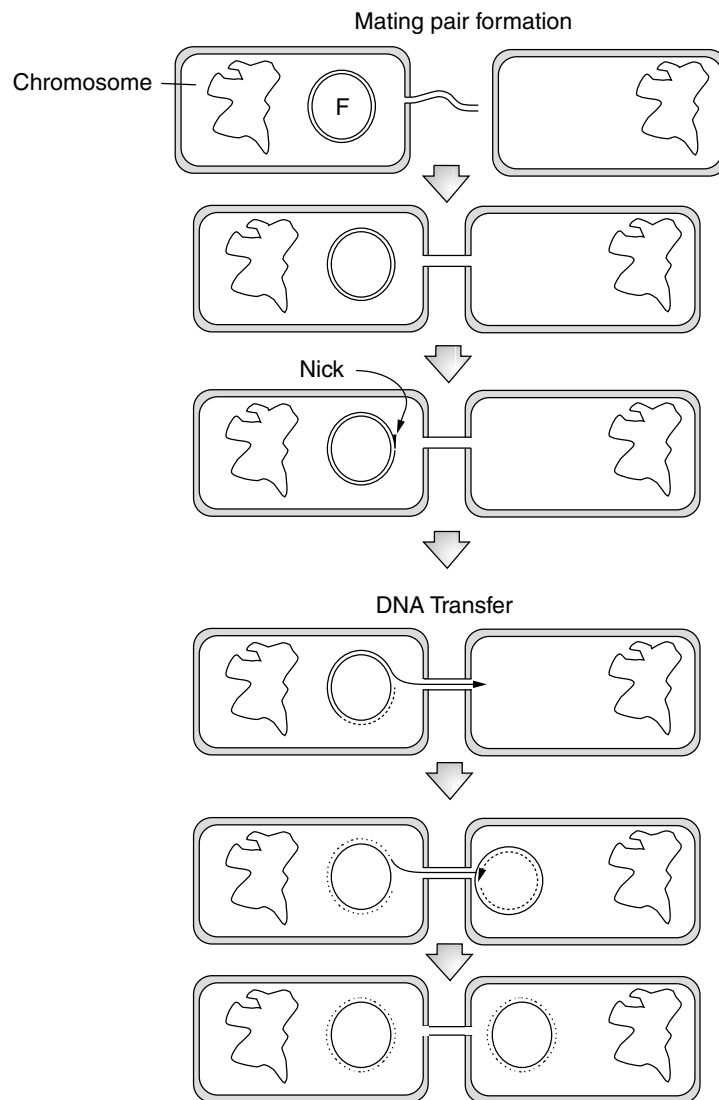


Fig. 3-5. Steps in F plasmid transfer. The process is initiated by cell-to-cell contact, mediated by the F-plasmid-coded pilus on the donor cell. A nick at the origin of transfer site, *oriT*, supplies the 5' terminus that invades the recipient cell (F DNA synthesis can, but need not, occur simultaneously on the intact covalently closed circular molecule of single-stranded DNA). Transfer of a genetic element capable of autonomous replication requires production of the complementary molecule on the transferred strand (continuous DNA synthesis is shown by dashed lines) and circularization.

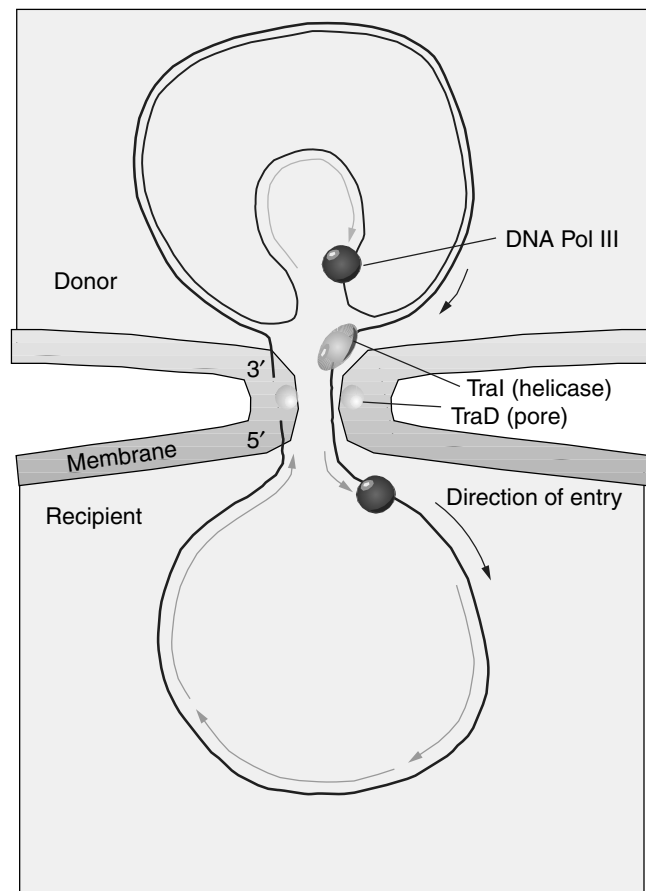


Fig. 3-6. Model for the conjugative transfer of F. A specific strand of the plasmid (thick line) is nicked at *oriT* by the *traI* endonuclease and transferred in the 5' to 3' direction through a pore, involving the TraD protein, formed between the juxtaposed donor and recipient cell envelopes. The plasmid strand retained in the donor cell is shown by a thin line. The termini of the transferred strand are attached to the cell membrane by a complex that includes the endonuclease. DNA helicase I (*traI* gene product) migrates on the strand, undergoing transfer to unwind the plasmid duplex DNA; if the helicase is, in turn, bound to the membrane complex during conjugation, the concomitant ATP hydrolysis might provide the motive force to displace the transferred strand into the recipient cell. DNA transfer is associated with synthesis of a replacement strand in the donor cell and a complementary strand in the recipient cell (broken lines); both processes require de novo primer synthesis and the activity of DNA polymerase III holoenzyme. The model assumes that a single-strand-binding protein coats DNA, to aid conjugal DNA synthesis; depending on the nature of the pore, this protein might even be transferred from donor to recipient cell, bound to the DNA.

nicked DNA, containing TraI as a pilot protein, travels through the TraD pore at the membrane bridge, not through the pilus itself, as originally believed.

The 5'-ended strand is transferred to the recipient via a rolling circle type of replication with the intact strand serving as a template. The *traI* gene product (DNA helicase I) unwinds the plasmid duplex (1000 bp/s), pumping one strand into the

recipient cell. DNA pol III synthesizes a replacement strand in the donor. The 5' end of the strand, upon entering the recipient, becomes anchored to the membrane. As the donor strand is transferred into the recipient cell, it, too, will undergo replication and become circularized. The circularization process does not rely on the host RecA synaptase (see “Recombination”) but may require a plasmid-encoded recombination system. At this point, the recipient becomes a donor cell capable of transferring F to another cell. Figure 3-5 illustrates the overall process of conjugation while Figure 3-6 presents a more detailed analysis of the proteins involved.

Barriers to Conjugation. It has been known for some time that cells carrying an F factor are poor recipients in conjugational crosses (termed **surface exclusion** or entry exclusion). A related phenomenon, incompatibility (see above), operates after an F' element enters into a recipient cell already carrying an F factor and is expressed as the inability of the superinfecting F' element and the resident F factor to coexist stably in the same cell (see “F' Formation” for an explanation of F').

Two genes, *traS* and *traT*, are required for surface exclusion, with TraT being an outer membrane protein. It is believed that the TraT protein might block mating pair stabilization sites or, alternatively, might affect the synthesis of structural proteins (other than OmpA) necessary for stabilization. If mating pair formation occurs in spite of the efforts of TraT, TraS is thought to block DNA-strand transfer, although how it does this is not known. An odd phenomenon never fully explained is that growth of F⁺ cells into late stationary phase imparts a recipient ability almost equivalent to an F⁻ cell. These cells are called F⁻ phenocopies and are nonpiliated.

Fertility Inhibition. Most conjugative plasmids transfer their DNA at a markedly reduced rate as compared with F because these plasmids (e.g., R100) possess a regulatory mechanism (the FinOP fertility inhibition system) that normally represses their *tra* genes. The *finO* and *finP* products interact to form a FinOP inhibitor of *tra* gene expression. In contrast to these other plasmids, the F factor is *finO* *finP*⁺ (due to an IS3 insertion in *finO*; see “Insertion Sequences and Transposable Elements”) and so is naturally derepressed for conjugation. However, if another plasmid that is *finO*⁺ resides in the same cell as F, then the F *finP* gene product and the coresident plasmid's *finO* gene product can interact, producing the FinOP_F inhibitor. This will act as a negative regulator of the F factor's *traJ*, which itself is the positive regulator for the other *tra* genes. Consequently, the fertility of F will be inhibited.

FinP is a 79 nucleotide untranslated RNA molecule that is complementary (antisense) to the untranslated leader region of *traJ* mRNA. FinP antisense RNA is thought to form an RNA/RNA duplex with *traJ* mRNA, occluding the *traJ* ribosome-binding site and thereby preventing translation of the TraJ protein. The FinO protein (21 kDa) in F-like plasmids other than F stabilizes FinP antisense RNA by preventing degradation by RNase E. This increases the in vivo concentration of the antisense RNA. In addition, FinO binds to the FinP and TraJ RNA molecules, promoting duplex formation and thereby enhancing inhibition of TraJ translation.

Hfr Formation. The F factor is an example of a plasmid that can exist autonomously in a cell or can integrate into the bacterial chromosome. A cell that contains an integrated F factor is referred to as an **Hfr cell** (high frequency of recombination). The frequency of insertion occurs at about 10⁻⁵ to 10⁻⁷ per generation—that is,

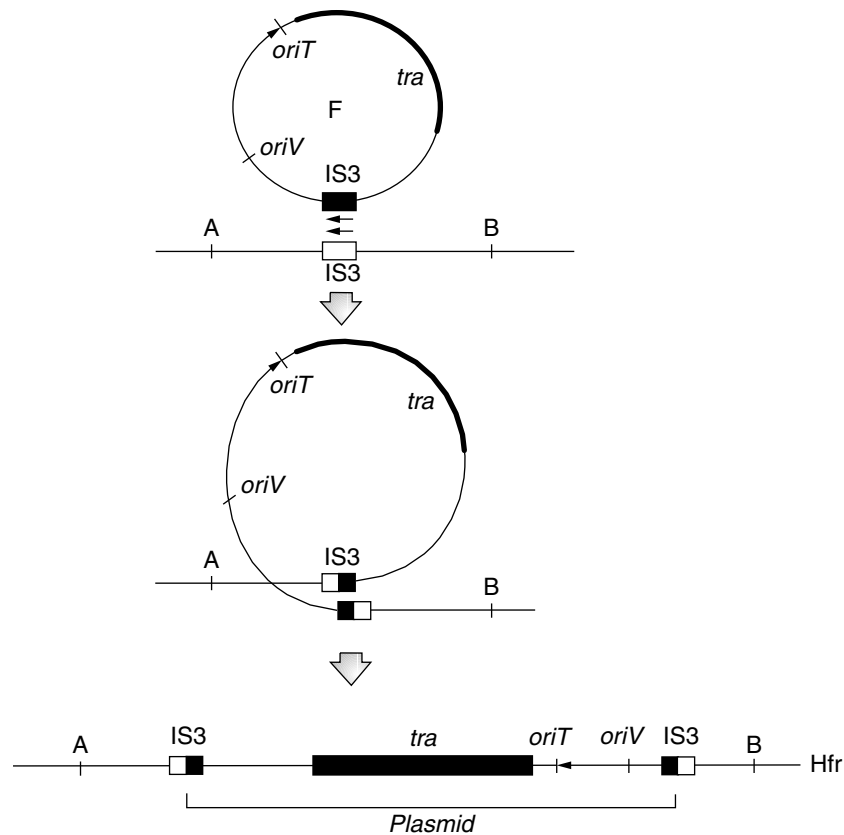


Fig. 3-7. Mechanism of integration of the F factor into the bacterial chromosome. *IS3* is an insertion element present on the *E. coli* chromosome and F factor.

among a population of 10^7 F^+ cells, 1 to 100 cells will have an integrated F. The mechanism of integration is illustrated in Figure 3-7. Integration involves homologous recombination between two covalently closed circular DNA molecules forming one circular molecule that contains both of the original DNA structures. It is thought that the insertion sequences (*IS*) present in the F genome (F carries two *IS3* and one *IS2* sequence) and those in the host chromosome (*E. coli* contains five each of *IS2* and *IS3*) serve as regions of homology for the insertional event. F integration is predominantly RecA dependent (see "Recombination"), but rare RecA-independent Hfr's can be formed based on the transposition functions of the insertion elements. Consequently, Hfr formation is mainly a nonrandom event, primarily occurring in regions of the chromosome containing an insertion sequence element.

Once integrated, the F DNA is replicated along with the host chromosome. However, in situations where the host *dnaA* gene is inactive, replication of the entire chromosome can initiate from an integrated F in a process called **integrative suppression**.

An integrated F factor still has active transfer functions such that an Hfr to F^- cross will transfer host DNA to the recipient where the donor DNA can recombine with the recipient DNA. In Figure 3-7, note that the direction of transfer from *oriT* is such that the *tra* genes are always transferred last. Thus, there is directionality to conjugal

DNA transfer. In this illustration, the proximally transferred host gene is B while the distally transferred gene is A (the tip of the arrowhead represents the 5' leading end of the transferred strand). If the orientation of the chromosomal IS element was in the opposite direction, then the resulting Hfr would transfer gene A as a proximal marker and gene B would be one of the last genes transferred. Since conjugal transfer of the host chromosome in an Hfr cell is time dependent (it takes approximately 100 minutes to transfer the entire *E. coli* chromosome), a gene can be mapped relative to the position of the integrated F factor simply by determining how long it takes for the gene to be transferred to a recipient. The map position assignments for genes are often given in minutes; an assignment is based on the 100 minutes required for conjugal transfer.

F' Formation. As indicated in Figure 3-7, integration of F factor is a reversible process. Normally, excision of the F factor restores the host chromosome to its original state. However, improper or aberrant excision can occur at a low frequency, forming a plasmid containing both F and bacterial DNA. This type of plasmid is called an F-prime (F'). There are two types of faulty genetic exchanges that can result in F' formation. The first involves recombination between a region on the bacterial chromosome and one within the integrated F factor. The resulting F' has lost some F sequences but now carries some host DNA originally located at one or the other side of the integrated F. This is a type I F'.

As an alternative, host sequences located on both sides of the integrated F can undergo genetic exchange. The F' formed in this situation (type II) contains all of F plus some host DNA from both sides of the point where F was integrated. In both situations the host DNA contained in the F' is deleted from the host chromosome. However, when F' (type I or type II) is transferred to a new host, a partially diploid (merodiploid) situation occurs for the host genes contained on the F'. By constructing merodiploids, information can be derived regarding dominance of certain mutations over wild-type alleles of specific genes (see "cis/trans Complementation Test").

cis/trans complementation Test

When conducting a genetic analysis of mutations that affect a given phenotype, the investigator wants to know whether closely linked mutations reside in a single gene or in different genes of an operon and whether a given mutation resides within a control region (promoter/operator) or within a structural gene. Often these questions can be answered by performing complementation tests using merodiploids. For example, two different mutations are considered to lie in the same gene if their presence on duplicate genes in a merodiploid (one mutation on the plasmid, the other on the chromosome) fails to restore the wild phenotype—that is, the two mutations fail to complement each other in trans.

If the mutations lie in two different genes, then they should complement each other in trans—that is, an F' with the genotype $trpA^+ trpB^-$ (two genes involved in tryptophan biosynthesis) will produce a functional TrpA protein that can complement a host with the genotype $trpA^- trpB^+$. This premise depends, of course, on whether the gene produces a diffusible gene product—that is, protein or RNA. Regulatory genes such as operators or promoters do not produce diffusible products and so mutations in these genes cannot be complemented in trans. An exception occurs

with intragenic complementation. Two different mutations in a gene whose product functions as a multimeric enzyme (homodimer or tetramer) can sometimes complement each other in trans. The resulting active enzyme is composed of two different mutant polypeptides where the defect in one polypeptide sequence is compensated by the presence of the correct sequence on the other polypeptide. A wonderful example of this is called **α complementation** with β -galactosidase. Here a small N-terminal piece of β -galactosidase (called the α peptide) can form an active multimeric enzyme complex with the C-peptide part of β -galactosidase even when the genes encoding each peptide reside on separate plasmids.

Conjugation and Pheromones in Enterococci

For a long time it was believed that conjugation only occurred in the Enterobacteriaceae. Evidence accumulating since 1964, however, has shown that the streptococci also possess conjugative plasmids and conjugative transposons. A number of conjugative plasmids have been characterized, some of which can mobilize nonconjugative plasmids and even some chromosomal markers.

In *Enterococcus hirae* (*E. faecalis*) conjugative plasmids can be placed into two general categories. Members of the first group (e.g., pAD1, pOB1, pPD1, pJH2, pAM1, pAM2, pAM3) transfer at a relatively high frequency in broth (10^{-3} to 10^{-1} per donor cell). Members of the second group transfer poorly in broth but are fairly efficient when matings are carried out on filter membranes (e.g., pAC1, pIIP501, and p5M15346). The reason for this difference involves the production of sex pheromones by streptococci. Plasmids that transfer efficiently in broth use the pheromones to generate cell-to-cell contact. Recipient cells excrete small, soluble peptides (seven to eight amino acids long) that induce certain donor cells to become adherent. This adherence property facilitates formation of donor–recipient mating aggregates that arise from random collisions. Once the aggregate forms, transfer of the plasmid begins. The genes encoding the pheromone peptides have not been unambiguously identified, but computer searches of genomic databases have found several pheromone sequences located within the signal sequence segments of putative lipoproteins (all of unknown function).

The model for induction of pAD1 conjugation starts with a product of pAD1, TraA, repressing many other pAD1 genes (Fig. 3-8). The cAD1 pheromone produced from a potential recipient binds to the PrgZ surface component of the donor cell's oligopeptide permease. The peptide is then transported into the donor cell through a transmembrane channel formed by two other members of the permease: OppB and OppF. Once inside, cAD1 is thought to bind TraA, causing the repressor to release from its DNA-binding site. Transcription then proceeds through TraE, a positive regulator of many pCD1 transfer genes including the *asa1* gene (a.k.a. *prgB* in pCF10) encoding aggregation substance. Aggregation substance (AS) is a cell-surface microfibrillar protein that will recognize a compound known as binding substance (BS) produced on all potential recipient cells. The interaction between AS and BS causes aggregation between donor and recipient and stimulates conjugal transfer.

Once the plasmid is received, how does the new cell keep from stimulating itself? After all, it is still making the cAD1 pheromone. Autostimulation is prevented because the plasmid also contains a cAD1-inducible gene (*iad*; *prgY* in pCF10) that encodes an inhibitor of AD1 peptide. The iAD1 peptide binds to extracellular cAD, effectively neutralizing its effect. However, this only works at low concentrations of cAD1. In

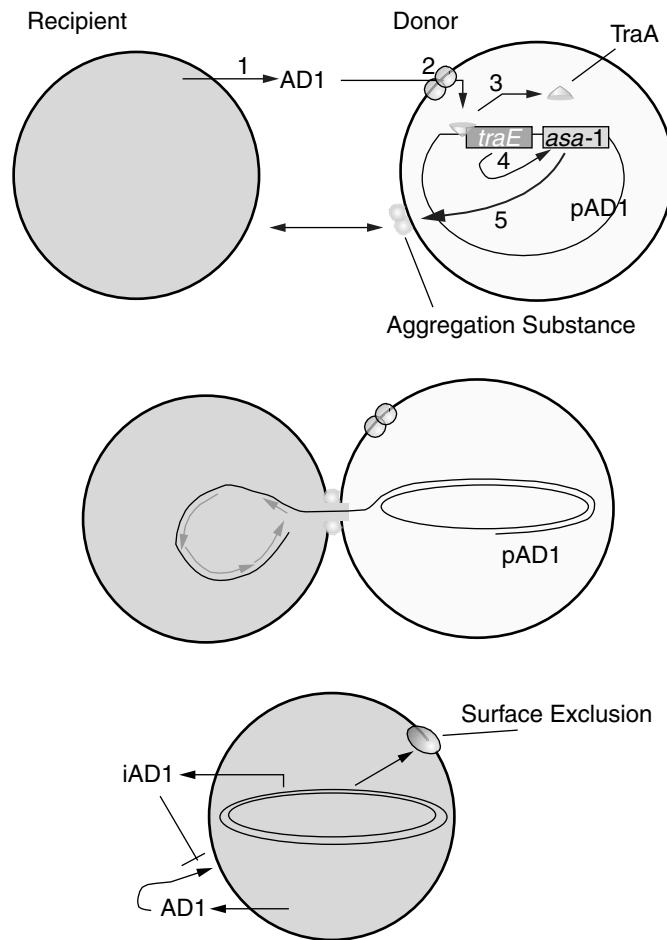


Fig. 3-8. Model of sex pheromone-induced conjugation of pAD1 in streptococci.

addition, cAD1 autoinduces genes required for cell surface exclusion, so a cell already containing a pAD1 plasmid will not receive another copy from a second cell.

It should be apparent that the role of pheromones goes beyond simple aggregate formation. This can be illustrated through donor-donor matings with two donors each having a different plasmid. When only one of the donors is induced, transfer occurs only in the direction from the induced strain to the uninduced strain. Thus, the pheromone also helps trigger transfer of the plasmid.

Conjugation, Cell-Cell Signaling, and Bacterial-Induced Tumors

The Ti plasmid in *Agrobacterium tumefaciens* provides an excellent example whereby environmental conditions influence conjugative functions. The Ti plasmid is very large (1.2×10^8 daltons) but can be transferred from this bacterium to dicotyledonous plants, resulting in production of a plant tumor (e.g., crown gall disease). When wounded, plant cells release specific phenolic signal molecules called opines (e.g., acetosyringone) that are sensed by a two-component signal transduction system, VirA/VirG (see Chapter 5).

In addition, monosaccharide and acid pH in the environment potentiate expression by the VirA/VirG system. Activation of the VirA/VirG system signals transcription of other *vir* genes whose products transfer a defined segment of the Ti plasmid to the plant nuclear genome, transforming the plant cell and leading to tumorous growth. Conjugal transfer of Ti between bacteria or between bacteria and plants is also regulated by cell–cell signaling. Donor bacteria induced by the conjugal opines release a soluble homoserine lactone signal molecule (autoinducer) that interacts with TraR (a homolog of the *Vibrio fischeri* LuxR) and induces conjugal functions (see “Quorum Sensing” in Chapter 5).

TRANSFORMATION

Not all bacteria are capable of conjugation, but that does not stop them from exchanging genetic information. The first demonstrated system of gene transfer actually did not require cell-to-cell contact. Griffith first discovered the phenomenon, called transformation, in 1928 in the course of his investigations of *Streptococcus pneumoniae* (pneumococcus). Capsule-producing pneumococci were shown to be virulent for mice, while nonencapsulated strains were avirulent. Griffith discovered that if mice were injected with mixtures of heat-killed encapsulated (smooth = S) and live uncapsulated (rough = R) cells, a curious phenomenon occurred:

$$\begin{array}{c} \text{Living} \\ \text{R cells} \end{array} + \begin{array}{c} \text{Heat-killed} \\ \text{S cells} \end{array} \xrightarrow[\text{into mice}]{\text{Injected}} \begin{array}{c} \text{Dead mice} \\ \text{(recovered living, virulent S cells)} \end{array}$$

The R cells recovered the ability to produce capsules and regained the capacity of virulence! There are different antigenic types of capsular material produced by different strains of pneumococci. Consequently, Griffith showed that if the avirulent R cells injected into the mice were derived from capsular type II cells and the heat-killed cells were of capsular type III, the viable, encapsulated cells recovered from the dead mice were of capsular type III. This indicated a transformation took place, so type II R cells now produced capsules of antigenic type III.

Later, in 1944, Avery, MacLeod, and McCarty demonstrated that it was DNA from the heat-killed encapsulated strain that was responsible for the transformation. They found that if living, rough type II cells were exposed to DNA isolated from type III encapsulated pneumococci, viable type III encapsulated organisms that were virulent for mice could be recovered. These findings were of exceptional importance because they showed that DNA had the ability to carry hereditary information. Conventional wisdom prior to this time held that hereditary traits were more likely to be borne by protein molecules.

Transformation occurs in many bacterial genera including *Haemophilus*, *Neisseria*, *Xanthomonas*, *Rhizobium*, *Bacillus*, and *Staphylococcus*. Considerable effort has been exerted to elucidate the nature of competence for transformation on the part of recipient cells. Competence is defined as a physiological state that permits a cell to take up transforming DNA and be genetically changed by it. Organisms that undergo natural transformation can be divided into two groups based on development of the competent state. Some organisms become transiently competent in late exponential phase — for example, *Streptococcus pneumoniae*. Others, such as *Neisseria*, are always

competent. These different patterns of competence development belie a complex series of regulatory processes required to control this process. The specific physiological and genetic factors involved are discussed in relation to specific groups of organisms.

Transformation processes in different microorganisms can be categorized into two main DNA uptake routes known as the *Streptococcus*–*Bacillus* model (gram-positive bacteria) and the *Haemophilus*–*Neisseria* model (gram-negative bacteria). Keep in mind that separation into these two basic models is artificial, since some of the bacteria in one group share features of the transformation process typically associated with the other group.

Gram-Positive Transformation

***Streptococcus*.** Transformation in streptococci has been studied primarily in *Streptococcus pneumoniae* and in members of the streptococci belonging to serological group H. The competent state is transient and persists for only a short period during the growth cycle of a culture of recipient bacteria. The competent state in pneumococci is induced by a specific protein, the **competence-stimulating peptide (CSP)** (17 amino acids). Binding of this activator protein to receptors on the plasma membrane triggers the synthesis of at least a dozen new proteins within minutes. After induction by CSP, cells develop the capacity to bind DNA molecules. After binding to recipient cell membranes (there are approximately 30 to 80 binding sites per cell), donor DNA molecules are acted upon by a **translocasome** (also known as **transformasome**) complex located at the cell surface of competent recipient cells. The translocasome includes the endonuclease EndA. This nuclease attacks and degrades one strand of DNA while facilitating the entry of the complementary strand into the cell (Fig. 3-9). After entry, the single-stranded DNA can be recombined into the chromosome of the transformed cell.

The development of competence in *S. pneumoniae* is a wonderful example of a phenomenon known as **quorum sensing** (see Chapter 5). Under appropriate conditions, the 17 amino acid CSP (encoded by *comC*) is processed, exported from the cell, and accumulates in the growth medium until a certain cell density is reached. Once CSP accumulates to a certain level, it binds to the membrane-localized ComD histidine kinase, which autophosphorylates and subsequently transphosphorylates its cognate response regulator protein ComE. ComE-P then activates transcription of competence-specific genes, including the *comCDE* operon in an autocatalytic feedback loop. ComE-P also appears to trigger a regulatory cascade that induces expression of an alternative σ factor ComX (σ_H), which in turn drives the expression of numerous other operons. Some of these new expressed gene products form the DNA uptake transformasome while other products are thought to remodel the cell wall around the transformasome. Other genes regulated by σ_H , such as those encoding single-strand binding protein and the recombination enzyme RecA, act on DNA after its entry.

Hex Mismatch Repair. Following recombination between single-stranded donor DNA and the recipient chromosome, nucleotide mismatches occur. A mismatch repair system called Hex (analogous to the MutLS system of *E. coli*; see below) repairs different base–base mismatches with different efficiencies. Thus, the efficiency of integration of genetic markers into the genome of the recipient cell varies drastically with different types of mismatches. Transition mutations (G/T or

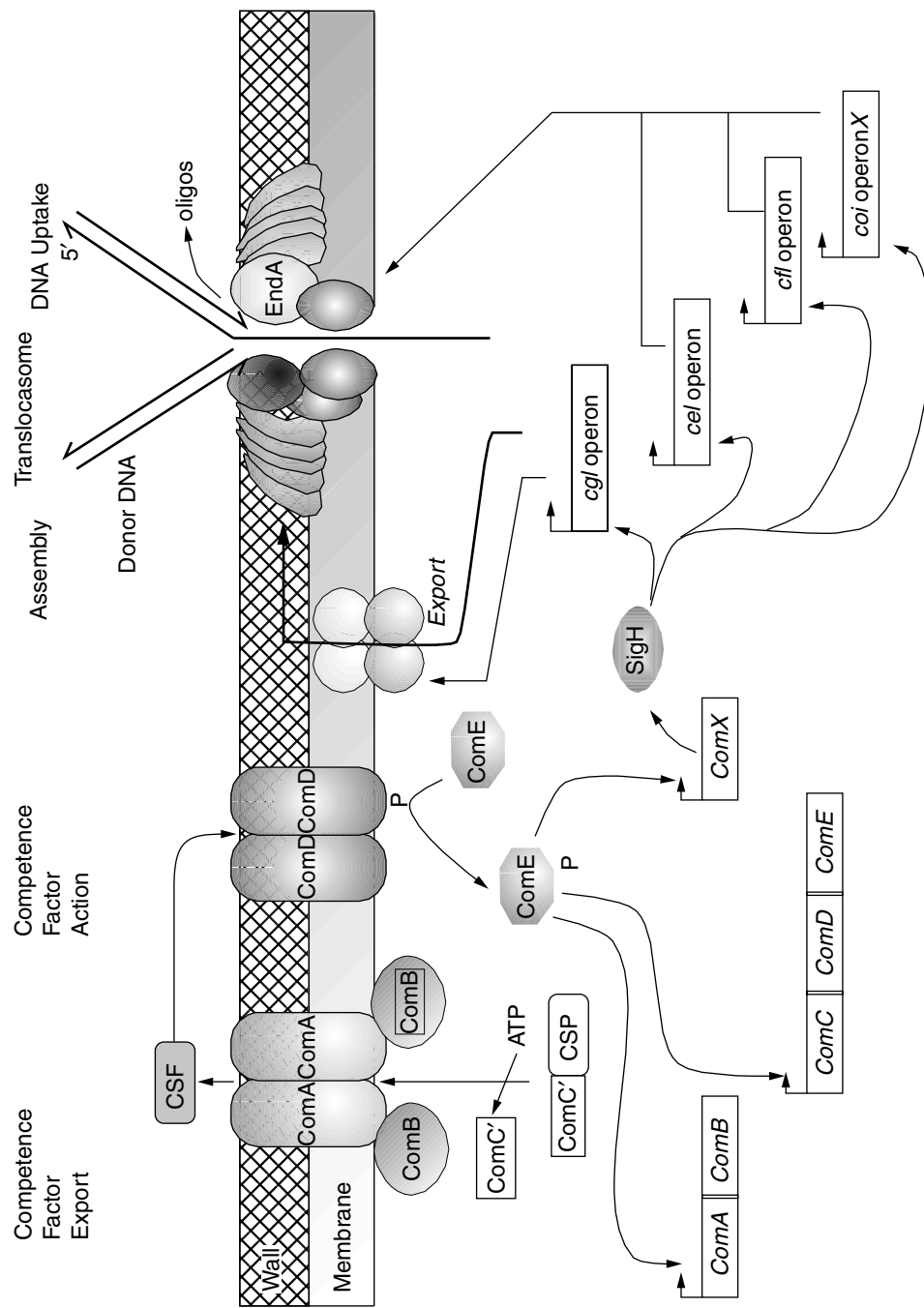


Fig. 3-9. Schematic view of transformation in streptococci.

A/C mismatches) exhibit a low efficiency of transformation (LE markers) because the Hex system is very efficient at repairing the mutations. The HexA protein recognizes the mismatch in heteroduplex DNA resulting from transformation and selectively removes the entire donor strand, which can be as large as up to several kilobases. It is also proposed that HexB protein is activated by binding HexA–DNA mismatch complexes and in some manner increases the efficiency of the system.

In contrast to LE transition mutations, transversion mutations (G/C or C/C mismatches) are not recognized by the Hex system. DNA fragments containing these mutations exhibit 10- to 20-fold higher transformation efficiency and are referred to as high-efficiency (HE) markers. How does the Hex system recognize the donor and recipient strands? Strand discrimination may be determined by identifying the presence of DNA ends in the donor DNA fragments.

***B. subtilis*.** Like *Streptococcus*, *Bacillus* spp can indiscriminantly bind any DNA to its surface. Donor DNA becomes associated with specific sites on competent cell membranes (ComEA protein involvement) and remains associated with these membranes until it is integrated. Competence development in *B. subtilis* is subjected to three types of regulation: nutritional, growth stage specific, and cell-type specific. The regulatory pathway of competence intersects with those of several other stress responses including sporulation, motility, and secretion of macromolecule-degrading enzymes. It is useful to think of competence development as consisting of two modules. Module 1 is a quorum-sensing machine whose major input is population density signaled by two pheromones (CSF and ComX). Once the pheromones are sensed, a small peptide, ComS, is produced. The second module senses ComS and produces the transcriptional regulator ComK in response. ComK activates transcription of the late competence genes required to import DNA and the recombination genes required to incorporate DNA into the chromosome.

As illustrated in Figure 3-10, the ComQ protein is required to process pre-ComX into the smaller peptide pheromone ComX, which is secreted into the medium. As cell density increases, so does the extracellular concentration of ComX. Once it reaches a critical concentration, ComX will interact with a membrane histidine kinase sensor protein called ComP. This triggers a regulatory cascade in which ComP autophosphorylates itself on a conserved histidine residue and then transphosphorylates a cytoplasmic regulatory protein called ComA. Phosphorylated ComA (ComA-P) is a transcriptional activator that activates the *comS* gene. Production of ComS completes the pheromone module.

The ultimate goal of this regulatory cascade is to increase the level of ComK, the transcriptional activator of the late competence genes. However, ComK is subject to degradation by the ClpCP protease. An increase in ComS will interfere with this degradation, allowing ComK to accumulate. Regulation of ComK proteolysis is actually very similar to what we described previously for the *E. coli* σ S protein. In *Bacillus*, an adaptor protein called MecA binds to ComS and to the ClpC ATPase chaperone of the ClpCP protease, thus expanding the substrate specificity of ClpC (Fig. 3-10). ComK is then fed into the ClpP barrel and degraded. ComS, however, competes with ComK for binding MecA. Thus, at high enough concentrations, ComS will preferentially bind ClpC and effectively prevent the degradation of ComK. ComK levels increase as a result and induce expression of the genes required for uptake and processing of DNA. The uptake system appears to be very similar to that of *Streptococcus*.

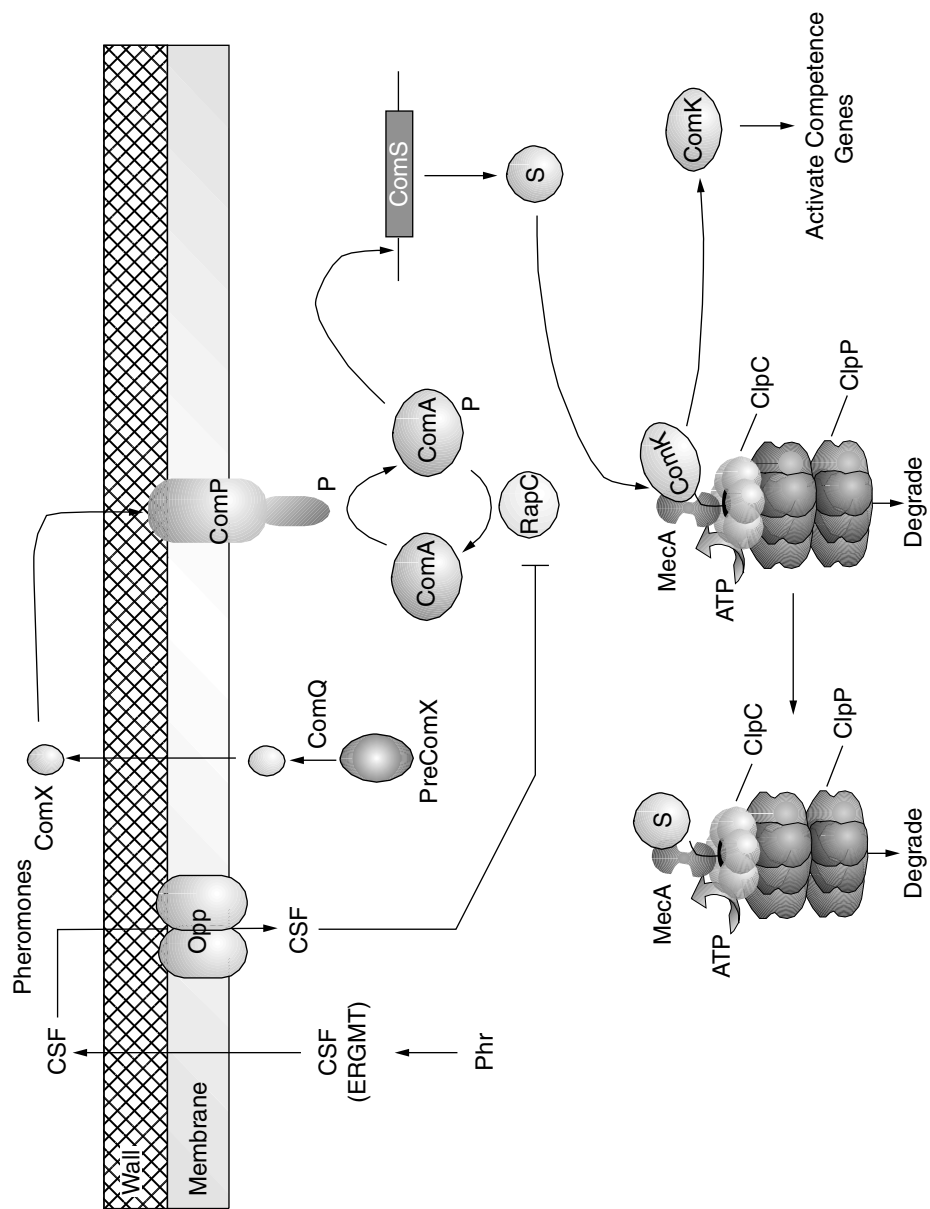


Fig. 3-10. Competence development in *Bacillus subtilis*.

Gram-Negative Transformation

In addition to needing a mechanism to transport DNA intact across the cytoplasmic membrane, gram-negative organisms have to move DNA through the outer membrane. The best-studied gram-negative systems are *Neisseria gonorrhoeae* and *Haemophilus influenzae*. Unlike gram positives, these organisms are naturally competent (no CSF), but competence is induced by entry into the stationary phase. In another departure from the gram-positive systems, transformation in gram-negative systems is sequence specific.

***Neisseria*.** DNA first nonspecifically binds to the neisserial cell surface by electrostatic forces (probably via the Opa outer membrane proteins). Then an unknown protein or protein complex recognizes a specific sequence (GCCGTCTGAA) within the bound DNA. This sequence, resembling a bar code, indicates what DNA is suitable for uptake. Transport of the tagged DNA across the membrane involves an amazing type IV pilus structure similar to that described previously for type II protein export (Chapter 2). The pilus biogenesis factors involved are PilD, T, F, and G, located at the inner membrane. The pilus, comprised of PilE protein, is assembled and protrudes through an outer membrane pore formed by PilQ (a.k.a. Omc). Another protein, PilC, helps gate the polymerized pili through the pore. This same pore serves as the import channel for DNA during transformation.

Recall that type IV pili polymerize and depolymerize, moving like a piston in and out of the outer membrane (see Fig. 2-31B in Chapter 2). One model for DNA uptake is that depolymerization of the pilus would somehow drag DNA into the periplasm. Once in the periplasm, the net-like peptidoglycan sacculus in the periplasm is likely to represent a significant barrier for incoming DNA. Two competence proteins (ComL and Tpc) bind to and cleave peptidoglycan, respectively. It is presumed that these proteins either transiently cleave murein or guide DNA through the sacculus net. The final stage, transport across the inner membrane, requires the inner membrane protein ComA.

***Haemophilus influenzae*.** If exponentially growing *H. influenzae* cells are shifted to nongrowth starvation conditions, 100% of the cells become competent. No secreted competence factors have been reported, so competence development is internally regulated. Changes in intracellular cyclic AMP (cAMP) levels play an important role in this process. As is discussed more fully in Chapter 5, cAMP is an intracellular signal molecule that accumulates as energy sources become scarce. A complex between this molecule and cAMP receptor protein serves as a potent gene regulator in many organisms including *Haemophilus*. As the cells begin to starve upon entering stationary phase, cAMP levels increase and trigger expression of a variety of genes including those for competence. Another competence regulatory gene in *Haemophilus* is called *sxy*. Mutations in this gene either eliminate competence or render it constitutive. An operon of six genes, *comA–comF*, is induced during competence development, but the activator of expression is unknown. Other genes associated with competence in this organism are Por, a periplasmic foldase that is probably involved in the maturation of periplasmic proteins required for transformation, and DprA, a protein required for DNA transport across the inner membrane.

In *H. influenzae*, changes in the cell envelope accompany the development of the competent state. Envelopes from competent cells exhibit elevated levels of lipopolysaccharide with a composition different from that of log-phase cell envelopes.

Six apparently new polypeptides are found in envelopes from competent cells. Most of the polypeptide changes are confined to the outer membrane, although one new polypeptide is associated with the inner cytoplasmic membrane. Structural changes in the envelope also occur in competent cells. Numerous vesicles called *transformasomes* bud from the surface and contain proteins that react specifically with conserved sequences (5'AAGTGCGGTCA3') present at 4 kb intervals on *Haemophilus* DNA. These vesicles appear to mediate the uptake of transforming DNA. The DNA uptake site is made up of two proteins: 28 kDa and 52 kDa. Several DNA receptor proteins are also produced that recognize the conserved DNA sequence. After binding, the receptor proteins present the donor DNA to the membrane-associated uptake sites. Consequently, DNA binding and uptake are very specific for *Haemophilus* DNA.

Why Competence? There are three proposals that attempt to explain the intrinsic cellular value of developing transformation competence. One model is that cells can use the DNA as an energy source. This may be possible for organisms such as *Streptococcus* and *Bacillus*, which indiscriminantly take up DNA from many different sources. However, the *H. influenzae* and *N. gonorrhoeae* systems exhibit substrate specificity with respect to DNA uptake. This does not suggest a food-gathering mechanism.

A second proposed function is that transformation plays a role in DNA repair where lysed cells provide a source of DNA that can be taken up and used for the repair of otherwise lethal mutations. The finding that DNA repair machinery is induced as part of competence regulon supports this proposal.

The third popular hypothesis proposes that transformation is a mechanism for exploring the "fitness landscape." For example, it is likely that transformation has played a role in horizontal gene transfer in which a gene from one species of bacteria becomes part of the genome of a second species. If these horizontally acquired genes improve the fitness of the organism, the genes will be retained and the more fit strain will ultimately dominate the population.

Transfection and Forced Competence

Transfection involves the transformation of bacterial cells with purified bacteriophage DNA. The transformed viral DNA will replicate and ultimately produce complete virus particles. Transfection has been demonstrated in a number of bacteria, including *B. subtilis*, *H. influenzae*, *Streptococcus*, *Staphylococcus aureus*, *E. coli*, and *Salmonella typhimurium*.

It should also be noted that organisms not considered naturally transformable (e.g., *E. coli* and *S. typhimurium*) can be transformed under special laboratory conditions. Alterations made in the outer membrane with CaCl_2 or through an electrical shock (electroporation) can be used to transfer DNA such as plasmids into cells. This has been an important factor in the success of recombinant DNA research.

TRANSDUCTION

Transduction is the transfer of bacterial genetic markers from one cell to another mediated by a bacteriophage. There are two types of transduction: generalized and specialized.

Generalized transduction is the phage-mediated transfer of any portion of a donor cell's genome into a second cell. The transducing viral particle contains only bacterial DNA, without phage DNA. During normal loading of nucleic acid into virus protein heads (see Chapter 6), the packaging apparatus occasionally makes a mistake, packaging chromosomal DNA into the phage rather than phage DNA. When a transducing bacteriophage binds to a bacterial cell, the donor DNA is injected into the bacterium by the phage and becomes integrated into the genome of the new cell through generalized recombination. Thus, all genes from a donor cell can potentially be transduced to a recipient cell population. Phages that can mediate generalized transduction include P1 (*E. coli*) and P22 (*Salmonella*).

A well-studied example of generalized transduction is that of *Salmonella* phage P22. To understand how generalized transduction occurs, you must have some understanding of how this phage replicates and packages DNA into its head. Phage DNA in P22 is linear but circularly permuted and terminally redundant. Thus, the termini of a P22 DNA molecule contain duplicate DNA sequences, but different P22 molecules within a population contain different terminal sequences. This seemingly odd chromosome structure is due to the packaging method P22 uses to insert DNA into its head. After infecting a cell, the linear P22 DNA circularizes through recombination between the terminal redundant ends. The circular molecule replicates via a rolling circle type of DNA synthesis, forming a long concatemer. A concatemer is a long DNA molecule containing multiple copies of the genome.

Packaging of P22 DNA into empty heads initiates at a specific region in P22 DNA called the *pac* site, where the DNA is first cut, and then proceeds along the concatemer. Once the P22 head is full (headful packaging), a second, nonsequence-specific cut is made. This second cut also defines the start of packing for the next phage head. However, to assure that each head receives a complete genome, the P22 system packs a little more than one genome length of P22 DNA. This is the source of the terminal redundancy of the P22 genome. One consequence of packing extra DNA is that the packing start site changes for each subsequent phage DNA molecule packaged from the concatemer. Each duplicated sequence at the ends of the P22 genome is different for each packed P22 molecule.

When the packaging system encounters a sequence in the bacterial chromosome that is similar to a *pac* site, it does not distinguish this site from P22 *pac* sites. It will use this homologous site to package chromosomal DNA such that progressive packaging will generate a series of phage particles that carry different parts of the chromosome. The size of packaged DNA is approximately 44 kb, which is about one-hundredth the size of the *Salmonella* chromosome. Therefore, a given P22 will package the equivalent of one conjugation minute worth of chromosomal DNA.

Cotransduction is the simultaneous transfer of two or more traits during the same transduction event, which enables the mapping of genes relative to each other. Cotransduction of two or more genes requires that the genes be close enough to each other on the host chromosome such that both genes can be packaged into the same phage head. The closer the two genes are to each other, the higher the probability they will be cotransduced. The only thing that could separate the two genes would be a recombinational event occurring between them. Thus, the closer the two genes are to each other, the smaller the recombinational target that could separate them (Fig. 3-11).

In **abortive transduction**, DNA that is transferred to the recipient cell does not become integrated into the genome of the recipient cell. Since this DNA is not

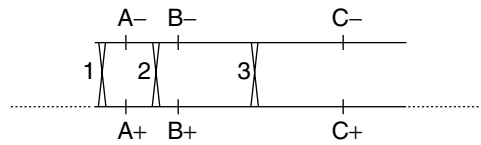


Fig. 3-11. Cotransduction of linked markers. Cross-over events 1,3 will occur more frequently than 1,2.

replicated, it is transmitted unilaterally from the original cell to only one of the daughter cells and only transiently expresses the function of the genetic information in these cells. It is not stably inherited, since passage of this DNA eventually is diluted out through subsequent cell divisions.

Specialized transduction is mediated by bacteriophage that integrate (lysogenize) into a specific site (*att*) on the bacterial chromosome. This specificity limits transfer of genetic material to host markers that lie within the immediate vicinity of this site. Lambda (λ) phage is the most actively studied phage of this type. λ phage almost invariably integrates at an *att* site near the galactose (*gal*) region of the chromosome and therefore can mediate specialized transduction of genes in that area to recipient cells (see Lambda Phage in Chapter 6).

Both low-frequency and high-frequency transduction have been described for λ (Fig. 3-12). An infecting λ DNA integrates into the bacterial chromosome (A). In low-frequency transduction (LFT), the integrated phage (prophage) may be induced through some forms of stress (e.g., DNA damage; see “SOS-Inducible Repair”) to enter a cycle of lytic infection. As a relatively rare event (10^{-5} – 10^{-6} per cell), a portion of the genome of the phage is replaced by a specific segment of the host chromosome (B). This occurs due to improper excision of integrated prophage DNA in a manner similar to the formation of type I F' factors (see Fig. 3-7). These are defective phages because

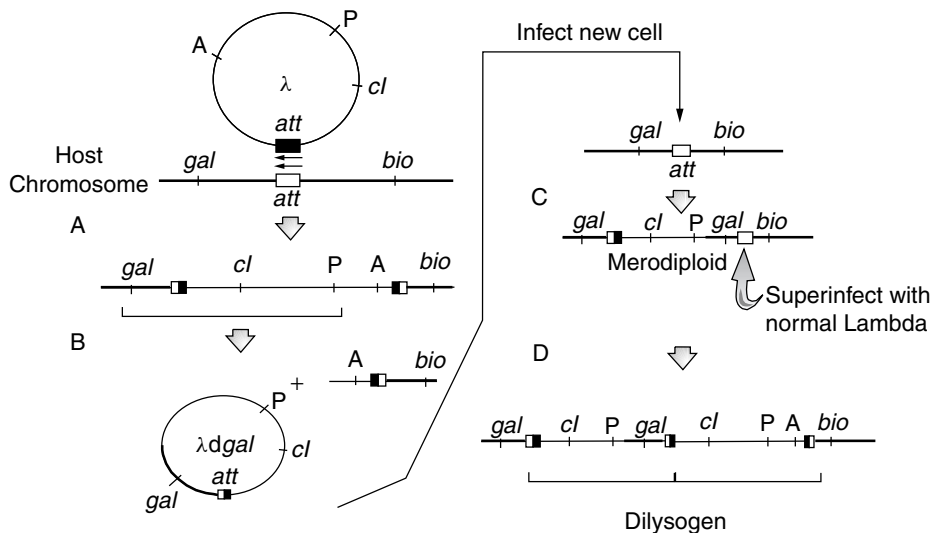


Fig. 3-12. Specialized transduction with λ phage showing the production of low-frequency transducing (LFT) and high-frequency transducing (HFT) lysates.